

***PM/APM UDC 6000
PM/APM UDC 6300
Integration Manual***

PM12-520

**Implementation
Process Manager - 1**

***PM/APM UDC 6000
PM/APM UDC 6300
Integration Manual***

**PM12-520
Release 500
08/04**

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About This Publication

This publication provides information about integrating UDC 6000 and UDC 6300 Process Controllers with a **TotalPlant** Solution (TPS) system Release 500 or later, through the Process Manager or the Advanced Process Manager. It covers data pertinent to the unique integration tasks and relies on the standard UDC 6000/6300 and TPS publications for individual routine operation information reference.

This publication supports **TotalPlant** Solution (TPS) system network Release 500 and later. TPS is the evolution of TDC 3000^X.

Table of Contents

SECTION 1 – INTRODUCTION TO INTEGRATION	1
1.1 Overview.....	1
1.2 PM/APM - UDC 6000/6300 Integration	2
1.3 Data Exchange Functions	4
1.4 Integration Components	7
1.5 Installation Checklist.....	11
SECTION 2 – INSTALLATION.....	13
2.1 Overview.....	13
2.2 Installation Tasks.....	14
2.3 Installing SDI IOP	15
2.4 Installing Power Adapter	16
2.5 Installing SDI/UDC 6000/6300 FTA	17
2.6 Wiring SDI IOP to Power Adapter	18
2.7 Wiring Power Adapter to SDI/UDC 6000/6300 FTA	19
2.8 Installing UDC 6000/6300 Controller.....	23
2.9 Wiring Digital Communications Link.....	24
2.10 Loading UDC 6000/6300 Graphics Software	27
SECTION 3 – SET UP	33
3.1 Overview.....	33
3.2 Configuration Process	34
3.3 Building a PM/APM Box Data Point	35
3.4 Identifying Mapped SDI Parameters	41
3.5 Building an SDI IOP Point	54
3.6 Configuring UDC 6000/6300 Controller	64
3.7 Configuring UDC 6000/6300 Schematics	65
3.8 Preparing SDI for Operation.....	106
SECTION 4 – PM/APM - UDC 6000/6300 CASCADE CONTROL INTEGRATION.....	111
4.1 PM/APM - UDC 6000/6300 Cascade Interface.....	111
4.2 UDC 6000/6300 SDI Point Connections	112
4.3 UDC 6000/6300 - .ii.PM/APM Interface Verification	114
4.4 UDC SDI Interface Parameters	115
4.5 UDC 6000/6300 Cascade Connections to PIDERFB Regulatory Points.....	116
4.6 PM/APM - UDC 6000/6300 Cascade Options	118
4.7 PM/APM - UDC 6000/6300 .ii.Cascade Options Configuration	120
4.8 SPCRSP Cascade (Supervisory using remote UDC setpoint)	121
4.9 SPCCSP Cascade (Supervisory using computer [TDC] setpoint).....	125
4.10 DDCLSP Cascade (Direct Control with local [UDC] setpoint)	128
4.11 DDCRSP Cascade (Direct Control with remote [UDC] setpoint)	131
4.12 DDCCSP Cascade (Direct Control with computer [TDC] setpoint).....	134
SECTION 5 – .II.OPERATION.....	139
5.1 Overview.....	139
5.2 UDC 6000/6300 Summary Display	140
5.3 UDC 6000/6300 Detail Display	143
5.4 Standard Detail Display	163
5.5 UDC 6000/6300 Group Display.....	165
5.6 UDC 6000/6300 Operator Change Zone Display	167
SECTION 6 – .II. SERVICE	171
6.1 Overview.....	171
6.2 Diagnostics	172
6.3 Removing a UDC 6000/6300 from an Active Link	176

Figures and Tables

Figure 1-1	Typical PM or APM - UDC 6000/6300 Integration Hierarchy	3
Figure 1-2	Mapped Parameters Are Basis for Data Exchange.....	4
Figure 1-3	SDI IOP Appears as an STI IOP to Other Components.....	5
Figure 1-4	Sixteen AI Points per SDI IOP	6
Figure 1-5	Identifying Integration Components	8
Figure 3-1	IOP Slot vs RS485 Address.....	42
Figure 3-2	Address vs Slot Configuration	43
Figure 4-1	UDC 6000/6300 to PM/APM Point Interface (Example).....	112
Figure 4-2	UDC 6000/6300 Loop in SPCRSP Configuration.....	121
Figure 4-3	UDC 6000/6300 SPCRSP Cascade w/ PM/APM Primary Block Diag	123
Figure 4-4	UDC 6000/6300 SPCRSP Cascade w/UDC Primary Block Diagram	124
Figure 4-5	UDC 6000/6300 Loop in SPCCSP Configuration.....	125
Figure 4-6	UDC 6000/6300 - SPCCSP / PIDERFB Interface	127
Figure 4-7	UDC 6000/6300 Loop in DDCLSP Configuration	128
Figure 4-8	UDC 6000/6300 - DDCLSP STIM SDI Point Connections	130
Figure 4-9	UDC 6000/6300 Loop in DDCRSP Configuration	131
Figure 4-10	Simplified UDC 6000/6300 DDCRSP Cascade Block Diagram	133
Figure 4-11	UDC 6000/6300 Loop in DDCCSP Configuration	134
Figure 4-12	Simplified UDC 6000/6300 DDCRSP Cascade Block Diagram	137
Figure 5-1	Detail Display Selections	143
Figure 5-2	PV Point Detail Display	147
Figure 5-3	Regulatory Schematic Display.....	151
Figure 5-4	Typical CONFIG PAGE for Standard Detail Display	163
Figure 5-5	Command Display for Reference.....	164

Table 1-1	Basic and Full Modes.....	7
Table 1-2	Part Numbers for Components Identified in Figure 1-5	9
Table 1-3	Cable Characteristics for Communications Link.....	10
Table 1-4	Component Checklist.....	11
Table 2-1	Installation Tasks Breakdown	14
Table 2-2	Installing an SDI IOP.....	15
Table 2-3	Installing a Power Adapter	16
Table 2-4	Installing an SDI/UDC 6000/6300 FTA	17
Table 2-5	Wiring SDI IOP to Power Adapter.....	18
Table 2-6	Wiring Power Adapter to SDI/UDC 6000/6300 FTA	19
Table 2-7	Installing UDC 6000/6300 Controllers	23
Table 2-8	Wiring Digital Communications Link	24
Table 2-9	Example of Selecting a Volume.....	28
Table 2-10	Loading UDC 6000/6300 Graphics Software	31
Table 3-1	Stages of the Configuration Process	34
Table 3-2	Building a PM/APM Box Data Point.....	35
Table 3-3	PV Point SDI Parameter Definitions	44
Table 3-4	OP Point SDI Parameter Definitions.....	49
Table 3-5	Building an STI IOP Point for a UDC 6000/6300 to SDI Point.....	55
Table 3-6	Configuring the Summary Display	65
Table 3-7	Adding a Point ID or Group Descriptor(s) to the Summary Display	76
Table 3-8	Configuring the Group Display.....	81
Table 3-9	Adding a Loop to the Group Display.....	93
Table 3-10	Adding an Operator Change Zone.....	97
Table 3-11	Preparing SDI for Operation	106
Table 4-1	UDC 6000/6300 RS485 Address to PM/APM IOP Point Mapping	113
Table 4-2	Basic SDI Parameters.....	115
Table 4-3	SDI Parameters Used for Cascade	115
Table 4-4	PIDERFB Input and Output Connections	116

Figures and Tables

Table 4-5	PIDERFB Related Parameters	117
Table 4-6	Cascade Terms and Definitions	118
Table 4-7	Configuring the Cascade Option	120
Table 4-8	Control Mode Definitions for SPCRSP Cascade	122
Table 4-9	Configure the UDC with PM/APM Primary	123
Table 4-10	Configure the UDC with UDC Primary	124
Table 4-11	Control Mode Definitions for SPCCSP Cascade	126
Table 4-12	SPCCSP - PIDERFB Point Build Connections	127
Table 4-13	Control Mode Definitions for DDCLSP Cascade	129
Table 4-14	DDCLSP - PIDERFB Point Build Connections	130
Table 4-15	Control Mode Definitions for DDCRSP Cascade	132
Table 4-16	DDCRSP - PIDERFB Point Build Connections	133
Table 4-17	Control Mode Definitions for DDCCSP Cascade	135
Table 4-18	DDCCSP - PIDERFB Point Build DDC (Output) Connections	136
Table 4-19	DDCCSP - PIDERFB Point Build SPC (Setpoint) Connections	136
Table 5-1	Calling Up Summary Display for Detail or Group Selection	140
Table 5-2	Display Targets	143
Table 5-3	Selecting OP Point Detail Display	144
Table 5-4	OP Point Detail Display Changeable Parameters	145
Table 5-5	Calling Up the Standard Detail Display	146
Table 5-6	PV Point Detail Display Changeable Parameters	148
Table 5-7	Calling Up the Standard Detail Display	149
Table 5-8	FIRST, OP DATA, and CONFIG Pages Summary	150
Table 5-9	Parameters on FIRST Page	152
Table 5-10	Adjusting Output Manually	153
Table 5-11	Calling Up OP DATA Page	155
Table 5-12	Changeable Parameters on OP DATA Page	156
Table 5-13	Calling Up CONFIG page	157
Table 5-14	Calling Up CONFIG Page for Loop 2	158
Table 5-15	Changing an Alarm Configuration Parameter	160
Table 5-16	Command Definitions for Mapped SDI Interface	164
Table 5-17	Changeable Parameters on Group Display	165
Table 5-18	Changing Set Point Value	165
Table 5-19	Calling Up a Custom Operator Change Zone Display	167
Table 5-20	Changeable Parameters on Change Zone Display	167
Table 5-21	Adjusting Output Manually from Change Zone Display	168
Table 6-1	Universal Station—System Status	172
Table 6-2	SDI IOP Diagnostics	172
Table 6-3	UDC 6000/6300 FTA Diagnostics	173
Table 6-4	STIM Point Detail Diagnostics	174
Table 6-5	Custom Schematic Diagnostics	175
Table 6-6	Removing a UDC from an Active Link	176

Acronyms

AUTO	Automatic Mode
CAS	Cascade Mode
Ddc	Direct Digital Control
DdcRsp	Direct Digital Control with Remote Setpoint Tracking
FTA	Field Termination Assembly
I/O	Input/Output
IOM	Input/Output Module
IOP	Input/Output Processor
LCN	Local Control Network
MAN	Manual Mode
NaN	Not a Number
NIM	Network Interface Module
PED	Parameter Entry Display
PM	Process Manager
PMM	Process Manager Module
PV	Process Variable
RTU	Remote Terminal Unit
SDI IOP	Serial Device Interface Input/Output Processor
SDI	Serial Device Interface
SP	Setpoint
SPC	Supervisory Control
STI	Smart Transmitter Interface
STIM	Smart Transmitter Interface Module
UCN	Universal Control Network
UDC 6000	Universal Digital Controller 6000
UDC 6300	Universal Digital Controller 6300
US	Universal Station

Parameters

C1	Output Override (UDC 6000/6300)
C2	PV Override (UDC 6000/6300)
CJTACT	Internal Cold Junction Temperature
COMMAND	Command for Data Transfers
CTLOPT	Control Processor Option
DAMPING	Transmitter Filter Lag Time
DECONF	DE Configuration Mode
EUDESC	Engineering Unit Descriptor
FAILOPT	Failure Option Outputs
INPTDIR	Input Direction
IOMTYP	Module Type
IOREDOPT	Redundancy Option
K	Overall Gain
KEYWORD	Point Keyword
LOCUTOFF	Low Signal Cutoff
LRV	Lower Range Value
MODNUM	Module Number
NAME	Tagname
NCTLSLOT	Number of Regulatory Control Slots
NFASTCTL	Number of Fast Regulatory Control Slots
NFASTPV	Number of Fast Regulatory PV Slots
NODENUM	Node Number
NODETYP	Node Type
NPVSLOT	Number of Regulatory PV Slots
NTWKNUM	Network Number
OVERVAL	Overview Value
PIUOTDCF	Open Thermocouple Detection Enable
PNTFORM	Point Form
PNTMODTYP	PM Module type
PTDESC	Point Descriptor
PVCHAR	PV Characterization
PVCLAMP	PV Clamping Option
PVEUHI	PV Range High
PVEULO	PV Range Low
PVEXEUHI	PV Range Extension High
PVEXEULO	PV Range Extension Low
PVFORMAT	PV Decimal Format
PVRAW	PV Raw Value
PVRAWHI	PV Raw High Range
PVRAWLO	PV Raw Low Range
PVSOURCE	PV Source
PVSRCOPT	PV Source Option
PVTV	PV Target Value
S1	General Purpose Parameters (UDC 6000/6300)
SCANRATE	Scan Rate
SECVAR	Secondary Variable
SENSORTYPE	Sensor Type
SERIALNO	Serial Number/PROM Number for Smart Transmitter
STATE	State
STIEU	STI Engineering Units
STISWVER	Software Revision Level of the Smart Transmitter
STITAG	Transmitter Tag ID
T1	Integral Time in Minutes
T2	Derivative Time in Minutes
TF	PV Filter Lag Time
UNIT	Unit ID
URL	Upper Range Limit
URV	Upper Range Value

References

Publication Title	Publication Number	Binder Number
<i>UDC 6000 Process Controller Product Manual</i>	51-52-25-32	N/A
<i>UDC 6300 Process Controller Product Manual</i>	51-52-25-45	N/A
<i>Process Manager Control Functions and Algorithms</i>	PM09-500	TPS 3040-1
<i>Advanced Process Manager Control Functions and Algorithms</i>	AP09-500	TPS 3042-1
<i>Process Manager Parameter Reference Dictionary</i>	PM09-540	TPS 3040-2
<i>Advanced Process Manager Parameter Reference Dictionary</i>	AP09-540	TPS 3042-2
<i>Advanced Process Manager Installation</i>	PM20-501	TPS 3043
<i>Process Manager/Advanced Process Manager Installation</i>		
<i>Advanced Process Manager Service</i>	PM13-501	TPS 3061-1
<i>Process Manager/Advanced Process Manager Service</i>		
<i>Engineer's Reference Manual</i>	SW09-505	TPS 3030-2
<i>Data Entity Builder Manual</i>	SW11-511	TPS 3032-1
<i>Picture Editor Reference Manual</i>	SW09-550	TPS 3032-2
<i>Picture Editor Data Entry</i>	SW11-550	TPS 3032-3
<i>Button Configuration Form Instructions</i>	SW12-570	TPS 3032-1
<i>Button Configuration Data Entry</i>	SW11-570	TPS 3032-1
<i>Process Operations Manual</i>	Sw11-501	TPS 3050

Section 1 – Introduction to Integration

1.1 Overview

Purpose of this section

This section provides an introduction to PM/APM - UDC 6000/6300 Integration, describes control modes, and identifies integration components.

Reader assumptions

- You are familiar with **TotalPlant** Solution (TPS) system components and have a TPS bookset on hand.
 - You are familiar with UDC 6000/6300 process controllers and have a UDC 6000 or UDC 6300 product manuals on hand.
-

1.2 PM/APM - UDC 6000/6300 Integration

Definition

PM/APM - UDC 6000/6300 Integration is a term used to describe the coupling of standalone UDC 6000/6300 process controllers to a TPS Process Manager (PM) or Advanced Process Manager (APM) through a serial digital communications link.

This integration lets operator's access UDC 6000/6300 operation and configuration data through Universal Station (US) displays as well as the operator interface for the UDC 6000/6300.

Communications link

The communications link provides a secure and reliable exchange of data, which requires no operator interpolation or special program entry. It uses a MODBUS EIA-485 remote terminal unit (RTU) protocol. This supports multidrop configuration so several controllers can be connected in series on a single link.

A single link supports up to four UDC 6000/6300 controllers connected through an SDI field termination assembly (FTA), and a power adapter to a serial device interface (SDI) input/output processor (IOP) in a Process Manager or an Advanced Process Manager. Each SDI IOP supports up to two FTAs or a maximum of eight loops.

Compatibility

The PM/APM - UDC 6000/6300 Integration is compatible with TPS control systems that have software release R500 or later.

Benefits

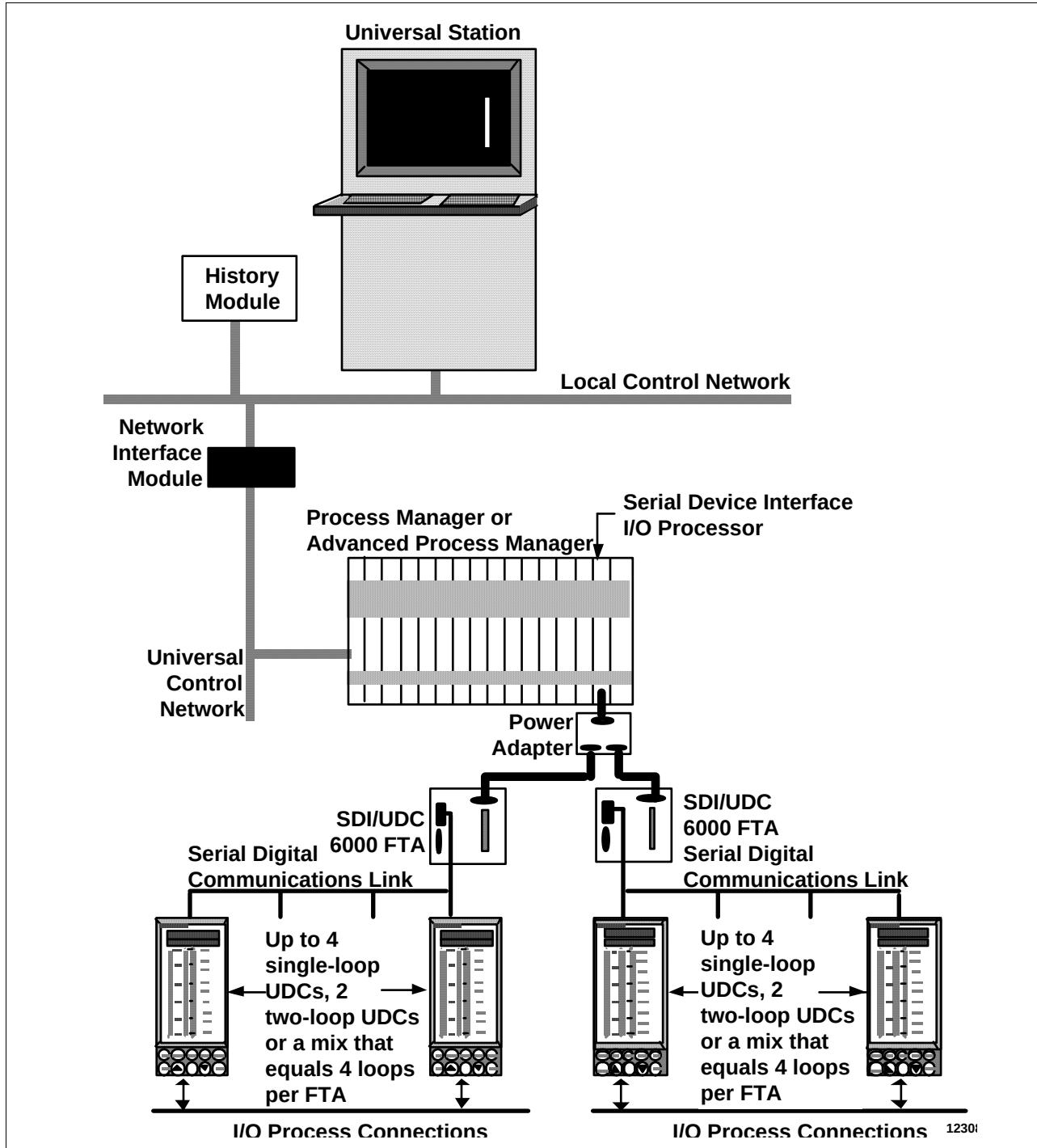
- Allows users to start small with single loop control and then expand to TPS.
 - Provides a cost-effective means to integrate standalone single-loop controllers into a distributed control system scheme.
 - Provides faster scan rate than Process Manager for "tighter" control and independent backup for greater integrity of non-redundant critical loops.
 - Provides Local or Remote display of PM or APM control loops with control shed functions.
 - Provides TPS alarm operations in addition to controller alarms for enhanced alarming functionality.
-

1.2 PM/APM - UDC 6000/6300 Integration, Continued

Diagram: typical integration hierarchy

Figure 1-1 shows a typical PM/APM - UDC 6000/6300 integration hierarchy that has up to eight single-loop UDC 6000/6300 controllers connected to the system through two SDI/UDC 6000/6300 FTAs, a power adapter, and an SDI IOP in the PM/APM.

Figure 1-1 Typical PM or APM - UDC 6000/6300 Integration Hierarchy



1.3 Data Exchange Functions

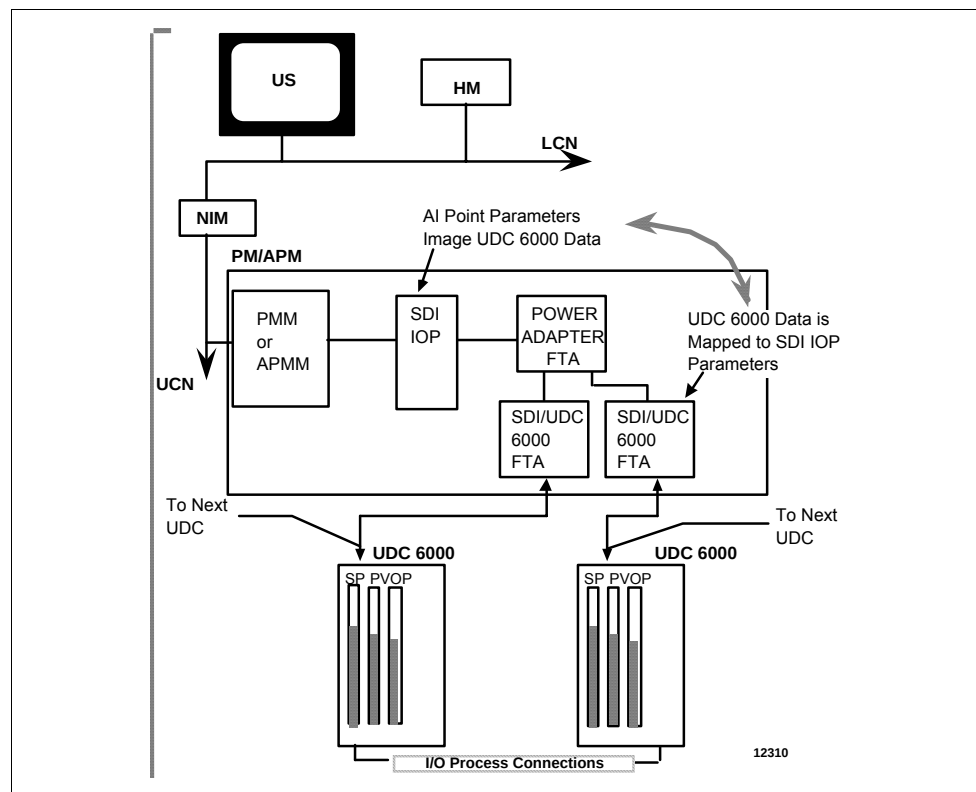
Introduction

The exchange of data over the bi-directional data path between the UDC 6000/6300 controller and the PM/APM is based on imaging UDC 6000/6300 data through the use of STIM Analog Input (AI) point parameters in the SDI IOP. This is done by mapping parameters from the SDI/UDC 6000/6300 FTA to the SDI IOP, and from the SDI IOP to the SDI/UDC 6000/6300 FTA as shown in Figure 1-2.

Note that the figure shows an example with one US, NIM, APM, etc., but you could have more than one.

This section discusses various functions that affect how the data is exchanged. Most of this information is for reference only, but some will be helpful when making point building decisions later. Refer to subsections 3.4 and 3.5 for details about mapped parameters and SDI IOP point building.

Figure 1-2 Mapped Parameters Are Basis for Data Exchange

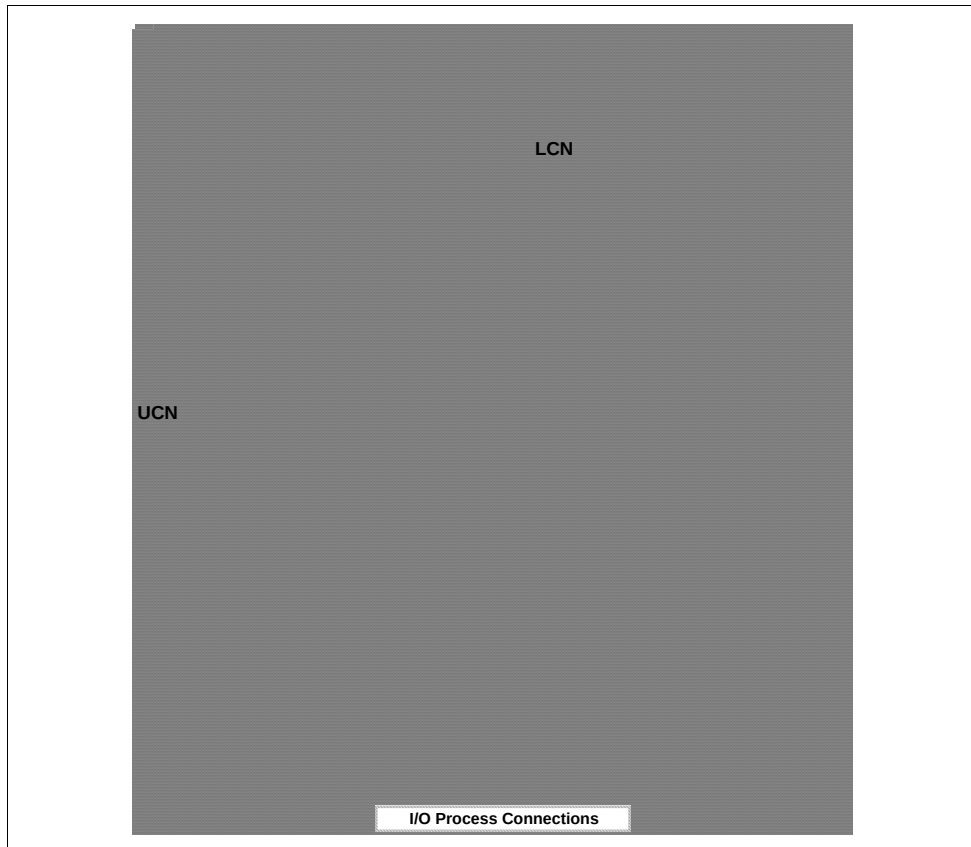


1.3 Data Exchange Functions, Continued

SDI IOP as STI IOP

The SDI IOP uses a Smart Transmitter Interface (STI) database structure. This means that the SDI IOP appears as an STI IOP to the Process Manager Module (PMM) or APMM and upper network components such as the US, Application Module (AM), and History Module (HM) as shown in Figure 1-3. For this reason, the SDI/UDC 6000/6300 FTA parameters are mapped to SDI IOP parameters with STI identifications.

Figure 1-3 SDI IOP Appears as an STI IOP to Other Components



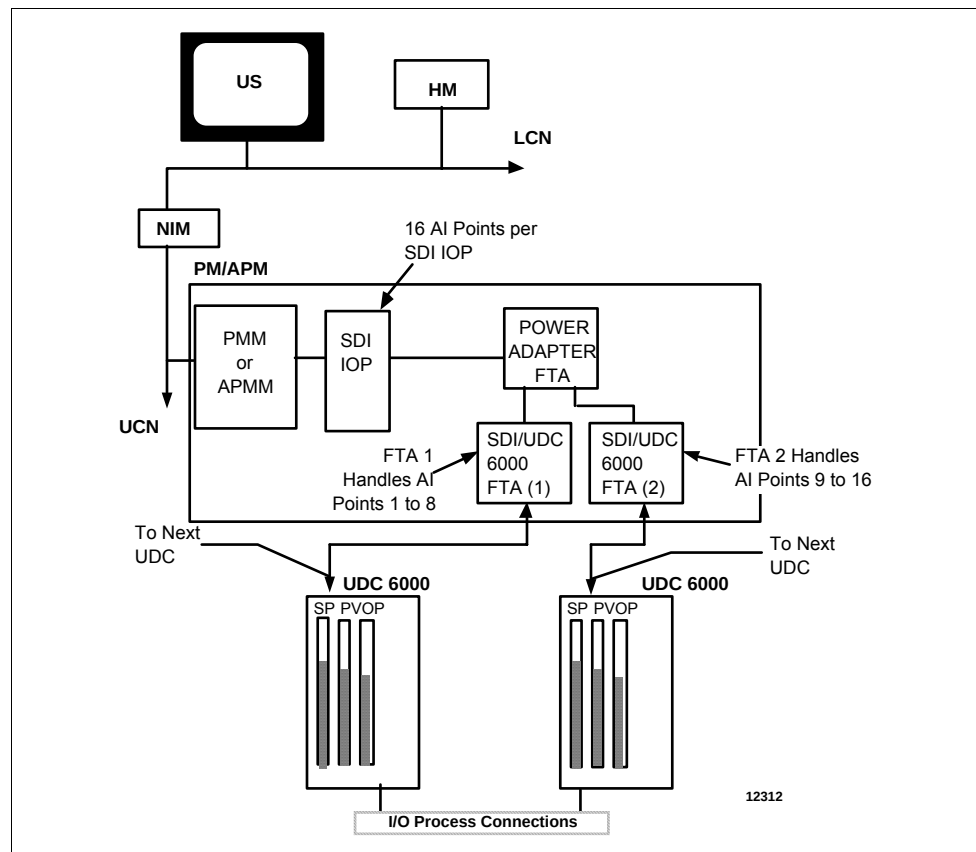
1.3 Data Exchange Functions, Continued

16 Points per SDI IOP

The SDI IOP contains sixteen AI points that are read/write accessible from the PMM or APMM and upper network components as shown in Figure 1-4. The UDC 6000/6300 parameters are mapped within these sixteen points. In this case, SDI IOP points one to eight are allocated to UDC 6000/6300 control loops connected to FTA 1; and IOP points nine to sixteen are allocated to UDC 6000/6300 control loops connected to FTA 2.

Note that points include the usual IOP PV processing parameters such as alarm limits, alarm hysteresis, and PV clamping, which are not mapped to or from the FTA.

Figure 1-4 Sixteen AI Points per SDI IOP



1.3 Data Exchange Functions, Continued

UDC 6000/6300 Box Mode

You can set the overall mode of operation for each UDC 6000/6300 as either Basic or Full. Box Mode is a configurable parameter maintained by the UDC 6000/6300.

Essentially, the Basic and Full modes determine the level of interaction allowed from the TDC to the UDC 6000/6300.

Table 1-1 Basic and Full Modes Comparison

WHEN a point is in ...	THEN ...
Basic mode	• permits only monitor operation from the TPS.
Full mode	• permits control operations including Mode changes and Output and Setpoint overrides.

Also, see subsection 3.4 for the SDI CTTACT point parameter usage.

1.4 Integration Components

Summary

You will need these hardware and software components to integrate UDC 6000/6300 controllers with a TPS system R500 or later.

- Serial Device Interface Input/Output Processor for PM/APM.
- Power adapter FTA.
- SDI/UDC 6000/6300 Field Termination Assembly (FTA).
- UDC 6000/6300 with communications card for 485 RTU/TDC communications.
- Cable for FTA to power adapter connector.
- Cable for communications link between UDC 6000/6300s and FTA.
- Diskette/cartridge with custom graphic software for Universal Station.

UDC field upgrade kits can be ordered to add TPS integration capability to an existing UDC 6000/6300. The field upgrade consists of two kits:

- An RS485 communication card P/N 30755865-503, and
- a separate field replaceable EEPROM upgrade kit:
 - UDC 6000 P/N 30757085-501
 - UDC 6300 P/N 51309350-501

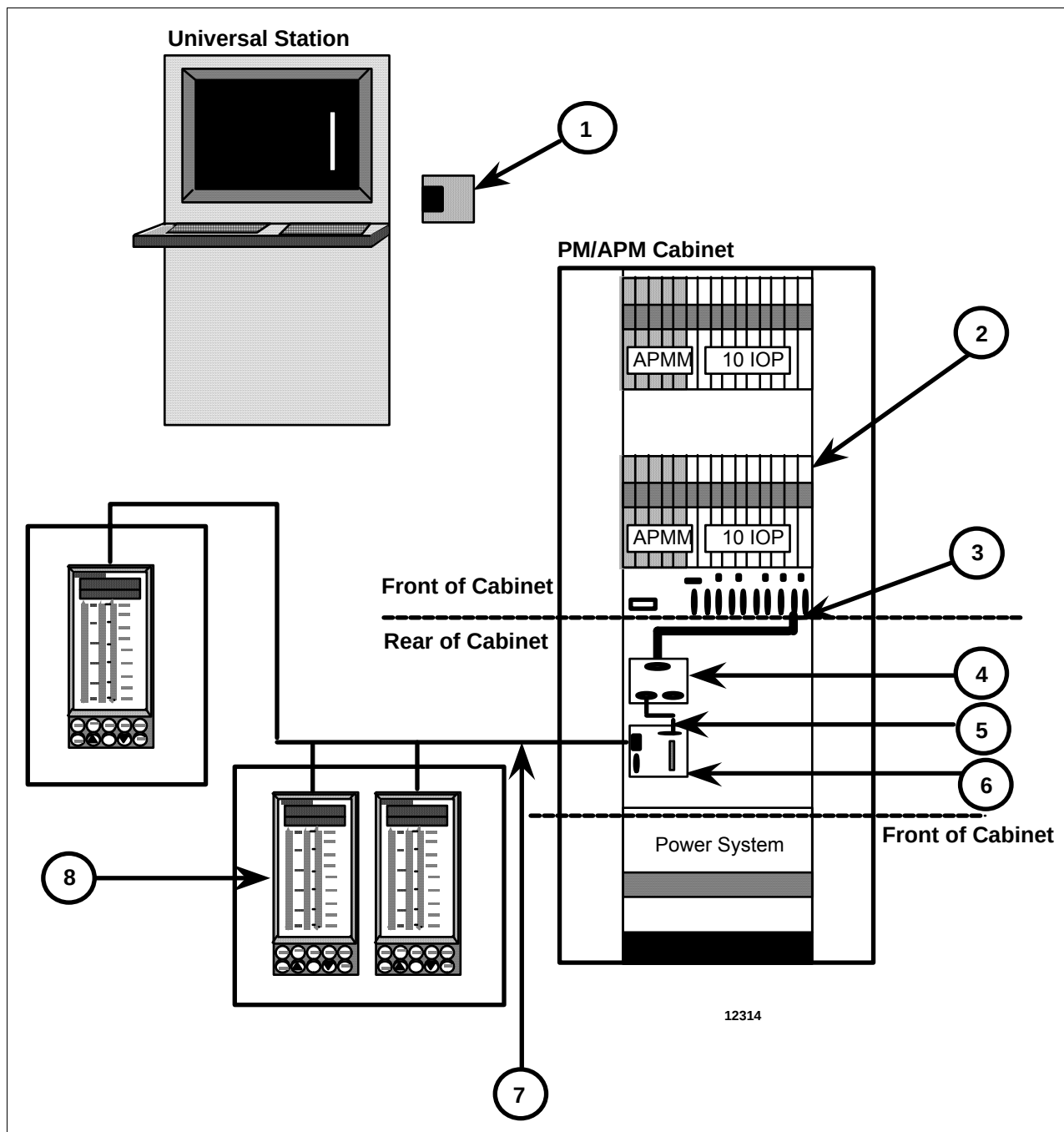
This means you can easily update existing controllers in the field to get the advantages of integration.

1.4 Integration Components, Continued

Component : identification

Figure 1-5 and Table 1-2 identify the integration components. Figure 1-5 identifies integration components by their relative location in the system and/or an associated component. These locations are shown for reference only, since the actual locations will vary based on the size and complexity of your process control scheme. Table 1-2 lists the part or model number for the components identified in Figure 1-5.

Figure 1-5 Identifying Integration Components



1.4 Integration Components, Continued

Component identification, continued

Table 1-2 Part Numbers for Components Identified in Figure 1-5

No.	Part/Model Number	Description
1	MP-SWUG1 MP-SWUG2 MP-SWUG3 MP-SWUG4 MP-SWUG5 MP-SWUG6	Custom graphics software for UDC 6000/6300 displays on the TPS system for Release 500. 8 Character Tags - Floppy Media - R300 8 Character Tags - Cartridge Media - R300 8 Character Tags - Floppy Media - R400 8 Character Tags - Cartridge Media - R400 16 Character Tags - Floppy Media - R400 16 Character Tags - Cartridge Media - R400
2	MU-PSDX02	SDI IOP for I/O slot in PM or I/O cardfile.
3	MU-KFTA00 (local cabinet use) MU-KTFAXX* *XX=Length	FTA I/O Cable - Cable with phone type connectors for IOP to power adapter connection. Lengths available from 5 to 50 meters in 5 meter increments; such as, MU-KFTA15 (15 mtrs).
4	MU-TLPA02 MU-GRPA01 MC-GRPA01 MU-TRPA01 MC-TRPA01	LLMUX Power adapter Remote Hardened MUX GI/IS Power Adapter (Div. 1, uncoated) Remote Hardened MUX GI/IS Power Adapter (Div. 1, conformally coated) Remote Hardened MUX GI/NI Power Adapter (Div. 1, uncoated) Remote Hardened MUX GI/NI Power Adapter (Div. 1, conformally coated)
5	MU-KLAMOO MU-KLX076 MU-KLX152 MU-KLX305	Twisted pair cable for power adapter to FTA connection Local cabinet use Shielded Cable (18 ga.) 250 feet Shielded Cable (18 ga.) 500 feet Shielded Cable (18 ga.) 1000 feet
6	MU-TSDU02	SDI FTA for UDC 6000/6300 Interface communications
7	MU-KSX030 MU-KSX152 MU-KSX305	Two-conductor shielded cable for communications link - FTA to UDC 6000/6300 30 meters - 100 ft. 152 meters - 500 ft. 305 meters - 1000 ft. See Cable selection guidelines for recommendation.
8	PC600X-U-3XX-XX-X-0X *X defines user selected options	UDC 6000/6300 Process Controller
	Not shown 30755865-504 30757085-501 51309350-501	Upgrade kits to convert existing UDC 6000/6300 to handle TPS Digital Integration. Communications Card UDC 6000 TPS Upgrade PROM UDC 6300 TPS Upgrade PROM

1.4 Integration Components, Continued

Cable selection guidelines

You must select a two-conductor shielded cable for the digital communications link between the UDC 6000/6300 and the FTA that satisfies your routing requirements as well as the local codes and ordinances for wiring.

There are three standard Honeywell Model Numbers for this cable:

- MU-KSX030 - 30 meters/100 ft.
- MU-KSX152 - 152 meters/500 ft.
- MU-KSX305 - 305 meters/1000 ft.

As an alternative, you can order Consolidated 24 gauge, shielded, signal cable catalog number 9940 or equivalent grade cable for the communications link.

Table 1-3 lists some characteristics of the recommended cable for comparison purposes.

Table 1-3 Cable Characteristics for Communications Link

Item	Description
Conductor	24 AWG stranded copper
Insulation	Color coded PVDF
Shield	100% coverage mylar/aluminum foil, 25% overlap, foil facing in with stranded tinned copper drain wire
Jacket	• <i>Composition:</i> Polymer alloy • <i>Temperature Range:</i> –80 to 75°C (–112 to 167°F) • <i>Nominal Outside Diameter:</i> 0.106 in (2.7 mm) • <i>Voltage:</i> 150 Volts
Approval Body	• UL listed for low smoke and fire characteristics per National Electrical Code Article 725-38(b)3 • UL Subject 13, Power Limited Circuit Cables, Type CL2P • UL Subject 910, Smoke and Flame Test • Passes UL VW-1 Vertical Flame Test • OSHA acceptable • Passes IEEE 383 70,000 BTU Flame Test

1.5 Installation Checklist

Introduction

Before attempting to install the hardware and software associated with a PM/APM - UDC 6000/6300 integration, make sure you have all the components listed in Table 1-4.

Table 1-4 Component Checklist

TPS — PROCESS MANAGER HARDWARE	
Description	Model Number
- SDI FTA for UDC 6000/6300 - Firmware revision ≥ “A”	MU-TSDU02
- SDI IOP - Firmware revision ≥ “B”	MU-PSDX02
FTA Power Adapter	MU-TLPA02
Interconnect Cables (as required)	
UNIVERSAL STATION CUSTOM GRAPHICS SOFTWARE (FOR 6000/6300/TDC INTEGRATION)	
Description	Model Number
8 character tags for R300 floppy OR	MP-SWUG1
8 character tags for R300 cartridge OR	MP-SWUG2
8 character tags for R400 floppy OR	MP-SWUG3
8 character tags for R400 cartridge OR	MP-SWUG4
16 character tags for R400 floppy OR	MP-SWUG5
16 character tags for R400 cartridge	MP-SWUG6
UDC 6000/6300 HARDWARE/FIRMWARE	
Description	Model Number
- UDC 6000/6300 with Modbus-RS485RTU - Firmware revision ≥ 7710 - OR - UDC 6000/6300 RS485 communications card upgrade kit to convert existing UDC 6000/6300 to handle TPS Integration	PC600X-U-3XX-XX-X-0X PC630X-U-3XX-XX-X-0X NOTE: “X” defines user selected options 30755865-504
PLUS	
- PROM upgrade kits - UDC 6000 - UDC 6300	30757085-501 51309350-501

Section 2 – Installation

2.1 Overview

Purpose of this section

This section defines the tasks and procedures for installing the hardware and software associated with a PM/APM - UDC 6000/6300 integration.

Reader assumptions

- You are familiar with the installation of **TotalPlant** Solution (TPS) system components or have read site planning and individual installation sections included in the TPS bookset.
- You are familiar with the installation of UDC 6000/6300 controllers or have read the installation section in the UDC 6000 or UDC 6300 product manual.

ATTENTION

ATTENTION—If you already have a PM/APM system with SDI installed, you can skip to page 24 for installation instructions.

2.2 Installation Tasks

Description

The installation tasks basically consist of physically installing and wiring the PM/APM - UDC 6000/6300 integration components. These tasks focus on functions that are associated with the integration feature only. You must supplement this information with data from the TPS system documentation as well as the UDC 6000 or UDC 6300 product manual.

WARNING

WARNING—The equipment in the PM/APM cabinet is subject to damage by Electrostatic Discharge (ESD). Wear your ESD wriststrap any time you handle printed circuit boards.

ATTENTION

ATTENTION—Please note that all power and I/O field wiring for the control loops goes directly to the UDC 6000/6300. The wiring diagram shown in task #6 “Installing UDC 6000/6300 Controller” is for reference only.

Please refer to the UDC 6000 or UDC 6300 product manual for specific wiring details.

Breakdown of tasks

Table 2-1 provides a breakdown of specific installation tasks in numerical order for reference. The order of the tasks is somewhat arbitrary. However, you must, in most cases, install a component before you can wire it.

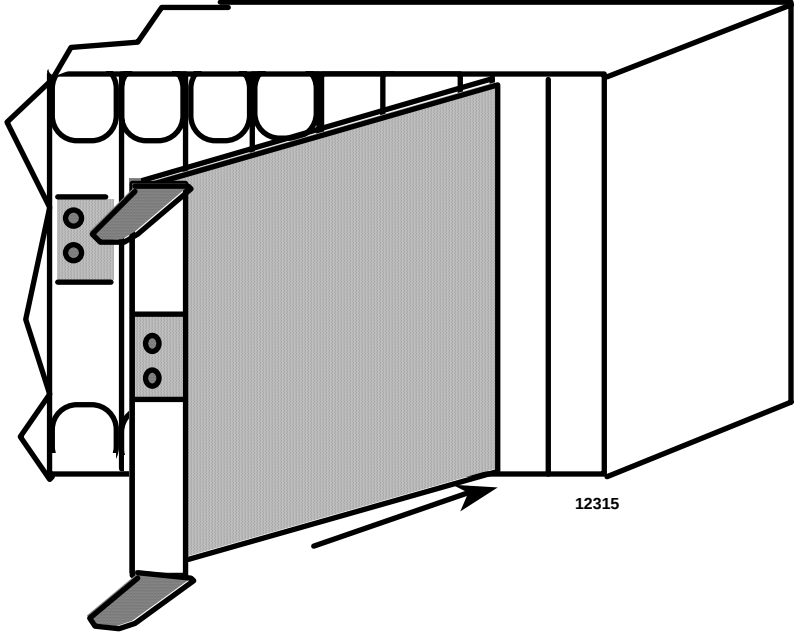
Table 2-1 Installation Tasks Breakdown

Task No.	Description
1	Installing SDI IOP
2	Installing power adapter
3	Installing SDI/UDC 6000/6300 FTA
4	Wiring SDI IOP to power adapter
5	Wiring power adapter to SDI/UDC 6000/6300 FTA
6	Installing UDC 6000/6300 controller
7	Wiring digital communications link
8	Loading UDC 6000/6300 graphics software

2.3 Installing SDI IOP

Background	The SDI IOP is the I/O processor used for handling the information exchanges between the UDC 6000/6300 and the PM/APM. It mounts in a slot in the PM/APM or I/O cardfile.
Prerequisite	You have an I/O slot in a PM/APM or I/O cardfile that can be used for SDI IOP operation.
Procedure	Perform the steps in Table 2-2 to install an SDI IOP in a PM/APM or I/O cardfile.

Table 2-2 Installing an SDI IOP

Step	Action
1	Locate an empty slot in PM/APM or I/O cardfile. ATTENTION The SDI IOP is designed for no redundant operation only.
2	With power ON or OFF, slide SDI IOP module into one slot until it is fully seated.
3	Close latches to lock module in place. <u>Example:</u> Installing module in cardfile slot. 

2.4 Installing Power Adapter

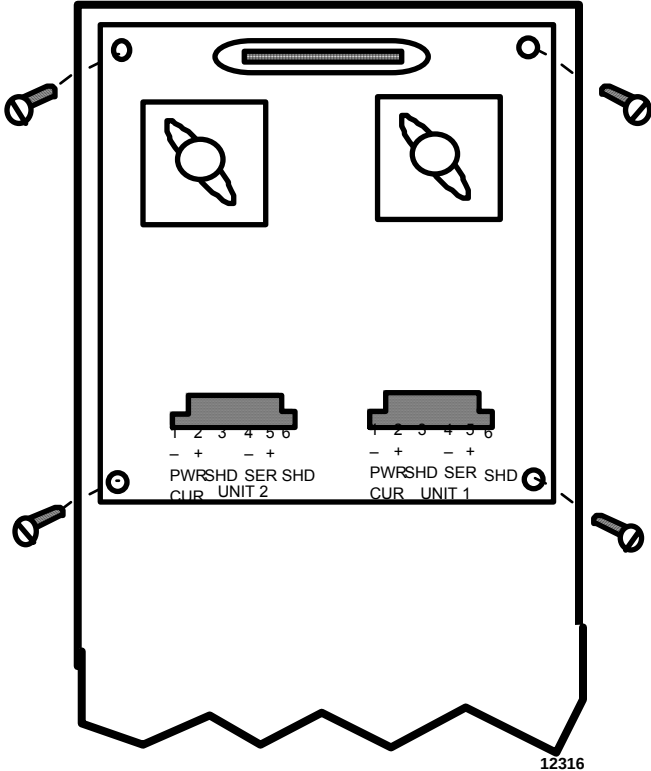
Background The power adapter accepts connections from up to two SDI/UDC 6000/6300 FTAs located up to 305 meters (1000 feet) away for communications with a single SDI IOP.

ATTENTION ATTENTION—If you are mounting SDI/UDC 6000/6300 FTAs in the same location, they should be located adjacent to the Power Adapter FTA.

Prerequisite You have a PM/APM cabinet with mounting channel for size A FTAs. They should be grouped with voltages less than 30 Vdc.

Procedure Perform the steps in Table 2-3 to install a power adapter in a PM/APM cabinet.

Table 2-3 Installing a Power Adapter

Step	Action
1	Locate a free space for size “A” FTAs in a mounting channel for FTAs under 30 Vdc in PM/APM cabinet.
2	Mount power adapter card in channel and secure with screws. <u>Example:</u> Mounting power adapter in mounting channel. 

2.5 Installing SDI/UDC 6000/6300 FTA

Background

The SDI/UDC 6000/6300 FTA provides communication drivers for the UDC 6000/6300 controllers connected to the RTU485/TDC communications link.

Prerequisite

You have a PM/APM cabinet with space available in a mounting channel for size A FTAs.

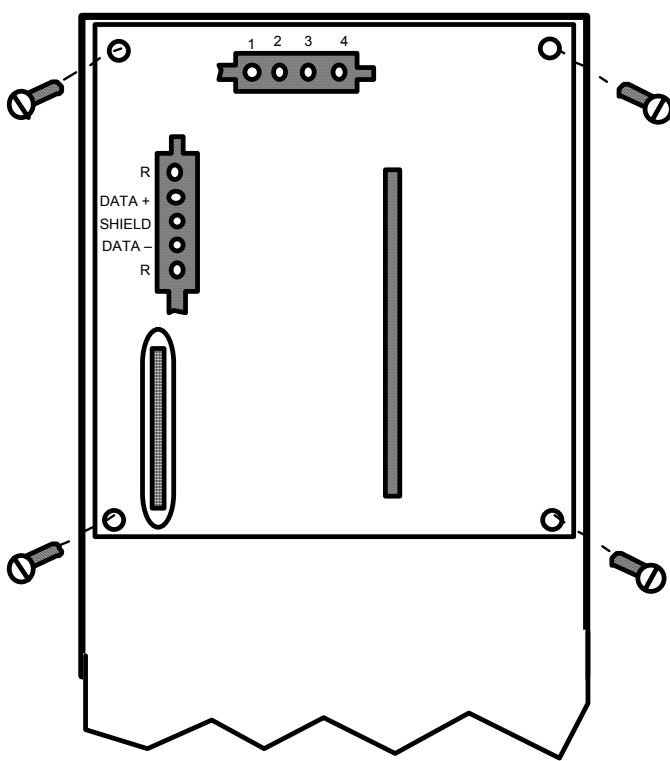
WARNING

WARNING—To maintain proper shielding for FTAs, always locate FTA mounting channels inside a PM/APM cabinet. If you must mount FTAs in another location, refer to the *Process Manager Installation* for guidelines.

Procedure

Perform the steps in Table 2-4 to install an SDI/UDC 6000/6300 FTA in a PM/APM cabinet .

Table 2-4 Installing an SDI/UDC 6000/6300 FTA

Step	Action
1	Locate a free space for size A FTAs, in a mounting channel for FTAs less than 30 Vdc, in PM/APM cabinet.
2	Mount SDI/UDC 6000/6300 FTA card in channel and secure with screws. <u>Example:</u> Mounting SDI/UDC 6000/6300 FTA in mounting channel. 

2.6 Wiring SDI IOP to Power Adapter

Background

The SDI IOP connects to a power adapter through a prefabricated FTA I/O cable that is available from Honeywell in lengths up to 50 meters (164 feet).

Prerequisite

You have FTA cable of appropriate length.

Procedure

Perform the steps in Table 2-5 to wire an SDI IOP to a power adapter.

Table 2-5 Wiring SDI IOP to Power Adapter

Step	Action
1	Plug connector on one end of cable into cardfile connector for IOP.
2	Route cable to power adapter.
3	Plug other end of cable into connector on power adapter. <u>Example:</u> Connecting cable between IOP and power adapter.

The diagram illustrates the connection between a Power Adapter FTA and a cardfile connector. The Power Adapter FTA is a rectangular unit with a cable extending from it. The cable is plugged into a cardfile connector on the IOP. The IOP is a larger unit with two cardfile connectors and two terminal blocks labeled UNIT 1 and UNIT 2. The terminal blocks have pins labeled 1, 2, 3, 4, 5, 6 and are also labeled PWR, SHD, SER, SHD, CUR, UNIT 1 and UNIT 2.

2.7 Wiring Power Adapter to SDI/UDC 6000/6300 FTA

Background

The power adapter connects to an SDI/UDC 6000/6300 FTA through a shielded, twisted pair, prefabricated cable that is available from Honeywell for internal wiring to FTA in PM/APM cabinet or external wiring to FTA in remote location up to 330 meters (1000 feet) away.

Prerequisite

You have appropriate cable for internal or external wiring.

Procedure

Perform the steps in Table 2-6 to wire a power adapter to an SDI/UDC 6000/6300 FTA in the same PM/APM cabinet or in a remote location.

Table 2-6 Wiring Power Adapter to SDI/UDC 6000/6300 FTA

Step	Action
1	Locate unit 1 connector on lower, right-hand side of power adapter card.
2	Identify four conductor leads and two shield leads on one end of prefabricated cable.

2.7 Wiring Power Adapter to SDI/UDC 6000/6300 FTA, Continued

Procedure, continued

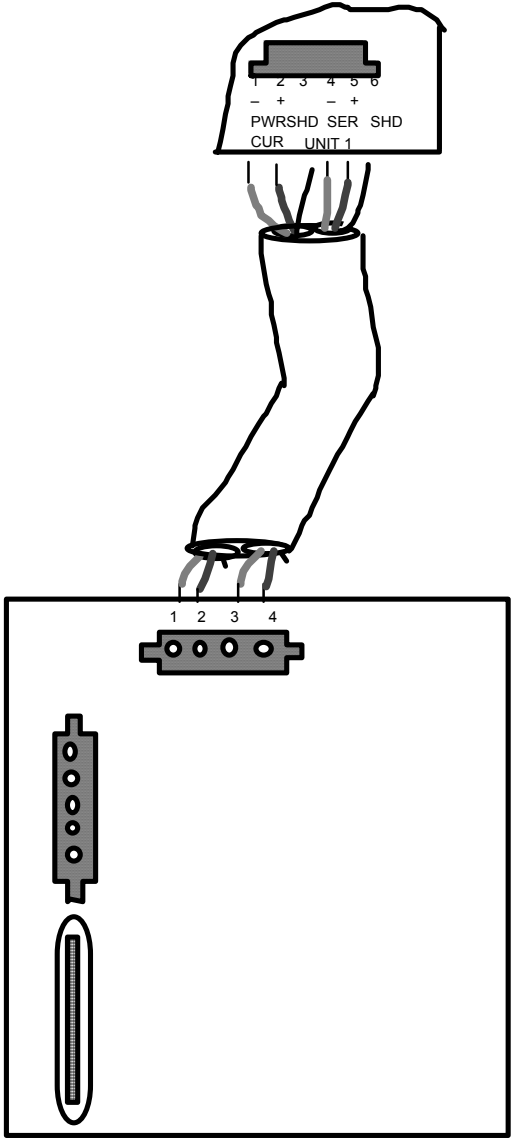
Table 2-6 Wiring Power Adapter to SDI/UDC 6000/6300 FTA,
Continued

Step	Action														
3	Insert stripped conductor lead ends and shields into appropriate compression fitting terminals in unit 1 connector as follows. <table border="1"> <thead> <tr> <th>Lead Type</th><th>Terminal</th></tr> </thead> <tbody> <tr> <td>conductor</td><td>1 –</td></tr> <tr> <td>conductor</td><td>2 +</td></tr> <tr> <td>shield</td><td>3 SHD</td></tr> <tr> <td>conductor</td><td>4 –</td></tr> <tr> <td>conductor</td><td>5 +</td></tr> <tr> <td>shield</td><td>6 SHD</td></tr> </tbody> </table> Note color-coding of conductor leads. <u>Example:</u> Making cable connections to power adapter. 12319	Lead Type	Terminal	conductor	1 –	conductor	2 +	shield	3 SHD	conductor	4 –	conductor	5 +	shield	6 SHD
Lead Type	Terminal														
conductor	1 –														
conductor	2 +														
shield	3 SHD														
conductor	4 –														
conductor	5 +														
shield	6 SHD														
4	Route cable to SDI/UDC 6000/6300 FTA.														
5	Trim shield leads on this end of cable close to insulation. You only connect shield leads to connector on power adapter.														

2.7 Wiring Power Adapter to SDI/UDC 6000/6300 FTA, Continued

Procedure, continued

Table 2-6 Wiring Power Adapter to SDI/UDC 6000/6300 FTA,
Continued

Step	Action
6	Observing color-coding noted in Step 3, insert stripped conductor lead ends of conductors connected to 1– and 2+ into matching numbered compression fitting terminals 1 and 2 in connector located near top center of FTA card. Insert lead ends of conductors connected to 4– and 5+ into terminals 3 and 4, respectively. <u>Example:</u> Making mating connections on FTA.  12320
7	Repeat Steps 1 through 6 to connect another FTA to unit 2 connector.

2.7 Wiring Power Adapter to SDI/UDC 6000/6300 FTA,

Continued

FTA wiring guidelines

-
- Keep total length of cable between the power adapter and the SDI/UDC 6000/6300 FTA equal to 305 meters (1000 feet) or less.
 - Run signal wiring in bundle with low level signals or separate conduit apart from AC power wiring.
 - Mount the FTAs in a mounting channel with voltages less than 30 Vdc.
-

2.8 Installing UDC 6000/6300 Controller

Background

The UDC 6000/6300 controller provides standalone loop control capability that can be completely integrated with the operation of a TPS control system through a digital communications link. All power and I/O wiring for the control loops goes directly to the UDC 6000/6300.

Prerequisite

You have a UDC 6000/6300 controller with a communications card for RTU485/TDC communications.

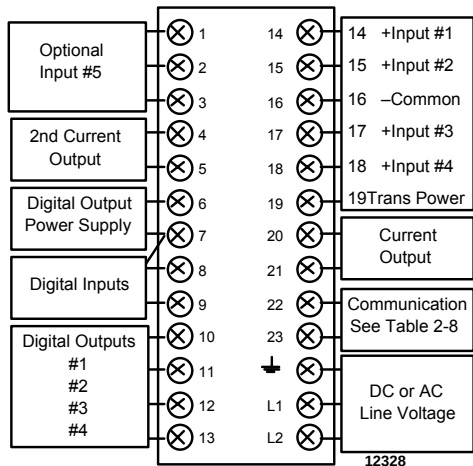
ATTENTION

ATTENTION—If the UDC 6000/6300's communications card supports a different protocol, an upgrade kit consisting of an RS485 communication card part number 30755865-504 and a field replaceable EEPROM is available to upgrade the UDC 6000/6300 to TPS Integration (see page 12).

Procedure

Perform the steps in Table 2-7 to install a UDC 6000/6300 controller.

Table 2-7 Installing UDC 6000/6300 Controllers

Step	Action
1	Check power requirement. See Installation section in UDC 6000 or UDC 6300 product manual for details.
2	Mount controller in a panel using mounting kit supplied. See Installation section in UDC 6000 or UDC 6300 product manual for details.
3	Connect power and I/O wiring to rear of case screw terminals. See Installation section in UDC 6000 or UDC 6300 product manual for details. The diagram shown below is for reference only.  The diagram shows a terminal block with 23 terminals. On the left, terminals 1-13 are grouped into boxes: 'Optional Input #5' (1-3), '2nd Current Output' (4-5), 'Digital Output Power Supply' (6-7), 'Digital Inputs' (8-9), and 'Digital Outputs #1 #2 #3 #4' (10-13). On the right, terminals 14-23 are grouped into boxes: '+Input #1' (14), '+Input #2' (15), '-Common' (16), '+Input #3' (17), '+Input #4' (18), '19Trans Power' (19), 'Current Output' (20), 'Communication See Table 2-8' (21-23), and 'DC or AC Line Voltage' (L1, L2). A ground symbol is shown between terminals 11 and 12. The terminal block is labeled '12328' at the bottom right. CAUTION If you elect to power the UDC 6000/6300 with +24 Vdc, use a separate power supply as the source instead of the dc output from the PM/APM, unless you requested the factory to provide a PM power supply to handle this load when you placed your order.

2.9 Wiring Digital Communications Link

Background

The digital communications link serves as a secure and reliable data exchange path between a UDC 6000/6300 controller and the TPS system.

Prerequisite

- You have sufficient length of signal cable on hand.
- You have two 120 ohms, 10%, 1/4 watt resistors.
- You have installed an SDI/UDC 6000/6300 FTA.
- You have installed a UDC 6000/6300 with communications card for RTU485/TDC protocol.

Procedure

Perform the steps in Table 2-8 to wire the digital communications link between UDC 6000/6300 controllers and an SDI/UDC 6000/6300 FTA.

Table 2-8 Wiring Digital Communications Link

Step	Action
1	Strip 1/4 inch of insulation from end of signal cable conductor leads.
2	Insert one conductor lead and one 120 ohm resistor lead into DATA + compression fitting terminal in connector on upper left side of SDI/UDC 6000/6300 FTA.
3	Insert shield lead into SHIELD terminal in FTA connector.
4	Insert other conductor lead and other resistor lead into DATA – terminal in FTA connector. <u>Example:</u> Connecting cable to FTA connector terminals.

The diagram illustrates the connection of a signal cable to an FTA connector. The cable has four leads: DATA +, SHIELD, DATA -, and a 120 ohm resistor. The FTA connector has four terminals labeled 1, 2, 3, and 4. The wiring is as follows: DATA + to terminal 1, SHIELD to terminal 2, DATA - to terminal 3, and the 120 ohm resistor to terminal 4. A note indicates to note the color-coding of conductor leads.

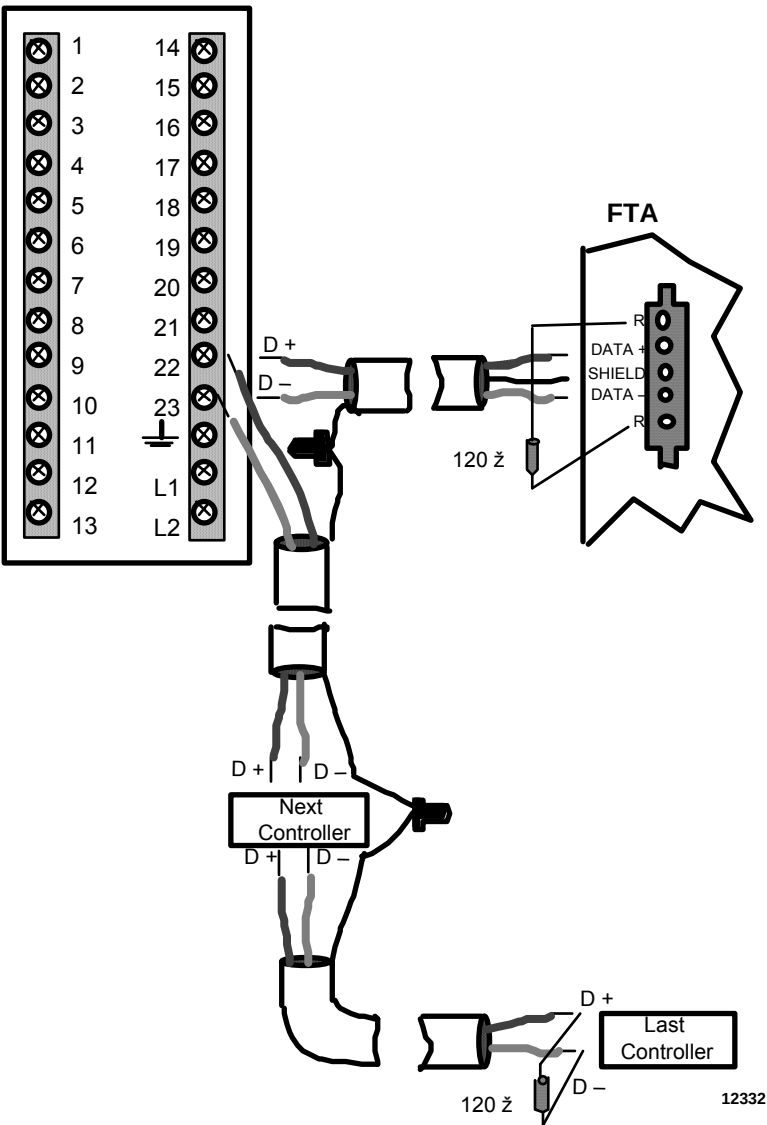
12329

Note color-coding of conductor leads.

2.9 Wiring Digital Communications Link, Continued

Procedure, continued

Table 2-8 Wiring Digital Communications Link, Continued

Step	Action
5	Route signal cable to first UDC 6000/6300.
6	Cut cable and strip 1/4 inch of insulation from conductor leads.
7	Observing color-coding noted in Step 4, connect conductor from DATA + to terminal 22 (D+) on rear of first UDC 6000/6300 in link. Connect conductor from DATA – to terminal 23 (D–).
8	To continue link to next UDC 6000/6300, take another length of signal cable and connect color-coded conductor leads to terminals 22 (D+) and 23 (D–) on first controller, respectively. Crimp shield wires together. <u>Example:</u> Connecting UDC 6000/6300 to link. 

2.9 Wiring Digital Communications Link, Continued

Procedure, continued

Table 2-8 Wiring Digital Communications Link, Continued

Step	Action
9	Route cable to next controller. Connect color-coded cable conductors to matching terminals 22 (D+) and 23 (D–) on rear of case.
10	Repeat Steps 8 and 9 until all UDC 6000/6300s on this link are wired together. ATTENTION Be sure to put a 120 ohms resistor across terminals 22 and 23 on the last controller in the link.
11	Repeat this procedure for additional links as required.

- Link wiring guidelines**
- Keep total length of FTA to UDC 6000/6300 communications link cable equal to 1,200 meters (3,937 feet) or less.
 - Terminate each end of the link with 120 ohm resistor across the conductors.
 - Run signal wiring in bundle with low level signals or a separate conduit apart from AC power wiring.

2.10 Loading UDC 6000/6300 Graphics Software

Background

The UDC 6000/6300 graphics software adds these displays to the Universal Station for dedicated UDC 6000/6300 interface.

- Point Summary
- Group
- Detail
- Operator Data
- Configuration Schematic
- Operator Change Zone

The software is supplied on a cartridge.

Prerequisite

- You have the correct graphics software to work with short (8-character) tagnames (R300/R400) or long (16-character) tagnames (R400 or later) in your TPS system.
 - You have a US with engineering personality function and a cartridge drive.
 - You have put the keyswitch on the US in the engineering (ENG) position.
 - You are familiar with the operation of the engineering personality utilities. If you are not familiar, refer to the *Utilities Operation Manual* for details.
-

Volume or directory selection

Before loading the software, select the existing directory in a volume of a history module (HM) on the network to which you will add the UDC 6000/6300 graphics. Base this selection on the directory/volume that has the most room to handle the additional schematic source files (*.DS) and schematic object files (*.DO) that make up the graphics software.

ATTENTION

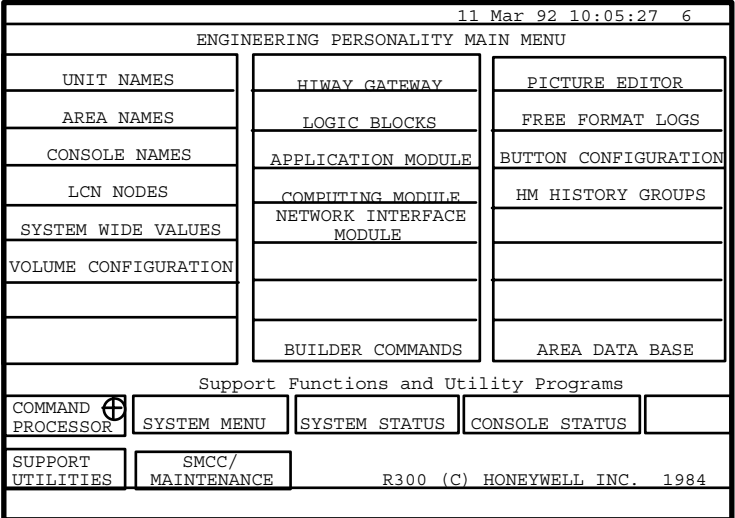
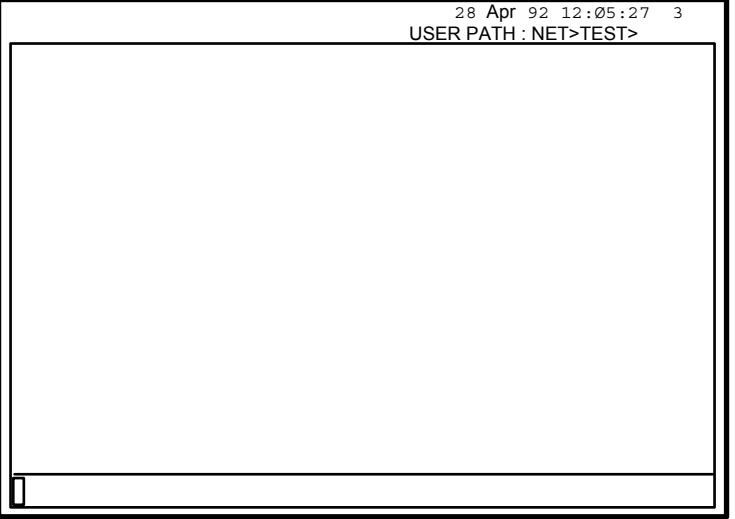
ATTENTION—How to determine file size and create another directory/volume, if required, are beyond the scope of this document. Refer to the *Engineer's Reference Manual* for information if the graphics files will not fit in an existing directory/volume.

2.10 Loading UDC 6000/6300 Graphics Software, Continued

Example: selecting a volume

The procedure in Table 2-9 outlines the steps for selecting an HM volume, which includes a directory with the name of DISP for the addition of UDC 6000/6300 graphics files as an example only.

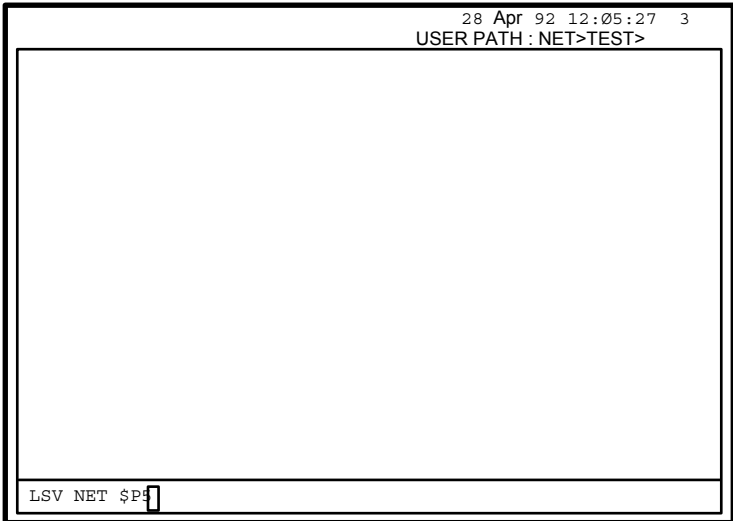
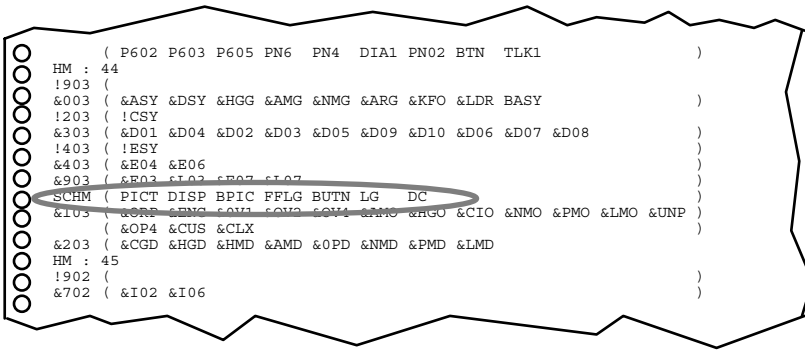
Table 2-9 Example of Selecting a Volume

Step	Action
1	Press [CTL] and [HELP] keys to call up the Engineering Personality Main Menu. <u>Example:</u> Calling up Engineering Personality Main Menu. 
2	Select COMMAND PROCESSOR target to call up display. <u>Example:</u> Calling up Command Processor display. 

2.10 Loading UDC 6000/6300 Graphics Software, Continued

Example: selecting a volume, continued

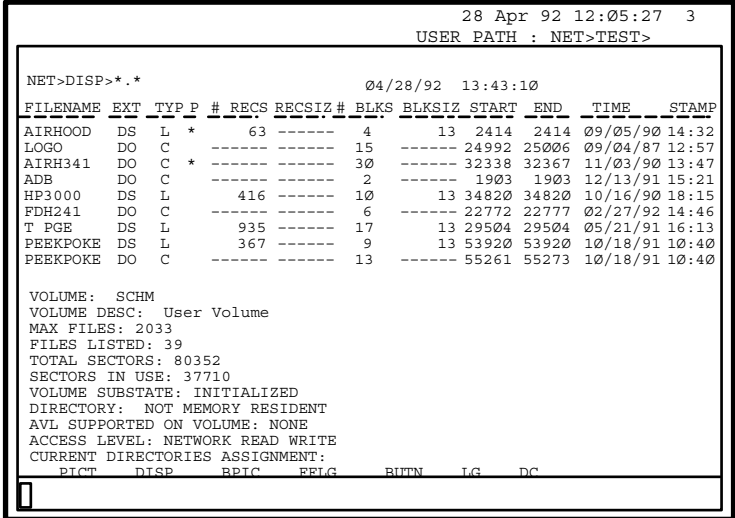
Table 2-9 Example of Selecting a Volume, Continued

Step	Action
3	Key in [LSV NET \$P5] and press [ENTER] to initiate a printout that lists volumes on HM nodes at printer number 5. <u>Example:</u> Initiating a volume printout.  12335
4	Check printout for possible existing volumes that include schematics. In this example, a directory named DISP is chosen from a volume named SCHM in HM node 44.  12336

2.10 Loading UDC 6000/6300 Graphics Software, Continued

Example: selecting a volume, continued

Table 2-9 Example of Selecting a Volume, Continued

Step	Action
5	Key in [LS NET>DISP>*.]*] and press [ENTER] key to initiate list of files in this directory on display to view how many files are already in this directory. <u>Example:</u> Initiating list of files for a directory.  <pre> 28 Apr 92 12:05:27 3 USER PATH : NET>TEST> NET>DISP>*. 04/28/92 13:43:10 FILENAME EXT TYP P # RECS RECSIZ# BLKS BLKSIZ START END TIME STAMP AIRHOOD DS L * 63 ----- 4 13 2414 2414 09/05/90 14:32 LOGO DO C ----- 15 ----- 24992 25006 09/04/87 12:57 AIRH341 DO C * ----- 30 ----- 32338 32367 11/03/90 13:47 ADB DO C ----- 2 ----- 1903 1903 12/13/91 15:21 HP3000 DS L 416 ----- 10 13 34820 34820 10/16/90 18:15 FDH241 DO C ----- 6 ----- 22772 22777 02/27/92 14:46 T PGE DS L 935 ----- 17 13 29504 29504 05/21/91 16:13 PEEKPOKE DS L 367 ----- 9 13 53920 53920 10/18/91 10:40 PEEKPOKE DO C ----- 13 ----- 55261 55273 10/18/91 10:40 VOLUME: SCHM VOLUME DESC: User Volume MAX FILES: 2033 FILES LISTED: 39 TOTAL SECTORS: 80352 SECTORS IN USE: 37710 VOLUME SUBSTATE: INITIALIZED DIRECTORY: NOT MEMORY RESIDENT AVL SUPPORTED ON VOLUME: NONE ACCESS LEVEL: NETWORK READ WRITE CURRENT DIRECTORIES ASSIGNMENT: BICT DISP BPTC ERLG BUTN LG DC </pre>
6	Record NET>DISP as selected directory pathname.

2.10 Loading UDC 6000/6300 Graphics Software, Continued

Loading software

Use the procedure in Table 2-10 to load the UDC 6000/6300 graphics software. Note that this procedure uses cartridge drive number 5 and directory pathname NET>DISP for example purposes only—Just substitute your cartridge drive number and selected directory pathname where appropriate.

Table 2-10 Loading UDC 6000/6300 Graphics Software

Step	Action
1	Insert graphics software cartridge in cartridge drive for US.
2	Press [CTL] and [HELP] keys to call up the Engineering Personality Main Menu.
3	Select COMMAND PROCESSOR target to call up display. <u>Example:</u> Calling up command processor display. 28 Apr 92 12:05:27 3 USER PATH : NET>TEST> EC \$F5>VID>INSTALL.EC NET>DISP> 12338
4	Key in command line [EC \$F5>VID>INSTALL.EC NET>DISP>=] and press the [ENTER] key. Where:

Section 3 – Set Up

3.1 Overview

Purpose of this section

This section explains how to set up the **TotalPlant** Solution (TPS) system for integration operation. It includes building points at the Universal Station and identifying mapped parameters, configuring displays, configuring the UDC 6000/6300 database and preparing the SDI for operation.

Reader assumptions

- You know how to interact with the TPS system using the US touch screens and keyboard. If you do not know, refer to the *Process Operations Manual* for details.
 - You are familiar with the “point” building concept for the PM/APM and the UCN and LCN networking schemes. If you are not familiar, refer to the *Data Entity Builder Manual* for information.
 - You know how to interact with the UDC 6000/6300 controller using its front face displays and keys. If you do not know, refer to the *UDC 6000* or *UDC 6300 Process Controller Product Manual* for details.
-

3.2 Configuration Process

About the process

The configuration process is a matter of confirming PM/APM node data, setting IOP (or IOM) parameters, building two SDI points for each UDC 6000/6300 loop on the link, configuring custom graphic displays, configuring the UDC 6000/6300 controllers, and preparing the SDI for operation.

Stages of the process

Table 3-1 lists the stages of the configuration process as determined by the tasks required. The order of the stages is for reference only so they can be accomplished out of sequence.

Table 3-1 Stages of the Configuration Process

Stage	Description
1	Build PM/APM box data point so system will recognize SDI IOP.
2	Identify mapped SDI parameters to help make SDI IOP point configuration selections.
3	Build two SDI IOP points so system will recognize loop connected through these points.
4	Configure UDC 6000/6300 controller through its interface.
5	Configure UDC 6000/6300 schematics.
6	Prepare SDI for operation.

3.3 Building a PM/APM Box Data Point

Background

A box data point identifies the physical location of an IOP by mapping its file and card number to one of the 40 conceptual slots allotted to the primary PMM/APMM in each PM/APM. This integrates the point into the PM/APM's control functions and makes it easy to reference the IOP in the various UCN/PM-APM status displays.

If an IOP is added to a PM/APM, a box data point must be built for it.

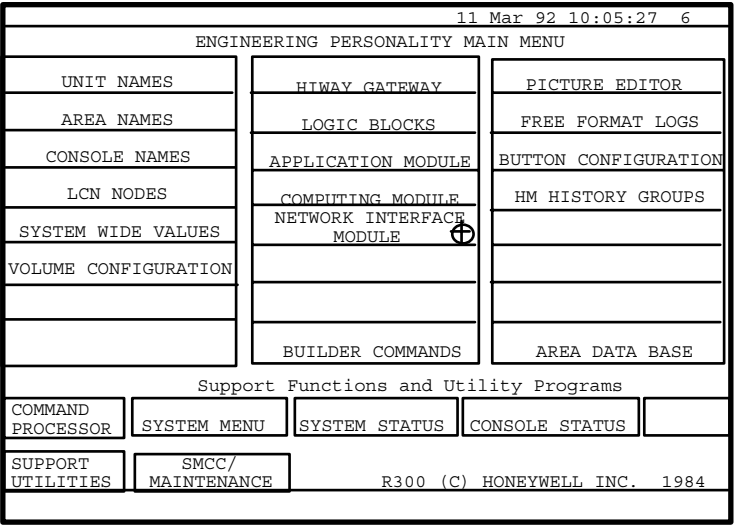
Prerequisites

- You are familiar with the overall network arrangement including node type, network number, and node number assignments.
- You know the physical location of the installed SDI IOP in terms of file number and card number.
- You have a US in engineering personality and have put the keyswitch on the US in the engineering (ENG) position.
- You know that the NIM and PM/APM are operating and that the NIM status is "OK."

Procedure

Use the general steps in Table 3-2 to build a box data point. The parameter values used in these steps are for example purposes only.

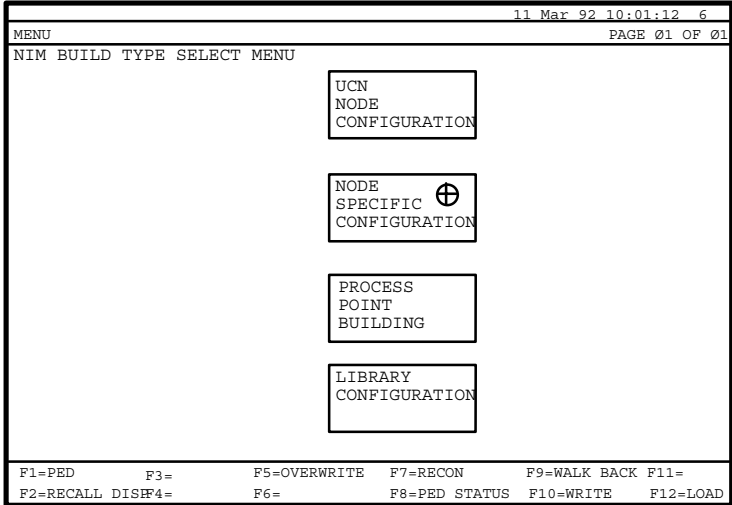
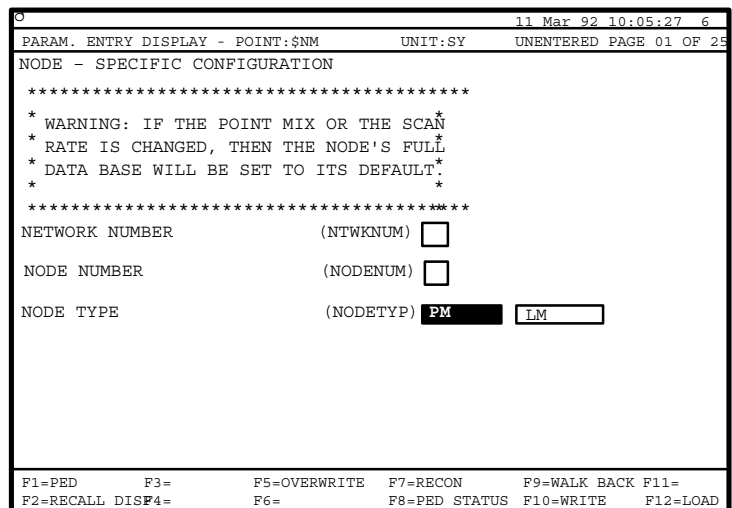
Table 3-2 Building a PM/APM Box Data Point

Step	Action
1	Press [CTL] and [HELP] keys to call up Engineering Personality Main Menu. <u>Example:</u> Calling up Engineering Personality Main Menu. 

3.3 Building a PM/APM Box Data Point, Continued

Procedure, continued

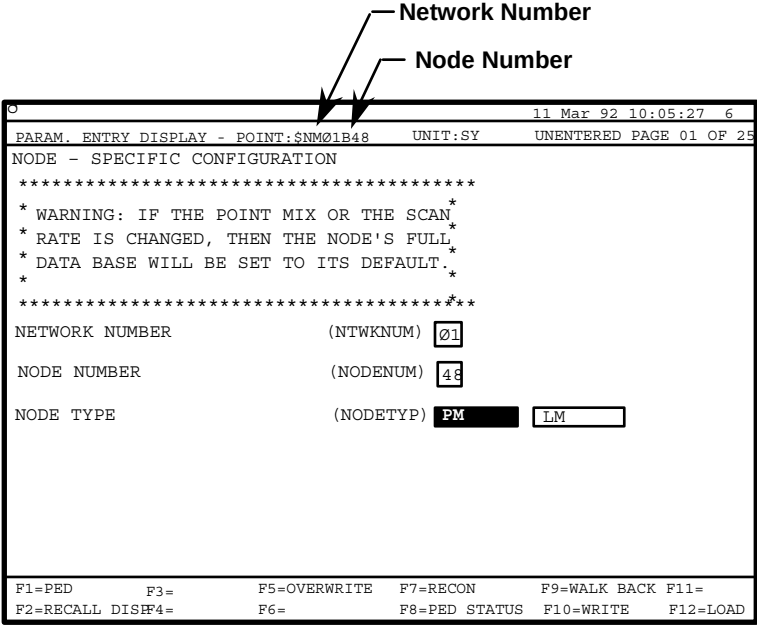
Table 3-2 Building a PM/APM Box Data Point, Continued

Step	Action
2	Select NETWORK INTERFACE MODULE target to call up NIM Build Type Select Menu. <u>Example:</u> Calling up NIM Build Type Select Menu.  12340
3	Select NODE SPECIFIC CONFIGURATION target to call up Parameter Entry Display (PED). <u>Example:</u> Calling up PED for node specific configuration.  12341
4	Key in [01] as desired network number and press [ENTER] key.

3.3 Building a PM/APM Box Data Point, Continued

Procedure, continued

Table 3-2 Building a PM/APM Box Data Point, Continued

Step	Action
5	Key in [48] as desired node number and press [ENTER] key.
6	Select PM as node type and press [ENTER] key.
7	Check that point name for display reflects the entered network and node numbers. <u>Example:</u> Checking point name.  The screenshot shows a terminal window titled 'PARAM. ENTRY DISPLAY - POINT:\$NM01B48 UNIT:SY UNENTERED PAGE 01 OF 25'. The screen displays a warning message and configuration fields. Two arrows point to the '01' in the 'NETWORK NUMBER' field and the '48' in the 'NODE NUMBER' field, with labels 'Network Number' and 'Node Number' respectively. The 'NODE TYPE' field shows 'PM' selected and 'LM' as an alternative. At the bottom, a row of function keys is listed: F1=PED, F3=, F5=OVERWRITE, F7=RECON, F9=WALK BACK, F11=, F2=RECALL DISF4=, F6=, F8=PED STATUS, F10=WRITE, F12=LOAD.

3.3 Building a PM/APM Box Data Point, Continued

Procedure, continued

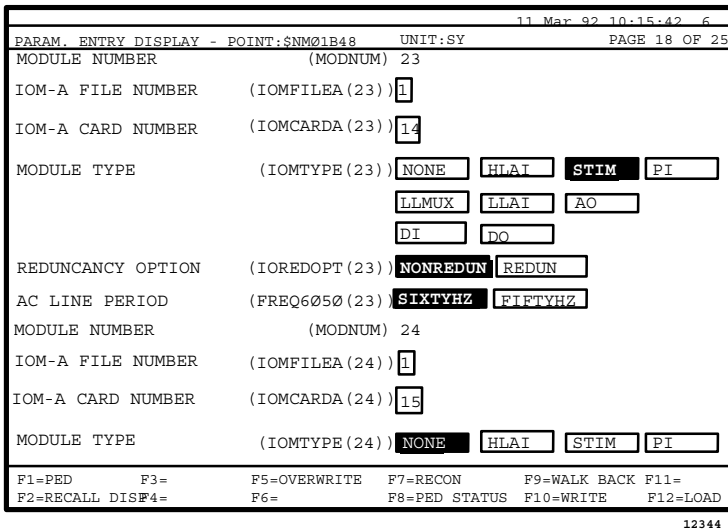
Table 3-2 Building a PM/APM Box Data Point, Continued

Step	Action
8	Use [PAGE FWD] key to call up pages 02 and 03 of PED and fill in other PM/APM configuration data as appropriate. Refer to <i>Process Manager Parameter Reference Dictionary</i> for details. Note that you can leave your point mix and scan rate the same when adding an SDI IOP. 12343 The figure shows two screenshots of the PED configuration interface. The top screenshot is 'PAGE 03 OF 25' and displays the following parameters: SEQUENCE PROCESSING (SEQPROC) with values 1_PU and 2_PU; NUMBER OF PROCESS MODULE SLOTS (NPMSLOT) set to 0; and NUMBER OF NUMERICS (NNUMERIC) set to 256. The bottom screenshot is 'PAGE 02 OF 25' and displays the 'NODE - SPECIFIC CONFIGURATION (FOR PM)' screen. It includes: CONTROL CARD OPTION (CTLOPT) with ON and OFF buttons; NUMBER OF REG. CONTROL SLOTS (NCTLSLOT) set to 100; NUMBER OF FAST REG CONTROL SLOTS (NFASTCTL) set to 0; NUMBER OF REG. PV SLOTS (NPVSLLOT) set to 20; NUMBER OF FAST REG. PV SLOTS (NFASTPV) set to 48; NUMBER OF LOGIC SLOTS (NLOGSLOT) set to 25; NUMBER OF FAST LOGIC SLOTS (NFASTLOG) set to 0; NUMBER OF DIG. COMPOSITE SLOTS (NDCSLOT) set to 150; and NUMBER OF FAST DIGITAL COMPOSITE SLOTS (NFASTDC) set to 0. A function key legend at the bottom lists: F1=PED, F2=RECALL DISF, F3=, F4=, F5=OVERWRITE, F6=, F7=RECON, F8=PED STATUS, F9=WALK BACK, F10=WRITE, F11=, F12=LOAD.
9	Use [PAGE FWD] key to page through I/O Module (IOM) Configuration pages of PED until assigned SDI IOP module number appears. ATTENTION The term IOM is used interchangeably with IOP in US displays. Both terms refer to the same hardware.

3.3 Building a PM/APM Box Data Point, Continued

Procedure, continued

Table 3-2 Building a PM/APM Box Data Point, Continued

Step	Action								
10	Enter this IOM configuration data for SDI IOP; - card file number (1 to 8), - card number (slot 1 to 15), - module type (STIM), and - redundancy option (NONREDUN). ac line period (not applicable - parameter is ignored by SDI) You must press [ENTER] key to write information to file. <u>Example:</u> Entering IOM configuration data.  ATTENTION The Smart Transmitter Interface Module (STIM) is equivalent to an SDI IOP for configuration purposes. However, the SDI IOP only supports nonredundant operation.								
11	Press [CTL] and [F12] to load Box data point information into file. <table border="1"> <thead> <tr> <th>If...</th><th>Then...</th></tr> </thead> <tbody> <tr> <td>Box data point loads</td><td>go to next step.</td></tr> <tr> <td>overwrite prompt appears</td><td>press [CTL] and [F5] keys to overwrite existing file.</td></tr> <tr> <td>error message appears</td><td>refer to <i>Data Entity Builder Manual</i> for error recovery data and recheck your work including hardware installation and wiring connections.</td></tr> </tbody> </table>	If...	Then...	Box data point loads	go to next step.	overwrite prompt appears	press [CTL] and [F5] keys to overwrite existing file.	error message appears	refer to <i>Data Entity Builder Manual</i> for error recovery data and recheck your work including hardware installation and wiring connections.
If...	Then...								
Box data point loads	go to next step.								
overwrite prompt appears	press [CTL] and [F5] keys to overwrite existing file.								
error message appears	refer to <i>Data Entity Builder Manual</i> for error recovery data and recheck your work including hardware installation and wiring connections.								
12	Press [CTL] and [HELP] keys to return to Engineering Personality Main Menu. You may be prompted to repeat this keystroke action to confirm PED changes.								

3.3 Building a PM/APM Box Data Point, Continued

Using reconstitute function

You can use the reconstitute function ([**CTL**] + [**F7**] or command) to change parameters in an existing Node Specific Point (PM/APM Box Data Point) to add an SDI IOP. Refer to the *Data Entity Builder Manual* for details.

Be sure you identify the correct PM/APM through the entry of these parameters on the first page of the PED.

- NETWORK NUMBER (NTWKNUM)
 - NODE NUMBER (NODENUM)
 - NODE TYPE (NODETYP)
-

3.4 Identifying Mapped SDI Parameters

Overview

As discussed in subsection 1.3, parameters are mapped through AI points in the SDI IOP to image UDC 6000/6300 data as the basis for the PM/APM - UDC 6000/6300 Integration data exchange. Because the SDI IOP appears as an **STI IOP** to the system, the mapped parameters retain their STI names but have different definitions as indicated in Table 3-3 and 3-4. The specific definition depends on the point for which it is being used.

Point 1 - PV (Process Variable) Table 3-3

Point 2 - OP (Output) Table 3-4

The STI names are:

C1	PV RAW LO
C2	PVTV
CJTACT	SI-1
COMMAND	SI-2
DAMPING	SI-3
DECONF	SI-4
INPTDIR	SECVAR
LRL	SERIAL NO
LRV	SFISWVER
PIUOTDCF	STATE
PVEUHI	STI_EU
PVEULO	STI TAG
PVRAW	URL
PVRAWHI	URV

3.4 Identifying Mapped SDI Parameters, Continued

STI Names and SDI IOP definitions

These parameters are of the view and non-configuration mapped data types. Some of the parameters offer several selections that affect the functional characteristics of the point. You determine the desired characteristic by the selection you make when building the SDI IOP point as described in the next section. In some cases, you will also be able to change certain selections through US displays.

More detailed parameter information follows in Tables 3-4 and 3-5 to help you make the right SDI IOP point building selections and to serve as a cross-reference for the *Process Manager Parameter Reference Dictionary*.

Mapped Data Types

The three basic types of mapped data are:

- *View or Read Only Type* for parameters that cannot be written to the IOP from upper network components, but that can be used for passing data from the SDI/UDC 6000/6300 FTA to the SDI IOP.
- *Configuration Type* for parameters that can be written to the SDI IOP only when the “point” is inactive.
- *Non-Configuration Type* for parameters that can be written to the IOP from the PM/APM or the SDI/UDC 6000/6300 FTA at any time.

UDC 6000/6300/SDI Interface

The UDC 6000/6300 is interfaced to the Serial Device Interface (SDI) IOP through a Field Termination Assembly (FTA) and a Power Adapter. Each SDI IOP maintains 16 SDI IOP slots, which are divided into 2 sets of 8 and directed through 2 independent RS485 communication links to 2 FTAs.

IOP slots 1 through 8 are interfaced through FTA 1 and IOP slots 9 through 16 are interfaced through FTA 2.

This provides each FTA with 8 SDI slot connections. The FTA’s map the IOP slots into RS485 multidrop addresses as shown in Figure 3-1.

Figure 3-1 IOP Slot vs. RS485 Address

FOR FTA 1		FOR FTA 2	
IOP SLOT	RS485 ADDRESS	IOP SLOT	RS485 ADDRESS
1	1	9	1
2	2	10	2
3	3	11	3
4	4	12	4
5	5	13	5
6	6	14	6
7	7	15	7
8	8	16	8

12345

3.4 Identifying Mapped SDI Parameters, Continued

SDI IOP points

UDC 6000/6300 controllers are connected by a multidrop EIA RS485 connection to an FTA. Each UDC 6000/6300 control loop requires two SDI IOP points for data transfer to the PM/APM to support the UDC 6000/6300 custom graphics.

The first point of each UDC loop is the **Process Variable (PV) point**. It provides access to PV related parameters including setpoint adjustment. The PV point is also used for device configuration operations.

The second point of each UDC loop is the **Output (OP) point**. It provides access to output related parameters. The OP point also provides access to limits and raw analog inputs.

The IOP slots for a particular UDC are selected at the UDC 6000/6300 front panel through setup of the RS485 PV (Process Variable) and OP (Output) addresses. These addresses are the FTA RS485 addresses shown in Figure 3-1.

Example

Four single loop UDC 6000/6300 controllers (UDC 1 to UDC 4) are to be connected. Two UDCs are connected to FTA 1 and two are connected to FTA 2 on a common IOP.

An example of a typical configuration is shown in Figure 3-2.

Figure 3-2 Address vs. Slot Configuration

RS485 ADDRESS		SLOT NUMBER
UDC 1/FTA 1	PV1 ADDR = 1	1
	OP1 ADDR = 2	2
UDC 2/FTA 1	PV1 ADDR = 3	3
	OP1 ADDR = 4	4
UDC 3/FTA 2	PV1 ADDR = 1	9
	OP1 ADDR = 2	10
UDC 4/FTA 2	PV1 ADDR = 3	11
	OP1 ADDR = 4	12

12346

3.4 Identifying Mapped SDI Parameters, Continued

SDI names and SDI IOP definitions

UDC 6000/6300 control variables and parameters are transferred to the IOP through SDI parameters, which are referenced through the PM/APM by STI parameter tagnames. These parameters are mapped into UDC 6000/6300 definitions and are dependent on the UDC point type (PV or OP) being mapped.

Tables 3-3 (PV) and 3-4 (OP) provide the UDC usage of SDI (STIM) parameters for PV and OP mappings and defines the purpose of each parameter when it is used in SDI IOP point data for UDC 6000/6300 communications. Note that column heading PtRes stands for Point Resident, and NaN means Not a Number.

PV point SDI definitions

Table 3-3 provides the UDC usage of SDI (STIM) parameters for PV mapping.

Table 3-3 PV Point SDI Parameter Definitions

Parameter	UDC Use	Type/Source	Lock	Default	PtRes	Range
C1	Cascade Setpoint Indicates the state of the override variable which is SP. It provides status feedback for automatic control schemes for avoiding windup or enabling back initialization of variables.	Real/IOP	Supr	None	PM/APM	
C2	PV Override	Real/IOP	Supr	None	PM/APM	
CJTACT	PV Port Override Enable	Boolean/UDC	Engr	N/A	PM/APM	ON - TDC has control of UDC through schematics OFF - Control is not allowed through schematics.

3.4 Identifying Mapped SDI Parameters, Continued

PV point SDI definitions, continued

Table 3-3 PV Point SDI Parameter Definitions, Continued

Parameter	UDC Use	Type/Source	Lock	Default	PtRes	Range
COMMAND (STI)	IOP Request - Allows access to operation and configuration data in UDC through IOP, using UDC 6000/6300 schematics.	Enumeration/IOP	Oper	None	PM/APM	0 - None 1 - DnLoadDb (Write parameter to UDC) 2 - UpLoadDb (Read parameter from UDC) 3 - Set_LRV - sets UDC control mode to BCAS 4 - Set_URV - sets UDC control mode to AUTO 5 - Cor_LRV - CAS 6 - Cor_URV - Manual 7 - Cor_Inpt - Not Used 8 - RstCor (- resets communications (UDC) database)
DAMPING (STI)	Alarm Status - Provide status of UDC alarms.	Real/UDC	View	None	PM/APM	0 - No UDC alarms 1 - UDC alarm 1 ON 2 - UDC alarm 2 ON 3 - Both alarms 1 and 2 ON
DECONF (STI)	Data Transfer Configuration - for UDC interface. This parameter should be set to PV_SV_DB.	Enumeration/IOP	Engr	PV_SV_DB	PM/APM	4 - Pv-Sv-Db

3.4 Identifying Mapped SDI Parameters, Continued

PV point SDI definitions, continued

Table 3-3 PV Point SDI Parameter Definitions, Continued

Parameter	UDC Use	Type/Source	Lock	Default	PtRes	Range
INPTDIR	Not Used					
LRL	Configuration Lower Range Limit - Provides floating point value equivalent to lower range limit of given UDC configuration parameter.	Real/UDC	View	NaN	PM/APM	
LRV	Configuration Read Value Configuration Write Value Used to read and write UDC parameters.	Real/UDC Real/IOP	Supr	NaN	PM/APM	
PIUOTDCF	Monitor/Control Flag	Boolean/UDC	Engr		PM/APM	ON - Monitor OFF - Control
PVEUHI	PVCALC/PVTV High Range in Engineering Units - Allows access to IOP PVRAW high range scaling value. Combines with PVEULO to set range used by NIM to check limits of the PVCALC (PVRAW) and PVTV values.	Real/IOP	Engr	100.0	PM/APM	Reflects limits set at UDC. -9999.9 to 9999.9
PVEULO	PVCALC/PVTV Low Range in Engineering Units - Allows access to IOP PVRAW low range scaling value.	Real/IOP	Engr	0.0	PM/APM	Reflects limits set at UDC. -9999.9 to 9999.9 Used in combination with PVEUHI
PVRAW	PV Raw Value - Process variable processed by UDC and sent through IOP as PVCALC value.	Real/UDC	View	NaN	UDC	UDC PV Range

3.4 Identifying Mapped SDI Parameters, Continued

PV point SDI definitions, continued

Table 3-3 PV Point SDI Parameter Definitions, Continued

Parameter	UDC Use	Type/Source	Lock	Default	PtRes	Range
PVRAWHI	Not Used					
PVRAWLO	Not Used					
PVTV	Control Target Value - Is the target value of the PVRAW parameter. It represents the SP for the point.	Real/IOP	Oper	NaN	PM/APM	From PVEUHI to PVEULO
S1_1	Transmitter Scratch Pad - Parsed for Tuning Set and Tuning Constants	String of up to 32/FTA	View		PM/APM	
S1_2	Transmitter Status String - Parsed for CASCADE, UDC STATUS, BOX MODE, and UDC PROMPT	String of up to 32/FTA	View		PM/APM	
S1_3	Setpoint Data	String of up to 32/FTA	View		PM/APM	
S1_4	Data Discrepancy Message - Display parameters that fail data integrity tests and diagnostic data.	String of up to 32/IOP	View		PM/APM	
SECVAR	UDC Set Point Track Value - Provides access to current UDC target setpoint value for display and tracking.	Real/UDC	View	NaN	PM/APM	UDC Setpoint Range
SERIALNO	Algorithm - Used to determine algorithm type at UDC.	String of up to 8/UDC	View		PM/APM	ON-OFF PID-A PID-B PD+MR 3PSTEP MA STN

3.4 Identifying Mapped SDI Parameters, Continued

PV point SDI definitions, continued

Table 3-3 PV Point SDI Parameter Definitions, Continued

Parameter	UDC Use	Type/Source	Lock	Default	PtRes	Range
STATE	SDI State - Indicates the state of SDI to FTA communications.	Enumeration/ UDC	View		PM/ APM	1 - Update (Configuration) 2 - Loading (Configuration) 3 - Loadcomp (Configuration complete) 4 - Loadfail (Configuration error) 7 - Calfail (Calibration error) 8 - OK (Normal Mode) 9 - DB change (UDC database change)
STI_EU	EU Description - Not used by the UDC.	Enumeration/ IOP	Engr		PM/ APM	
STISWVER	Loop and point type - Loop n PV.	String of up to 8/FTA	View			
STITAG	UDC Box Tagname - Enter UDC tagname through UDC Interface. See UDC 6000 or UDC 6300 product manual.	String of up to 8/IOP	Eng /PB		PM/ APM	
URL	Configuration Upper Range Limit - Provides floating point value equivalent to upper range limit of given UDC parameter.	Real/UDC	Eng	NaN	PM/ APM	
URV	Configuration Item Tag Number - Used to read or write selected UDC database parameter.	Real/ IOP	Supr	NaN	PM/ APM	

3.4 Identifying Mapped SDI Parameters, Continued

OP point SDI definitions

Table 3-4 provides the UDC usage of SDI (STIM) parameters for OP Point mapping.

Table 3-4 OP Point SDI Parameter Definitions

Parameter	UDC Use	Type/Source	Lock	Default	PtRes	Range
C1	DDC Output Indicates the state of the override variable, which is output. It provides status feedback for automatic control schemes for avoiding windup or enabling back initialization of variables.	Real/IOP	Supr	None	PM/APM	
C2	DDC Mode Override Transfers mode from PM (APM) to the UDC while the UDC is in DDC Cascade.	Real/IOP	Supr	None	PM/APM	0.0 = Last 1.0 = Man 2.0 = Cas 3.0 = Auto 4.0 = Bcas 5.0 = Normal
CJTACT (Note 1)	OP Point Override Enable	Boolean/UDC	Engr	N/A	PM/APM	ON - OP Override enabled OFF - OP Override disabled

Note 1. When the PV point for a loop has its CJTACT parameter set to ON, the OP point CJTACT will indicate if the output is allowed to be overridden by the TDC through the custom schematics. When the PV point CJTACT is OFF, the OP point CJTACT parameter can be set to ON to force the override of the UDC output mode and setpoint through CL programs or Logic.

3.4 Identifying Mapped SDI Parameters, Continued

OP point SDI definitions, continued

Table 3-4 OP Point SDI Parameter Definitions, Continued

Parameter	UDC Use	Type/Source	Lock	Default	PtRes	Range
COMMAND (STI)	IOP Request - Allows access to operation and configuration data in UDC through IOP, using UDC 6000/6300 schematics.	Enumeration/IOP	Oper	None	PM/APM	0 - None 1 - DnLoadDb (Write parameter to UDC) 2 - UpLoadDb (Read parameter from UDC) 3 - Set_LRV CAS 4 - Set_URV AUTO 5 - Cor_LRV - BCAS 6 - Cor_URV - MAN 7 - Cor_Inpt - Not Used 8 - RstCor - Reset UDC communications Database
DAMPING (STI)	Output Limits - Real value indication of the UDC output limit level.	Real/UDC	View	None	PM/APM	0 - No limit 1 - At low limit 2 - At high limit 3 - Local Manual
DECONF (STI)	Data Transfer Configuration - for UDC interface. This parameter should be set to PV_SV_DB.	Enumeration/IOP	Engr	PV_SV_DB	PM/APM	4 - Pv-Sv-Db

3.4 Identifying Mapped SDI Parameters, Continued

OP point SDI definitions, continued

Table 3-4 OP Point SDI Parameter Definitions, Continued

Parameter	UDC Use	Type/Source	Lock	Default	PtRes	Range
INPTDIR	Not Used					
LRL	Setpoint Lower Range Limit - Provides value equivalent to lower range limit.	Real/UDC	View	NaN	PM/APM	
LRV	Output Lower Range Limit Provides value equivalent to lower range limit.	Real/UDC	Supr	NaN	PM/APM	
PIUOTDCF	CAS Track Flag	Boolean/UDC	Engr		PM/APM	ON - Track On When ON, indicates to cascaded primary loop to back initialize its cascading variable(output) OFF - Track Off
PVEUHI	PVCALC/PVTV High Range in Engineering Units - Allows access to IOP PVRAW high range scaling value. Combines with PVEULO to set range used by NIM to check limits of the PVCALC (PVRAW) and PVTV values.	Real/IOP	Engr	100.0	PM/APM	Reflects limits set at UDC. -9999.9 to 9999.9
PVEULO	PVCALC/PVTV Low Range in Engineering Units - Allows access to IOP PVRAW low range scaling value.	Real/IOP	Engr	0.0	PM/APM	Reflects limits set at UDC. -9999.9 to 9999.9 Used in combination with PVEUHI
PVRAW	PV Raw Value - Output processed by UDC and sent through IOP as PVCALC value.	Real/UDC	View	NaN	UDC	UDC PV Range

3.4 Identifying Mapped SDI Parameters, Continued

OP point SDI definitions, continued

Table 3-4 OP Point SDI Parameter Definitions, Continued

Parameter	UDC Use	Type/Source	Lock	Default	PtRes	Range
PVRAWHI	Not Used					
PVRAWLO	Not Used					
PVTV	Control Target Value - Is the target value of the PVRAW Output parameter.	Real/IOP	Oper	NaN	PM/ APM	From PVEUHI to PVEULO
S1_1	Transmitter Scratch Pad - Input 1 to Input 4	String of up to 32/FTA	View		PM/ APM	
S1_2	Transmitter Status String - Parsed for CASCADE OPTION, UDC STATUS, and SOFTWARE VERSIONS	String of up to 32/FTA	View		PM/ APM	
S1_3	Input 5, Totalizer	String of up to 32/FTA	View		PM/ APM	
S1_4	Data Discrepancy Message - Display parameters that fail data integrity tests.	String of up to 32/IOP	View		PM/ APM	
SECVAR	UDC Remote Input -Provides access to current UDC Output tracking value (PV).	Real/UDC	View	NaN	PM/ APM	UDC Setpoint Range
SERIALNO	Control Mode - Used to determine control mode at UDC.	String of up to 8/UDC	View		PM/ APM	AUTO MAN AUTO C MAN C BCAS

3.4 Identifying Mapped SDI Parameters, Continued

OP point SDI definitions, continued

Table 3-4 OP Point SDI Parameter Definitions, Continued

Parameter	UDC Use	Type/Source	Lock	Default	PtRes	Range
STATE	SDI State - Indicates the state of SDI to FTA communications.	Enumeration/ UDC	View		PM/ APM	1 - Update (Configuration) 2 - Loading (Configuration) 3 - Loadcomp (Configuration complete) 4 - Loadfail (Configuration error) 7 - Calfail (Calibration error) 8 - OK (Normal Mode) 9 - DB change (UDC database change)
STI_EU	EU Description - Not used by the UDC. Can be set to any selection.	Enumeration/ IOP	Engr		PM/ APM	
STISWVER	Loop and point type - Loop n OP.	String of up to 8/FTA	View			
STITAG	UDC Devise Tagname - Enter UDC tagname through UDC Interface. See UDC 6000 or UDC 6300 product manual.	String of up to 8/IOP	Eng /PB		PM/ APM	
URL	Setpoint Upper Range Limit -	Real/UDC	Eng	NaN	PM/ APM	
URV	Output Upper Range Limit-	Real/ IOP	Supr	NaN	PM/ APM	

3.5 Building an SDI IOP Point

Background

Two SDI IOP points identify a given UDC 6000/6300 loop for connection to the TPS system as Smart Transmitter Interface (STI) points. Both SDI IOP points are built as STI IOP points. The SDI automatically maps UDC 6000/6300 parameters to established STI parameters to provide operator and engineer level interface functions through the Universal Station.

You must build two SDI IOP points for each UDC 6000/6300 loop as outlined in the following procedure.

Prerequisite

- You are familiar with the parameters associated with building data points for the PM/APM. Refer to the *Process Manager* or *Advanced Process Manager Parameter Reference Dictionary* and Table 3-3 (SP Point) and 3-4 (OP Point) in this section for definitions.
- You have a PM/APM A/I or STI Point form on hand for reference. Refer to the *Process Manager Configuration Forms* for a copy of the form.
- You have completed the previous PM/APM box data point build procedure.
- You have a US in engineering personality and have put the keyswitch on the US in the engineering (ENG) position.

ATTENTION

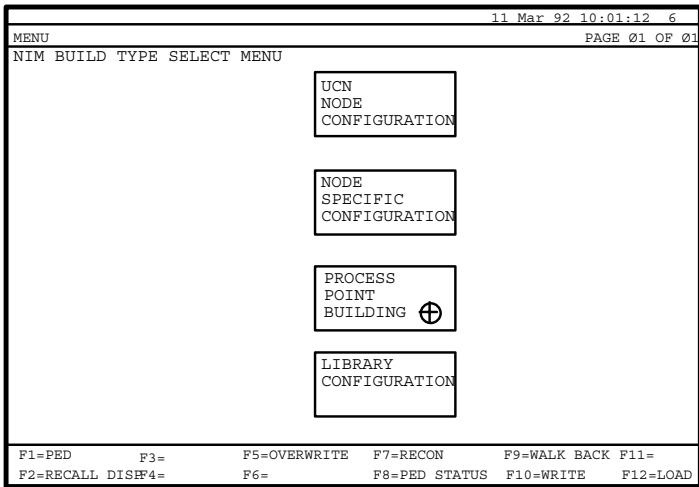
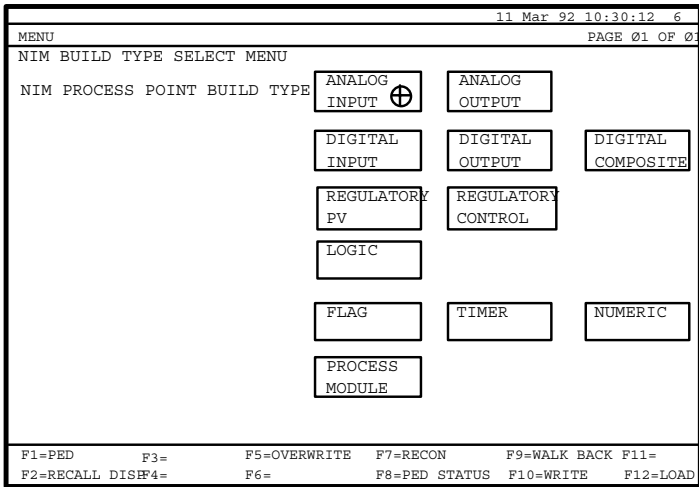
ATTENTION—If you must build a number of similar points, use the exception building method to save entry time. Refer to the *Data Entity Builder Manual* for details.

3.5 Building an SDI IOP Point, Continued

Procedure

Perform the steps in Table 3-5 to build an STI IOP point for a UDC 6000/6300 loop to be connected to the TPS system through an SDI IOP. For example purposes, the point is being built in SDI IOP in slot number 1 of PM/APM number 48 in UCN number 1.

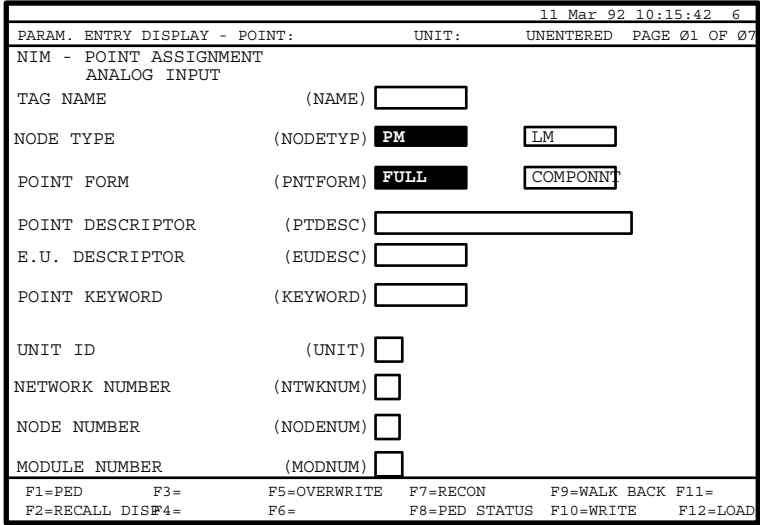
Table 3-5 Building an STI IOP Point for a UDC 6000/6300 to SDI Point

Step	Action
1	Select NETWORK INTERFACE MODULE target on Engineering Personality Main Menu to call up NIM Build Type Select Menu. <u>Example:</u> Calling up NIM Build Type Select Menu. 
2	Select PROCESS POINT BUILDING target to call up NIM Process Point Build Type menu. <u>Example:</u> Calling up NIM Process Point Build Type menu. 

3.5 Building an SDI IOP Point, Continued

Procedure, continued

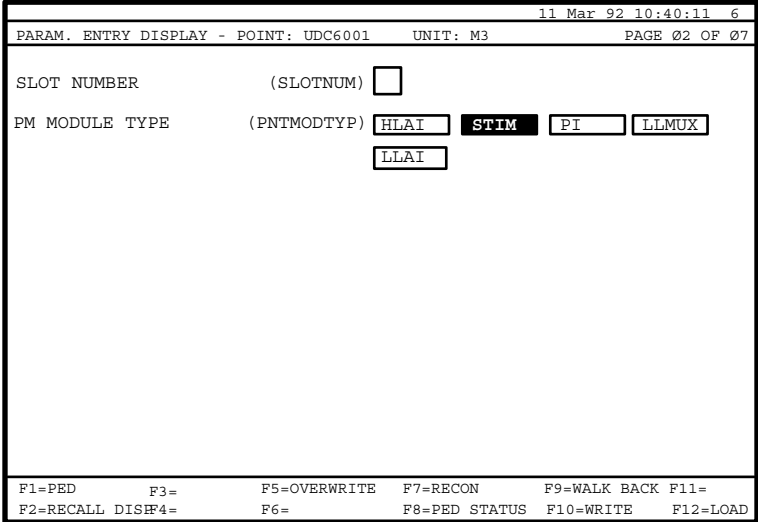
Table 3-5 Building an STI IOP Point for a UDC 6000/6300 to SDI Point, Continued

Step	Action
3	Select ANALOG INPUT target to call up PED for NIM Point Assignment display to configure an analog input. <u>Example:</u> Calling up NIM Point Assignment display for Analog Input.  The screenshot shows the 'NIM - POINT ASSIGNMENT ANALOG INPUT' screen. It has a title bar with the date and time '11 Mar 92 10:15:42 6'. Below the title bar, it says 'PARAM. ENTRY DISPLAY - POINT: UNIT: UNENTERED PAGE 01 OF 07'. The main area contains several fields: TAG NAME (NAME), NODE TYPE (NODETYP) with 'PM' selected, POINT FORM (PNTFORM) with 'FULL' selected, POINT DESCRIPTOR (PTDESC), E.U. DESCRIPTOR (EUDESC), POINT KEYWORD (KEYWORD), UNIT ID (UNIT), NETWORK NUMBER (NTWKNUM), NODE NUMBER (NODENUM), and MODULE NUMBER (MODNUM). At the bottom, there is a function key menu: F1=PED, F2=RECALL DISF, F3=, F4=, F5=OVERWRITE, F6=, F7=RECON, F8=PED STATUS, F9=WALK BACK, F10=WRITE, F11=, F12=LOAD. The number '12359' is in the bottom right corner.
4	Key in [UDC6001] as TAG NAME (NAME) or one of your choice and press [ENTER] key.
5	Select PM target as NODE TYPE (NODETYP).
6	Select FULL target as POINT FORM (PNTFORM).
7	Enter desired POINT DESCRIPTOR (PTDESC) such as [TEMPERATURE CONTROLLER] .
8	Enter desired E. U. DESCRIPTOR (EUDESC) such as [DEGF] .
9	Enter desired POINT KEYWORD (KEYWORD) such as [WATER] .
10	Enter UNIT ID (UNIT) for process unit point is assigned to such as [M3] .
11	Enter NETWORK NUMBER (NTWKNUM) for NIM's UCN that this point is in such as [01] .
12	Enter NODE NUMBER (NODENUM) for PMM's address on UCN that contains the SDI IOP such as [48] .
13	Enter MODULE NUMBER for IOP that handles this point such as [23] .

3.5 Building an SDI IOP Point, Continued

Procedure, continued

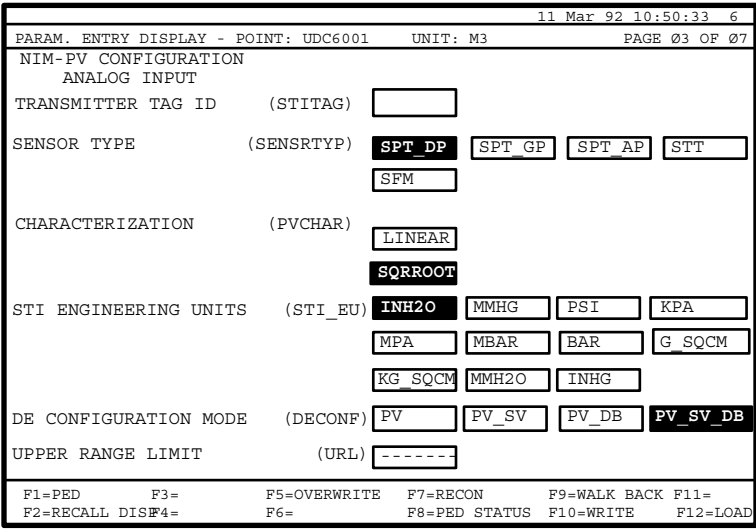
Table 3-5 Building an STI IOP Point for a UDC 6000/6300 to SDI Point, Continued

Step	Action
14	Press [ENTER] key to write entries to file and check for error indication. If entry error is signaled, recheck entries and correct as required.
15	Press [PAGE FWD] key to call up page 2 of point PED. <u>Example:</u> Calling up page 2 of PED for SDI point.  The screenshot shows a terminal window titled 'PARAM. ENTRY DISPLAY - POINT: UDC6001 UNIT: M3 PAGE 02 OF 07'. It contains two main input fields: 'SLOT NUMBER (SLOTNUM)' with an empty box, and 'PM MODULE TYPE (PNTMODTYP)' with a list of options: 'HLAI', 'STIM' (which is highlighted in black), 'PI', 'LLMUX', and 'LLAI'. At the bottom of the screen, there is a row of function keys: F1=PED, F3=, F5=OVERWRITE, F7=RECON, F9=WALK BACK, F11=, F2=RECALL, DISH4=, F6=, F8=PED STATUS, F10=WRITE, and F12=LOAD. The number '12360' is printed in the bottom right corner.
16	Enter SLOT NUMBER (SLOTNUM) for point such as [Ø1].
17	Select [STIM] target as point's I/O module type for PM MODULE TYPE (PNTMODTYP).

3.5 Building an SDI IOP Point, Continued

Procedure, continued

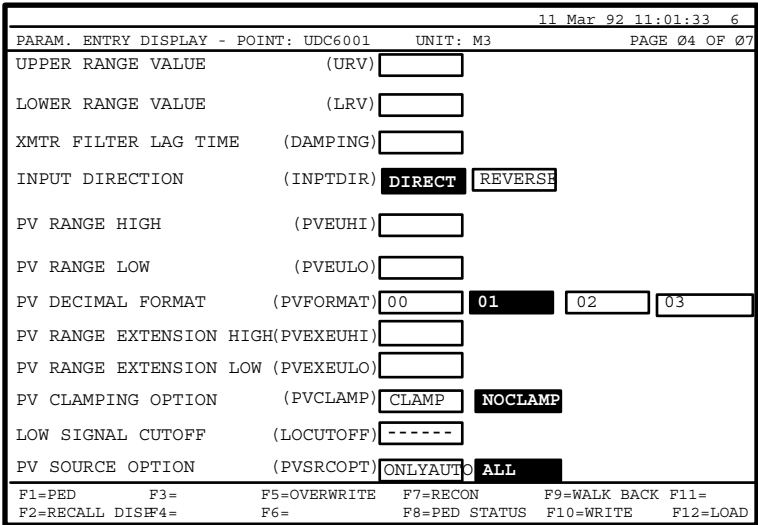
Table 3-5 Building an STI IOP Point for a UDC 6000/6300 to SDI Point, Continued

Step	Action
18	Press [PAGE FWD] key to call up page 3 of point PED. <u>Example:</u> Calling up page 3 of PED for SDI point.  12361 ATTENTION Display may look differently until SPT_DP is selected as the sensor type.
19	SDI IOP does not use SENSOR TYPE (SENSORTYP) parameter, but select SPT_DP target for other parameter configuration purposes.
20	SDI IOP does not use CHARACTERIZATION (PVCHAR) parameter since PV characterization is done in the UDC 6000/6300. Leave selection as is - it does not matter.
21	SDI IOP does not use STI ENGINEERING UNITS (STI EU) parameter. This parameter is undefined.

3.5 Building an SDI IOP Point, Continued

Procedure, continued

Table 3-5 Building an STI IOP Point for a UDC 6000/6300 to SDI Point, Continued

Step	Action
22	The SDI IOP does not use DE Configuration Mode. Set to pv_sv_db.
23	UPPER RANGE LIMIT (URL) is mapped to provide floating point value equivalent to upper range limit of given UDC parameter. So, you can skip this parameter entry also.
24	Press [PAGE FWD] key to call up page 4 of point PED. Example: Calling up page 4 of PED for SDI point.  12362
25	UPPER RANGE VALUE (URV) is mapped as a parameter ID to read or write selected UDC database parameter. Skip this parameter entry.
26	LOWER RANGE VALUE (LRV) is mapped as a read and writes UDC parameter value. Skip this parameter entry.
27	XMTR FILTER LAG TIME (DAMPING) is mapped as a read UDC alarm status parameter for SDI. It sets flags to show UDC alarms are ON or OFF, or at limits as applicable for given point format. Skip this parameter entry.

3.5 Building an SDI IOP Point, Continued

Procedure, Continued

Table 3-5 Building an SDI IOP Point for a UDC 6000/6300 to SDI Point, Continued

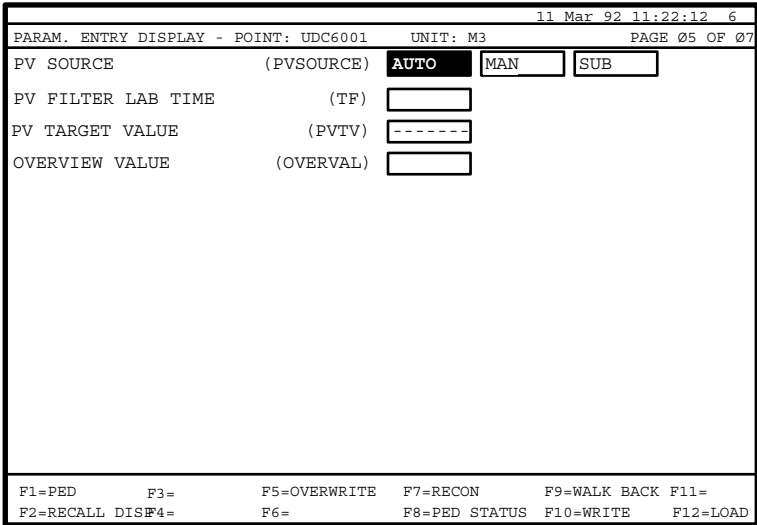
Step	Action
28	SDI IOP does not use INPUT DIRECTION (INPTDIR) parameter. Skip this parameter entry.
29*	Set desired IOP PVRAW high range scaling value as PV RANGE HIGH (PVEUHI) parameter value. It works in conjunction with PV range low to provide PV bar graph indication on display. Point 1 of each UDC interface requires that PVEUHI be configured to the UDC loop's Process Variable's high range. Point 2 of each UDC interface requires that PVEUHI be configured to the UDC loop's Output's high range (typically 100%).
30*	Set desired IOP PVRAW low range scaling value as PV RANGE LOW (PVEULO) parameter value. It works in conjunction with PV range high to provide PV bar graph indication on display. Point 1 of each UDC interface requires that PVEULO be configured to the UDC loop's Process Variable's low range. Point 2 of each UDC interface requires that PVEULO be configured to the UDC loop's Output's low range (typically 0%).
31	SDI IOP does not use PV RANGE EXTENSION HIGH (PVEXEUHI) parameter. PV limits are accessed from UDC 6000/6300 configuration. Skip this parameter entry.
32	SDI IOP does not use PV RANGE EXTENSION LOW (PVEXEULO) parameter. PV limits are accessed from UDC 6000/6300 configuration. Skip this parameter entry.
33	Select desired PV CLAMPING OPTION (PVCLAMP) parameter.
34	SDI IOP does not use LOW SIGNAL CUTOFF (LOCUTOFF) parameter. Skip this parameter entry.
35	Select ☐ ALL target as PV SOURCE OPTION (PVSRCOPT) parameter selection.
36	Press [ENTER] key to write entries to file and check for error indication. If entry error is signaled, recheck entries and correct as required.

* These parameters must be configured for proper UDC 6000/6300 operation.

3.5 Building an SDI IOP Point, Continued

Procedure, Continued

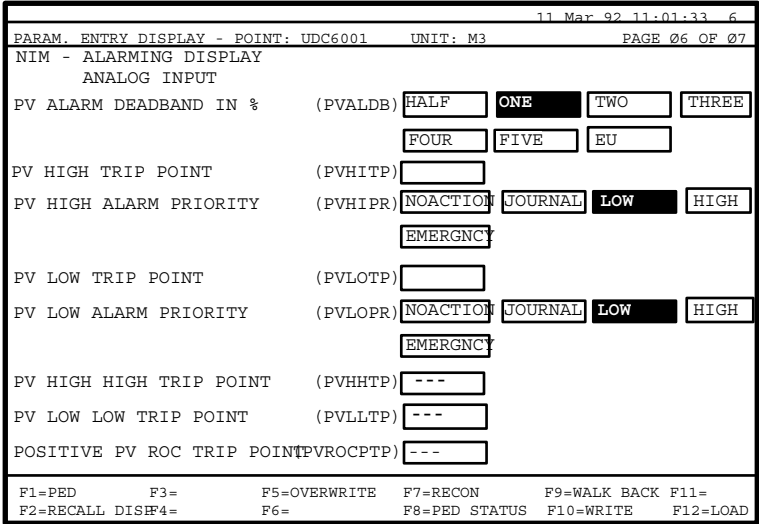
Table 3-5 Building an STI IOP Point for a UDC 6000/6300 to SDI Point, Continued

Step	Action								
37	Press [PAGE FWD] key to call up page 5 of point PED. <u>Example:</u> Calling up page 5 of PED for SDI point. 								
38	Select desired PV SOURCE (PVSOURCE) parameter. <table border="1"> <thead> <tr> <th>If source of PV is to be...</th><th>Then select...</th></tr> </thead> <tbody> <tr> <td>UDC 6000/6300</td><td> AUTO </td></tr> <tr> <td>operator entry</td><td> MAN </td></tr> <tr> <td>Application Module or another internal LCN node</td><td> SUB </td></tr> </tbody> </table>	If source of PV is to be...	Then select...	UDC 6000/6300	AUTO	operator entry	MAN	Application Module or another internal LCN node	SUB
If source of PV is to be...	Then select...								
UDC 6000/6300	AUTO								
operator entry	MAN								
Application Module or another internal LCN node	SUB								
39	SDI IOP does not use PV FILTER LAG TIME (TF) parameter. Skip this parameter entry.								
40	PV TARGET VALUE (PVTV) is mapped as a computer setpoint or output parameter for UDC. Skip this parameter entry.								
41	OVERVIEW VALUE (OVERVAL) parameter has no meaning. Skip this parameter entry.								
42	Press [ENTER] key to write entries to file and check for error indication. If entry error is signaled, recheck entries and correct as required.								

3.5 Building an SDI IOP Point, Continued

Procedure, Continued

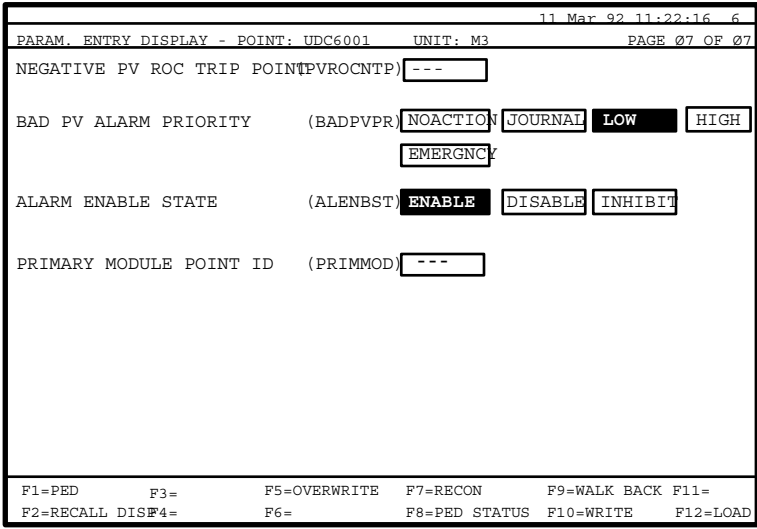
Table 3-5 Building an STI IOP Point for a UDC 6000/6300 to SDI Point, Continued

Step	Action
43	Press [PAGE FWD] key to call up page 6 of point PED. <u>Example:</u> Calling up page 6 of PED for SDI point.  The screenshot shows the 'PARAM. ENTRY DISPLAY - POINT: UDC6001 UNIT: M3 PAGE 06 OF 07'. The title is 'NIM - ALARMING DISPLAY ANALOG INPUT'. The parameters and their current settings are: - PV ALARM DEADBAND IN % (PVALDB): HALF (selected), ONE, TWO, THREE, FOUR, FIVE, EU - PV HIGH TRIP POINT (PVHITP): [] - PV HIGH ALARM PRIORITY (PVHIPR): NOACTION (selected), JOURNAL, LOW, HIGH, EMERGENCY - PV LOW TRIP POINT (PVLOTP): [] - PV LOW ALARM PRIORITY (PVLOPR): NOACTION (selected), JOURNAL, LOW, HIGH, EMERGENCY - PV HIGH HIGH TRIP POINT (PVHHTP): --- - PV LOW LOW TRIP POINT (PVL LTP): --- - POSITIVE PV ROC TRIP POINT (PVROCTP): --- Footer function keys: F1=PED, F2=RECALL DISF, F3=, F4=, F5=OVERWRITE, F6=, F7=RECON, F8=PED STATUS, F9=WALK BACK, F10=WRITE, F11=, F12=LOAD.
44	Enter desired IOP alarm function data for point through parameters on page 6 shown above. Refer to <i>PM/APM Analog Input Point form and Parameter Reference Dictionary</i> for details. ATTENTION Point 1 of each UDC 6000/6300 interface provides alarms for the UDC loop's Process Variable. Point 2 of each UDC 6000/6300 interface provides alarms for the UDC loop's Output. Only the Point 1 alarms are used in the custom schematic integration displays.

3.5 Building an SDI IOP Point, Continued

Procedure, Continued

Table 3-5 Building an STI IOP Point for a UDC 6000/6300 to SDI Point, Continued

Step	Action								
45	Press [PAGE FWD] key to call up page 7 of point PED. <u>Example:</u> Calling up page 7 of PED for SDI point. 								
46	Complete entry of alarm function data for point through parameters on page 7 shown above.								
47	Press [CTL] and [F12] to load point in file. <table border="1"> <thead> <tr> <th>If...</th><th>Then...</th></tr> </thead> <tbody> <tr> <td>point loads</td><td>go to next step.</td></tr> <tr> <td>overwrite prompt appears</td><td>press [CTL] and [F5] keys to overwrite existing file.</td></tr> <tr> <td>error message appears</td><td>refer to <i>Data Entity Builder Manual</i> for error recovery data and recheck your work including hardware installation and wiring connections.</td></tr> </tbody> </table>	If...	Then...	point loads	go to next step.	overwrite prompt appears	press [CTL] and [F5] keys to overwrite existing file.	error message appears	refer to <i>Data Entity Builder Manual</i> for error recovery data and recheck your work including hardware installation and wiring connections.
If...	Then...								
point loads	go to next step.								
overwrite prompt appears	press [CTL] and [F5] keys to overwrite existing file.								
error message appears	refer to <i>Data Entity Builder Manual</i> for error recovery data and recheck your work including hardware installation and wiring connections.								
48	Press [CTL] and [HELP] keys to call up Engineering Personality Main Menu.								
49	Repeat this procedure to build second point for this UDC 6000/6300 loop then, checkpoint your database.								

3.6 Configuring UDC 6000/6300 Controller

Background

Each UDC 6000/6300 controller stores its own configuration database. The database tells the UDC 6000/6300 how it should operate based on user-entered configuration selections and values.

You make configuration entries through the integral front panel operator interface of the UDC 6000/6300. When the UDC 6000/6300 is integrated with a TPS system, you can make changes in configuration entries through US displays.

Prerequisites

- Controller is powered. Note that the controller need not be installed to enter configuration data and it does retain its database when power is removed.
- You know how to use the UDC 6000/6300 operator interface.
- You have a copy of the UDC 6000 or UDC 6300 product manual on hand.

Procedure

Refer to the Configuration section in the *UDC 6000* or *UDC 6300 Process Controller Product Manual* to configure your controller's database to meet your particular control requirements. We recommend that you configure the original database through the controller's operator interface since the station address and communications link type can only be configured at the controller.

3.7 Configuring UDC 6000/6300 Schematics

Background

As previously mentioned, the UDC 6000/6300 graphics software provides templates for creating group and detail displays in the US dedicated to SDI/UDC 6000/6300 points. The graphics software also includes customized point summary, configuration schematics, and operator change zone displays.

Prerequisite

- You know how to use the US operator interface.
- You have a US in engineering personality and have put the keyswitch on the US in the engineering (ENG) position.
- You have loaded the UDC 6000/6300 graphics software and know the directory pathname.
- You have built two SDI points for each loop.

UDC Summary Display

The UDC Summary Display is the display used to access UDC 6000/6300 Detail Displays and the UDC 6000/6300 Group displays.

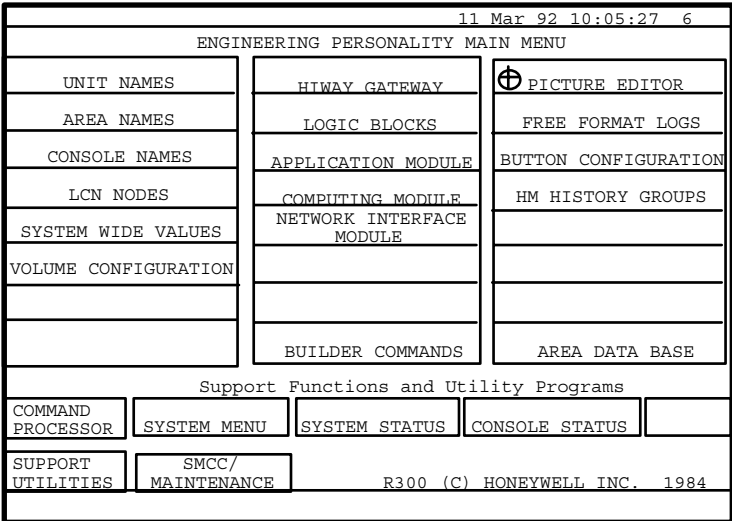
ATTENTION

- NET>DISP is used as the directory pathname in this procedure for example purposes only. Use the one you have selected. (See 2.10.)
- Schematic source files are shipped with WRITE protect attribute ON. Schematic source files must be unprotected before changes can be made.

Procedure for configuring the summary display

Use the steps in Table 3-6 to configure points to access UDC 6000/6300 Detail displays and group displays from the UDC 6000/6300 Summary display.

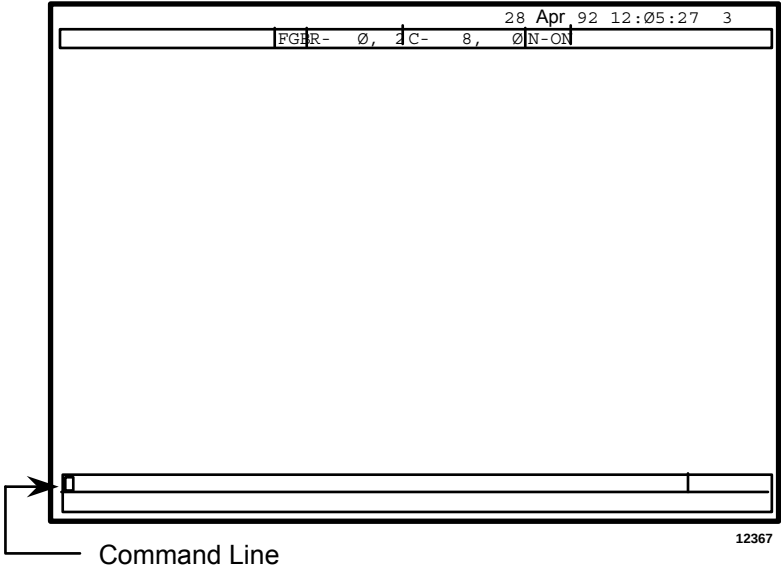
Table 3-6 Configuring the Summary Display

Step	Action
1	Press [CTL] and [HELP] keys to call up Engineering Personality Main Menu. The figure below is an example of what you will see. <u>Example:</u> Calling up Engineering Personality Main Menu. 

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for
configuring the
summary display,
continued

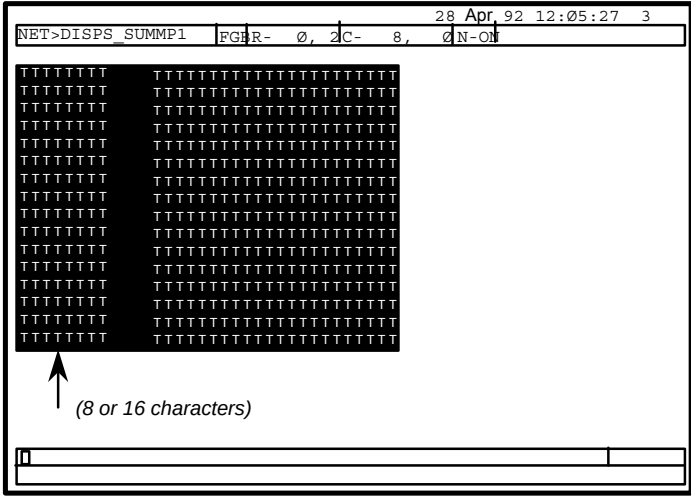
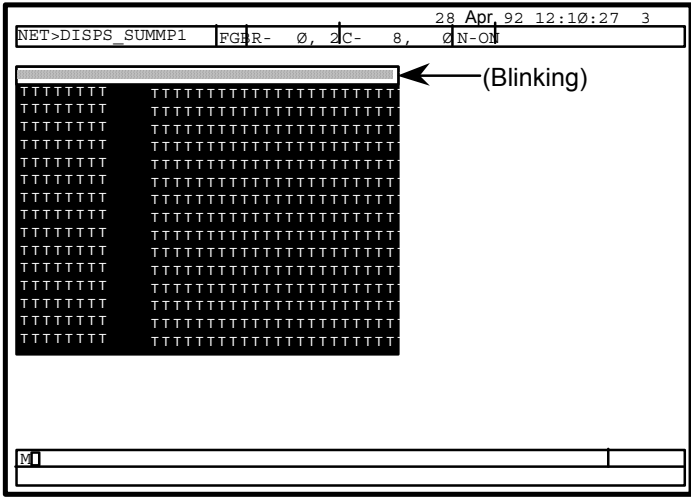
Table 3-6 Configuring the Summary Display, Continued

Step	Action
2	Select PICTURE EDITOR target on the Main menu to call up the display. The figure below is an example of what you will see. <u>Example:</u> Calling up Picture Editor.  Command Line
3	Enter your directory Pathname and Page number. Key in [R NET>DISP>S_SUMMP1] in command line. Where: R NET>DISP>S_SUMMP1 Page number 1 to 8 Source file for summary page Directory pathname for UDC 6000 graphics (For example only - Your directory was selected when the graphics were loaded - see 2.10) Read command 12368

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for
configuring the
summary display,
continued

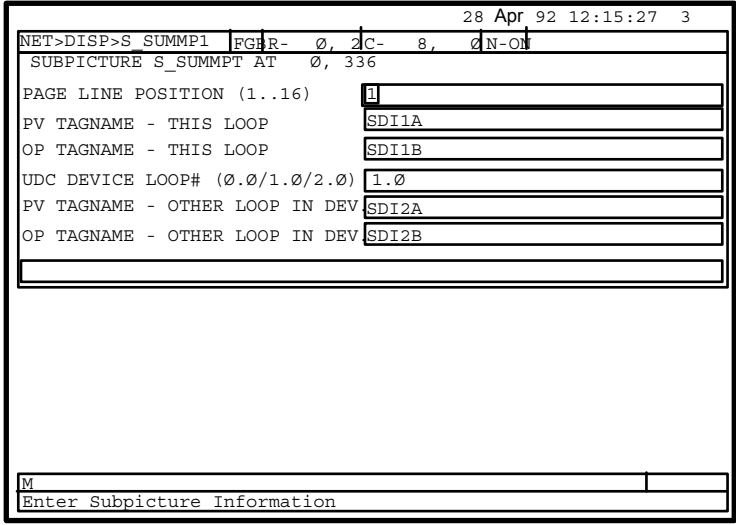
Table 3-6 Configuring the Summary Display, Continued

Step	Action
4	Press [ENTER] key to call up source file for page number 1. The example shows an 8-character tag field. It could be 16 characters. <u>Example:</u> Calling up source file for page 1. 
5	Move cursor to line 1 in source file display.
6	Press [SELECT] key to select line. Press [ENTER] key to activate line for modification. The line starts blinking. <u>Example:</u> Selecting the line for modification. 

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for
configuring the
summary display,
continued

Table 3-6 Configuring the Summary Display, Continued

Step	Action
7	Key in [M] in the command line and press [ENTER] key to call up the subpicture information for modification. The subpicture will show default data. Example: Calling up subpicture information.  The subpicture information will be modified (steps 8 thru 13) the same way for each <u>page line position</u> selected. Each loop has 2 point tagnames for you to identify. PV TAGNAME - THIS LOOP and OP TAGNAME - THIS LOOP. These point tagnames can be for: • The only loop in a single loop controller • Loop 1 of a 2 loop controller • Loop 2 of a 2 loop controller. Be aware of which Loop you are entering these tags for since you will make that selection in Step 11. If you are identifying the tagname for one of the loops in a 2 loop controller, you must also identify the tagnames for the other loop in the device.
8	In the first fill-in block, key in the PAGE LINE POSITION (1..16) for the POINT ID DESCRIPTOR line being modified.
9	Move the cursor to PV TAGNAME - THIS LOOP fill-in block. Key in the tagname for PV point for this loop. (example [SDI1A])
10	Move the cursor to OP TAGNAME - THIS LOOP fill-in block. Key in the tagname for the OP point for this loop. (example [SDI1B])

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for
configuring the
summary display,
continued

Table 3-6 Configuring the Summary Display, Continued

Step	Action								
11	In this step, you identify the device loop for the point tagnames you have entered in steps 9 and 10. Move cursor to UDC DEVICE LOOP# (Ø.Ø/1.Ø/2.Ø) fill-in block. <table border="1"> <thead> <tr> <th>If the points are for...</th><th>Then Key In</th></tr> </thead> <tbody> <tr> <td>the only loop in a one-loop UDC 6000/6300 controller</td><td>Ø.Ø</td></tr> <tr> <td>loop 1 in a two-loop UDC 6000/6300 controller</td><td>1.Ø</td></tr> <tr> <td>loop 2 in a two-loop UDC 6000/6300 controller</td><td>2.Ø</td></tr> </tbody> </table> In this example, key in [1.Ø] since it identifies loop 1 in a two-loop UDC 6000/6300.	If the points are for...	Then Key In	the only loop in a one-loop UDC 6000/6300 controller	Ø.Ø	loop 1 in a two-loop UDC 6000/6300 controller	1.Ø	loop 2 in a two-loop UDC 6000/6300 controller	2.Ø
If the points are for...	Then Key In								
the only loop in a one-loop UDC 6000/6300 controller	Ø.Ø								
loop 1 in a two-loop UDC 6000/6300 controller	1.Ø								
loop 2 in a two-loop UDC 6000/6300 controller	2.Ø								
12	In the next two steps, you will identify the point tagnames for the <i>other loop</i> in a 2 loop controller. These tagnames will be associated with the selections you made in steps 9 and 10. Move the cursor to PT1 TAGNAME - OTHER LOOP IN DEV. Refer to your entry in UDC DEVICE LOOP# (Ø.Ø/1.Ø/2.Ø) fill-in block. <table border="1"> <thead> <tr> <th>If the entry in step 11 was...</th><th>Then...</th></tr> </thead> <tbody> <tr> <td>Ø.Ø (The only Loop in a One-loop controller)</td><td>Key in the same PV TAGNAME you used for loop 1 and entered in step 9. (Example SDI1A)</td></tr> <tr> <td>1.Ø (Loop 1 in a 2-loop controller)</td><td>Key in the Tagname for the (PV Point) for loop 2 of the same controller <i>Loop 1- PV TAGNAME was identified in step 9.</i> (Example SDI2A)</td></tr> <tr> <td>2.Ø (Loop 2 in a 2-loop controller)</td><td>Key in the Tagname for the (PV Point) for loop 1 of the same controller- <i>Loop 2 - PV TAGNAME was identified in step 9.</i> (Example SDI1A)</td></tr> </tbody> </table>	If the entry in step 11 was...	Then...	Ø.Ø (The only Loop in a One-loop controller)	Key in the same PV TAGNAME you used for loop 1 and entered in step 9. (Example SDI1A)	1.Ø (Loop 1 in a 2-loop controller)	Key in the Tagname for the (PV Point) for loop 2 of the same controller <i>Loop 1- PV TAGNAME was identified in step 9.</i> (Example SDI2A)	2.Ø (Loop 2 in a 2-loop controller)	Key in the Tagname for the (PV Point) for loop 1 of the same controller- <i>Loop 2 - PV TAGNAME was identified in step 9.</i> (Example SDI1A)
If the entry in step 11 was...	Then...								
Ø.Ø (The only Loop in a One-loop controller)	Key in the same PV TAGNAME you used for loop 1 and entered in step 9. (Example SDI1A)								
1.Ø (Loop 1 in a 2-loop controller)	Key in the Tagname for the (PV Point) for loop 2 of the same controller <i>Loop 1- PV TAGNAME was identified in step 9.</i> (Example SDI2A)								
2.Ø (Loop 2 in a 2-loop controller)	Key in the Tagname for the (PV Point) for loop 1 of the same controller- <i>Loop 2 - PV TAGNAME was identified in step 9.</i> (Example SDI1A)								

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for
configuring the
summary display,
continued

Table 3-6 Configuring the Summary Display, Continued

Step	Action								
13	Move the cursor to OP TAGNAME - OTHER LOOP IN DEV. Refer to your entry in UDC DEVICE LOOP# (Ø.Ø/1.Ø/2.Ø) fill-in block. <table><tr><th>If the entry in step 11 was...</th><th>Then...</th></tr><tr><td>Ø.Ø <i>(The only Loop in a One-loop controller)</i></td><td>Key in the same OP TAGNAME you used for loop 1 and entered in step 10. (Example SDI1B)</td></tr><tr><td>1.Ø <i>(Loop 1 in a 2-loop controller)</i></td><td>Key in the Tagname for the (OP Point) for loop 2 of the same controller <i>Loop 1- OP TAGNAME was identified in step 10.</i> (Example SDI2B)</td></tr><tr><td>2.Ø <i>(Loop 2 in a 2-loop controller)</i></td><td>Key in the Tagname for the (OP Point) for loop 1 of the same controller <i>Loop 2 - OP TAGNAME was identified in step 10.</i> (Example SDI1B)</td></tr></table>	If the entry in step 11 was...	Then...	Ø.Ø <i>(The only Loop in a One-loop controller)</i>	Key in the same OP TAGNAME you used for loop 1 and entered in step 10. (Example SDI1B)	1.Ø <i>(Loop 1 in a 2-loop controller)</i>	Key in the Tagname for the (OP Point) for loop 2 of the same controller <i>Loop 1- OP TAGNAME was identified in step 10.</i> (Example SDI2B)	2.Ø <i>(Loop 2 in a 2-loop controller)</i>	Key in the Tagname for the (OP Point) for loop 1 of the same controller <i>Loop 2 - OP TAGNAME was identified in step 10.</i> (Example SDI1B)
If the entry in step 11 was...	Then...								
Ø.Ø <i>(The only Loop in a One-loop controller)</i>	Key in the same OP TAGNAME you used for loop 1 and entered in step 10. (Example SDI1B)								
1.Ø <i>(Loop 1 in a 2-loop controller)</i>	Key in the Tagname for the (OP Point) for loop 2 of the same controller <i>Loop 1- OP TAGNAME was identified in step 10.</i> (Example SDI2B)								
2.Ø <i>(Loop 2 in a 2-loop controller)</i>	Key in the Tagname for the (OP Point) for loop 1 of the same controller <i>Loop 2 - OP TAGNAME was identified in step 10.</i> (Example SDI1B)								
14	Press [ENTER]. Go to step 5 and move the cursor to a new POINT ID DESCRIPTOR line (Example - Line 2) and repeat steps 6 through 13 as required for the other loops.								
15	When the POINT ID DESCRIPTOR line modifications are complete, delete any lines from the screen that will not be used. Move the cursor to any line not being used. Press the [SELECT] key to select the line, then, press the [ENTER] key. The line starts blinking. Press the [DELETE] key, or key-in DEL, then press [ENTER].								

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for
configuring the
summary display,
continued

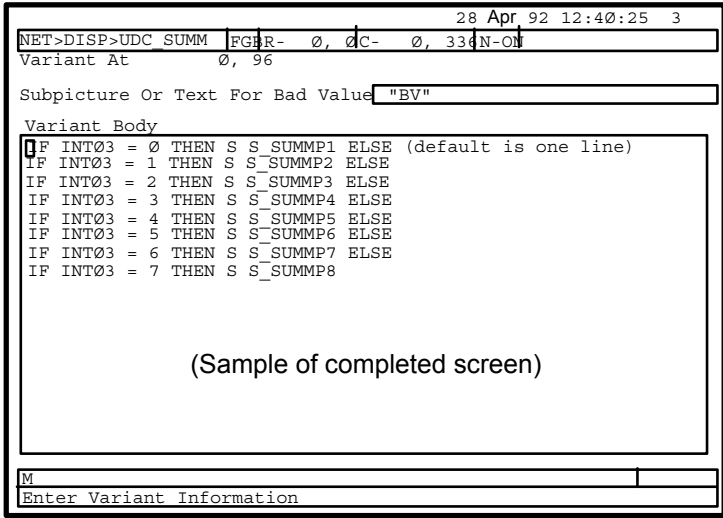
Table 3-6 Configuring the Summary Display, Continued

Step	Action
16	Write (save) the modified subpicture. Key in [W NET>DISP>S_SUMMP1] in command line. Where: W NET>DISP>S_SUMMP1 Press [ENTER] key. A prompt will read "Press ENTER to overwrite, CANCEL to abort write." Press required key.
17	Repeat steps 3 to 16 for each PAGE of the summary display as required.
18	Replace the existing subpictures in the summary display. Key in [R NET>DISP>UDC_SUMM] in command line and press [ENTER] key to call up Summary display in picture editor. <u>Example:</u> Calling up Summary display in picture editor.
19	Key in [REP SUB S_SUMMP1 S_SUMMP1] in command line and press [ENTER] key to replace subpicture for page 1 in Summary display.

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for configuring the summary display, continued

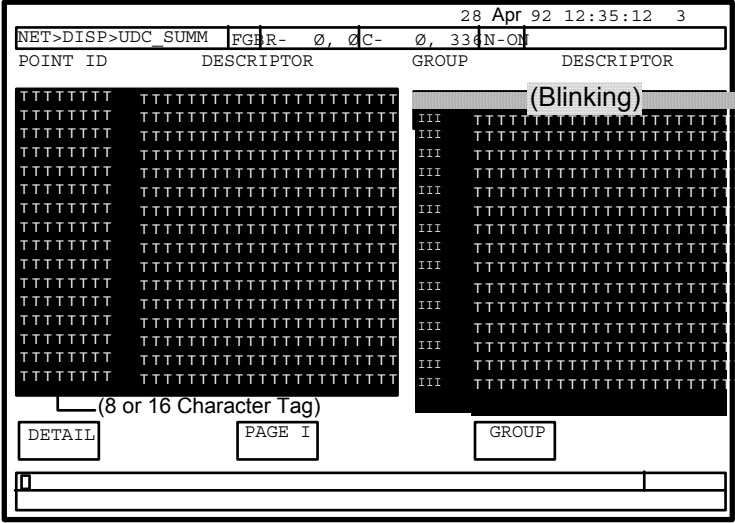
Table 3-6 Configuring the Summary Display, Continued

Step	Action
20	Repeat step 19 for each page modified, S_SUMMP1, S_SUMMP2, etc.
21	Move cursor to the first line in POINT ID/DESCRIPTOR field. Press [SELECT] key to select line. Press [ENTER] key to activate line for modification. Page starts blinking.
22	Key in [M] in command line and press [ENTER] key to call up variant at 0, 96 for modification. <u>Example:</u> Calling up variant at 0, 96 for modification.  (Sample of completed screen)
23	With cursor in Variant Body, key in this information for each page of summary as applicable. This example is for 8 pages. <pre> IF INT03 = 0 THEN S S_SUMMP1 ELSE IF INT03 = 1 THEN S S_SUMMP2 ELSE IF INT03 = 2 THEN S S_SUMMP3 ELSE IF INT03 = 3 THEN S S_SUMMP4 ELSE IF INT03 = 4 THEN S S_SUMMP5 ELSE IF INT03 = 5 THEN S S_SUMMP6 ELSE IF INT03 = 6 THEN S S_SUMMP7 ELSE IF INT03 = 7 THEN S S_SUMMP8 </pre>
24	Press [ENTER] key to enter variant modifications.

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for
configuring the
summary display,
continued

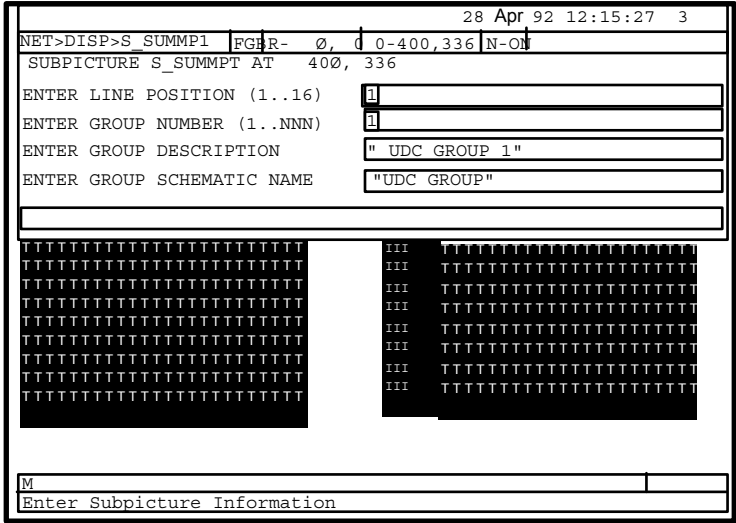
Table 3-6 Configuring the Summary Display, Continued

Step	Action
25	Modify the GROUP DESCRIPTORS. Move cursor to the first line in GROUP DESCRIPTOR field.
26	Press [SELECT] key to select line. Press [ENTER] key to activate line for modification. Line starts blinking. <u>Example:</u> Selecting A GROUP DESCRIPTOR line for modification. 

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for
configuring the
summary display,
continued

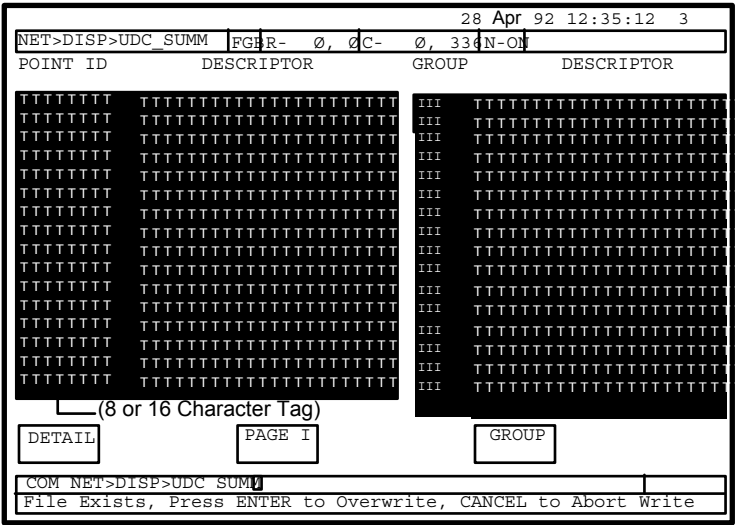
Table 3-6 Configuring the Summary Display, Continued

Step	Action
27	Key in [M] in the command line and press [ENTER] key to call up the subpicture information for modification. The subpicture will show default data. <u>Example:</u> Calling up subpicture information.  The Group Number is only a reference number that will appear on the summary display. The Group Description is a reference number also. Key in using " - - " quotes.
28	Key in [1] as LINE POSITION (1..16) for the GROUP DESCRIPTOR line being modified.
29	Move cursor to the first line in ENTER GROUP NUMBER (1...NNN) fill-in block. Key in the group number (reference only) that you want to appear on the summary display.
30	Move cursor to the first line in ENTER GROUP DESCRIPTION fill-in block. Key in a name you want to use for the group number selected in the previous step. Use quotes " - - " around the name. (example - "UDC GROUP 1")

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for
configuring the
summary display,
continued

Table 3-6 Configuring the Summary Display, Continued

Step	Action
31	Move cursor to the first line in ENTER GROUP SCHEMATIC NAME fill-in block. Key in a name you want to use for the group schematic associated with the above selections. (example - "UDCGROUP")
32	Press [ENTER] key. Go to step 25 and move the cursor to a new Group Descriptor line (example - Line 2) and repeat steps 27 thru 31 as required for the other groups.
33	When the Group Descriptor Line modification is complete, delete any blank lines from the screen that will not be used.
34	Key in [COM NET>DISP>UDC_SUMM] in command line. Press [ENTER] key to compile Summary display file. If file exists, prompt asks if you want to overwrite it or not. Press [ENTER] key to overwrite file. <u>Example:</u> Compiling Summary display.
	 12377
35	Key in [END] and press [ENTER] key to return to Engineering Personality Main Menu. A prompt will read "Picture modified." Press ENTER to exit, CANCEL to abort exit.
36	To view Summary and Group displays, press [SCHEM] key, key in [UDC_SUMM] as schematic name and press [ENTER] key to call up UDC 6000/6300 Summary display. Refer to 4.0 Operation for additional information.

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for adding a point ID or group descriptor to the summary display

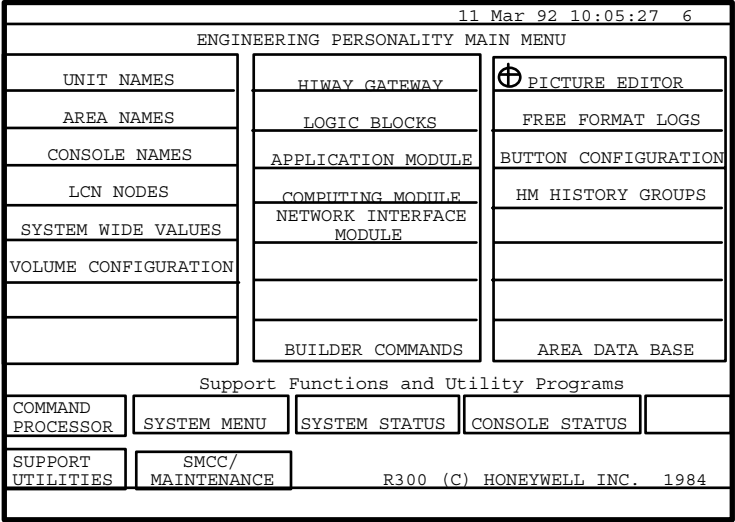
You can add a Point ID descriptor(s) or a Group descriptor(s) to the Summary display.

Use the steps in Table 3-7 to add a Point ID or Group descriptor(s).

ATTENTION

ATTENTION—NET>DISP is used as the directory pathname in this procedure for example purposes only. Use the one you have selected. (See 2.10)

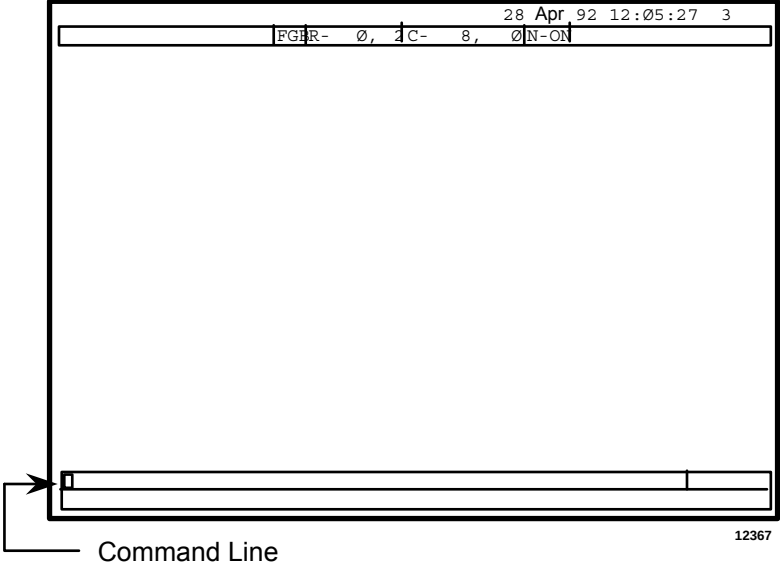
Table 3-7 Adding a Point ID or Group Descriptor(s) to the Summary Display

Step	Action
1	Press [CTL] and [HELP] keys to call up Engineering Personality Main Menu. The figure below is an example of what you will see. <u>Example:</u> Calling up Engineering Personality Main Menu. 

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for adding
a point ID or group
descriptor to the
summary display,
continued

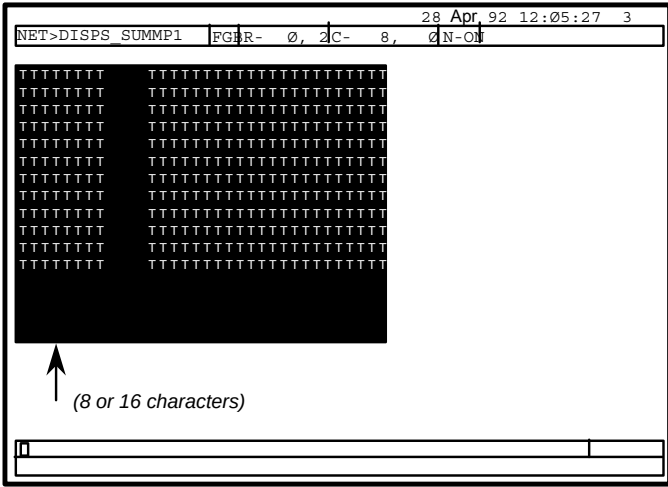
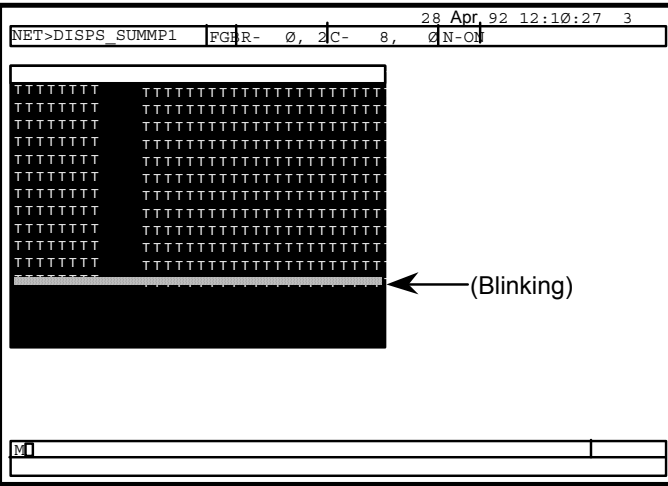
Table 3-7 Adding a Point ID or Group Descriptor(s) to the Summary
Display, Continued

Step	Action
2	Select PICTURE EDITOR target on the Main menu to call up the display. The figure below is an example of what you will see. <u>Example:</u> Calling up Picture Editor.  Command Line
3	Enter your directory Pathname and Page number. Key in [R NET>DISP>S_SUMMP1] in command line. Where: R NET>DISP>S_SUMMP1 Page number 1 to 8 Source file for summary page Directory pathname for UDC 6000 graphics (For example only - Your directory was selected when the graphics were loaded - see 2.10) Read command

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for adding
a point ID or group
descriptor to the
summary display,
continued

Table 3-7 Adding a Point ID or Group Descriptor(s) to the Summary Display, Continued

Step	Action
4	Press [ENTER] key to call up source file for page number 1. The example shows an 8-character tag field. It could be 16 characters. <u>Example:</u> Calling up source file for page 1.  12378
5	Move cursor to the bottom line in source file display.
6	Press [SELECT] key to select the line. Press [ENTER] key to activate line for copying. The line starts blinking. <u>Example:</u> Selecting the line for modification.  12379

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for adding
a point ID or group
descriptor to the
summary display,
continued

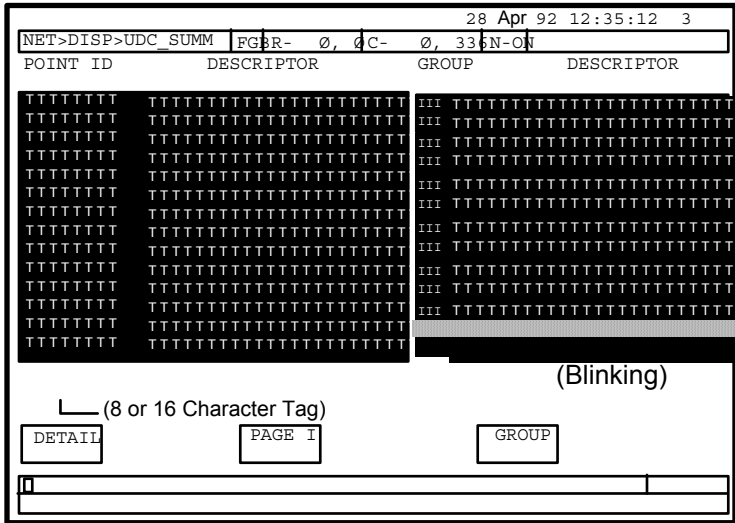
Table 3-7 Adding a Point ID or Group Descriptor(s) to the Summary Display, Continued

Step	Action
7	Copy the line. Press [C] then, press [ENTER]. A prompt on the command line states "Enter Copy Coordinates." There is a "COPY FROM" coordinate and a "COPY TO" coordinate.
8	Enter the "FROM" coordinate. Move the cursor to the blinking line. Press [SELECT]. The "FROM" coordinate is established. A + will appear at coordinate selected.
9	Enter the "TO" coordinate. Move the cursor to the next line <i>directly</i> below the "FROM" coordinate.
10	Press [SELECT], then [ENTER]. The line selected is copied to the next line.
11	Move the cursor to the added line. Select the line and modify the copied subpicture information with the information for the new point ID according to the instructions given in Table 3-6, steps 6 through 24.
12	Add a GROUP DESCRIPTOR(S). Move cursor to the last line in GROUP DESCRIPTOR field.

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for adding
a point ID or group
descriptor to the
summary display,
continued

Table 3-7 Adding a Point ID or Group descriptor(s) to the Summary Display, Continued

Step	Action
13	Press [SELECT] key to select line. Press [ENTER] key to activate line for copying. Line starts blinking. <u>Example:</u> Selecting A GROUP DESCRIPTOR line for modification.  12380
14	Enter the "FROM" coordinate. Move the cursor to the blinking line. Press [SELECT]. The "FROM" coordinate is established. A + will appear at coordinate selected.
15	Enter the "TO" coordinate. Move the cursor to the next line <i>directly</i> below the "FROM" coordinate.
16	Press [SELECT] then, press [ENTER]. The line selected is copied to the next line.
17	Move the cursor to the added line. Select the line and modify the copied subpicture information with the information for the new Group descriptor according to the instructions given in Table 3-6, steps 26 through 35.

3.7 Configuring UDC 6000/6300 Schematics, Continued

**Procedure for
configuring the group
display**

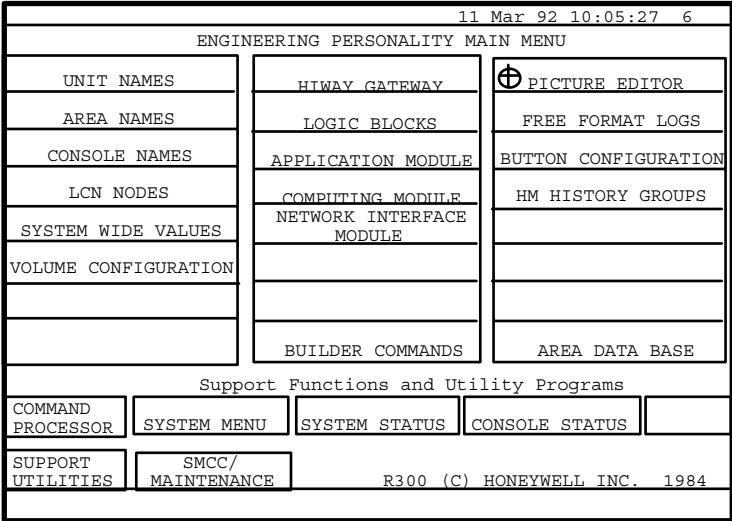
The UDC Group Display is provided containing the subpicture(s), which let you enter the specifications of each UDC loop that is part of a group. Each group requires a separate custom display, which can be copied from the one provided and modified accordingly.

Use the steps in Table 3-8 to configure the Group displays available from the UDC 6000/6300 Summary display.

ATTENTION

ATTENTION—NET>DISP is used as the directory pathname in this procedure for example purposes only. Use the one you have selected. (See 2.10)

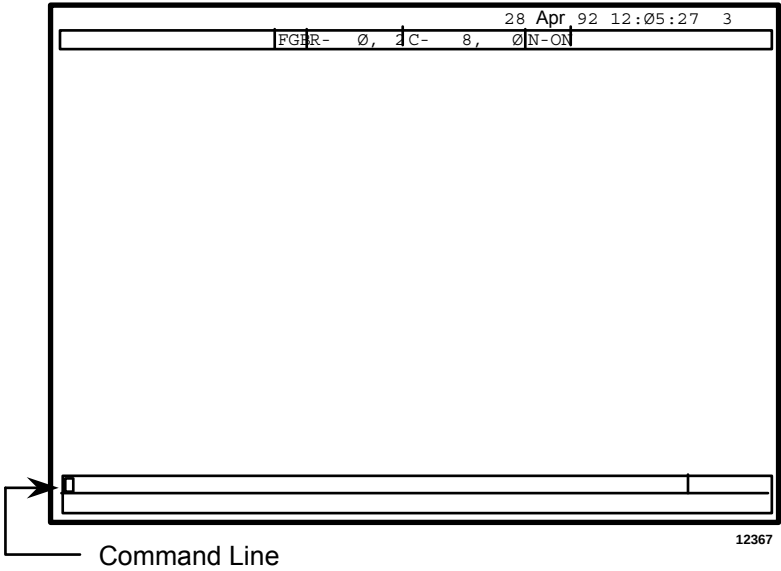
Table 3-8 Configuring the Group Display

Step	Action
1	Press [CTL] and [HELP] keys to call up Engineering Personality Main Menu. The figure below is an example of what you will see. <u>Example:</u> Calling up Engineering Personality Main Menu. 

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for
configuring the group
display, continued

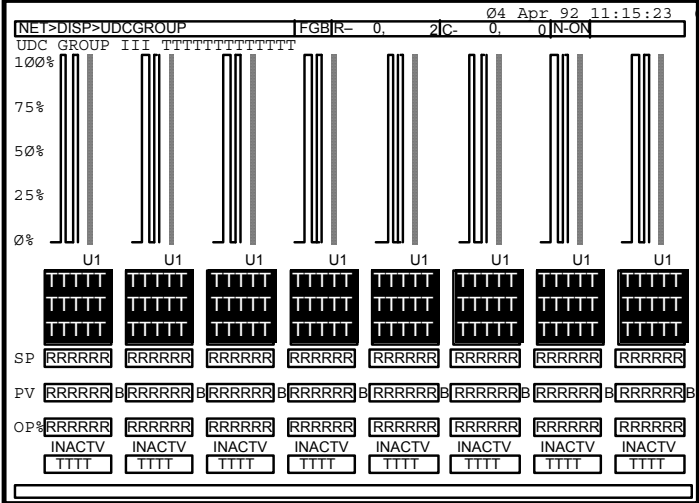
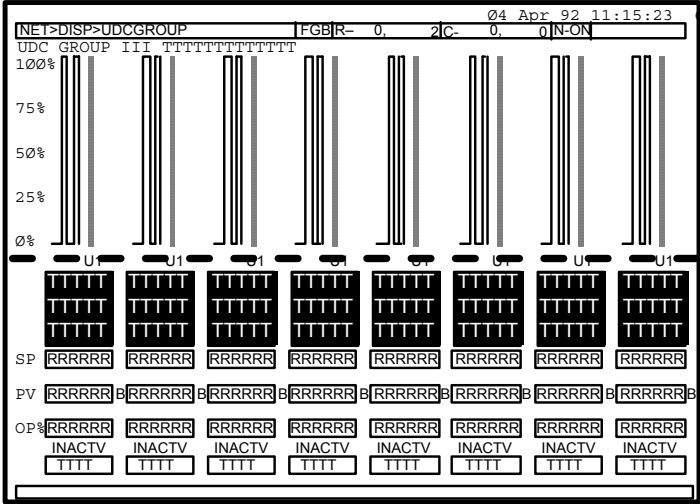
Table 3-8 Configuring the Group Display, Continued

Step	Action
2	Select PICTURE EDITOR target on the Main menu to call up the display. The figure below is an example of what you will see. <u>Example:</u> Calling up Picture Editor. 
3	Enter your directory Pathname and Group name. Key in [R NET>DISP>UDC GROUP] in command line. Where: R NET>DISP>UDCGROUP Read command Directory pathname for UDC 6000 graphics (For example only - Your directory was selected when the graphics were loaded - see 2.10) Group Name selected in Step 31 in Table 3-6 12381

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for
configuring the group
display, continued

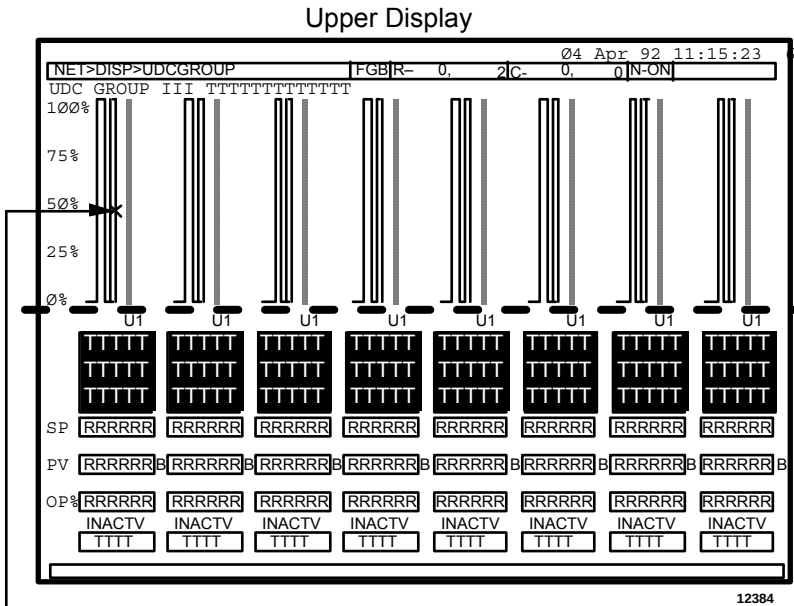
Table 3-8 Configuring the Group Display, Continued

Step	Action
4	Press [ENTER] key to call up source file for the group selected. <u>Example: Calling up source file for the group selected.</u>  12382
5	Configure each loop in the group in two parts using the upper and lower portions of the screen. The dashed line shown in the figure below divides the screen . Each individual loop will be configured the same way using steps 6 through 23. Upper Display  Lower Display 12383

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for configuring the group display, continued

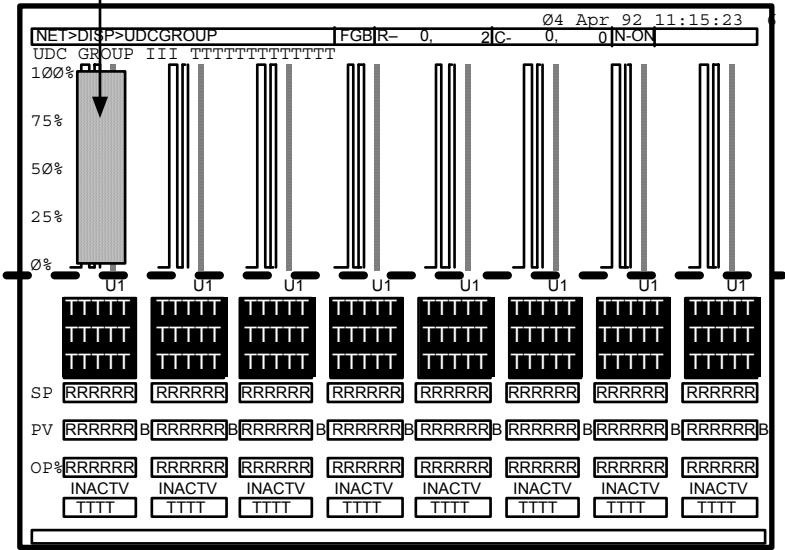
Table 3-8 Configuring the Group Display, Continued

Step	Action
6	- In the upper part, you identify the point tagnames for the loop. - In the lower part, the slot number is identified along with the tagnames and loop number that were identified on the summary display. Also configured in this part is the group and slot number of the loop so it can be used to invoke a standard TPS Group Trend display. Configure the upper part. Move cursor to the first loop bar graph in the display.  12384

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for
configuring the group
display, continued

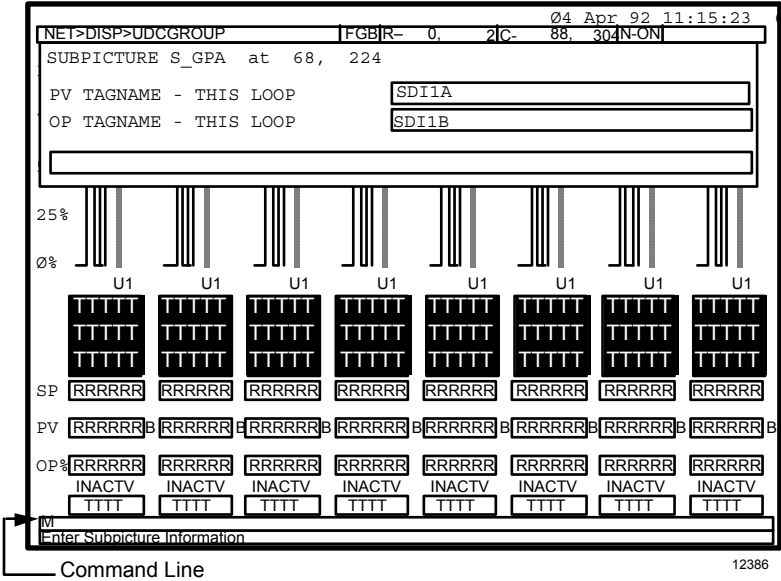
Table 3-8 Configuring the Group Display, Continued

Step	Action
7	Press [SELECT] key to select the loop bar graph selected. Press [ENTER] key to activate the bar graph for loop point tagname modification. The Bar graph starts blinking. <u>Example:</u> Selecting the bar graph for modification. blinking 

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for configuring the group display, continued

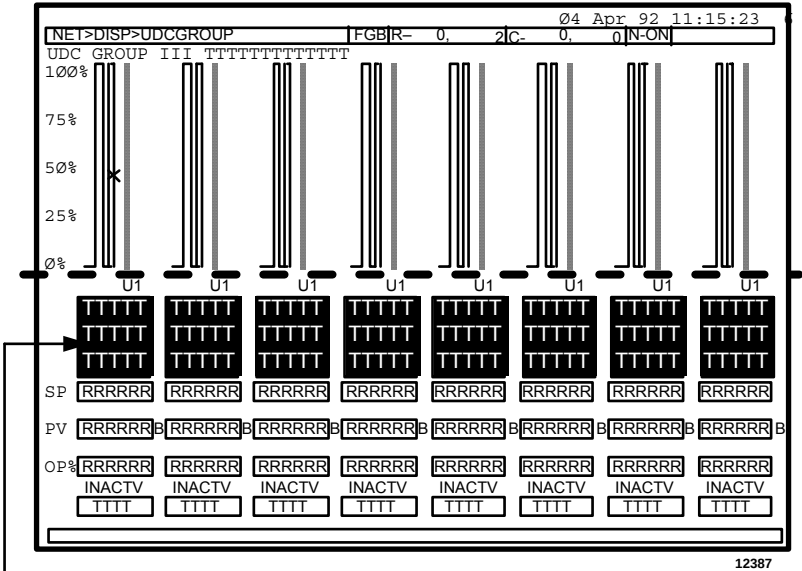
Table 3-8 Configuring the Group Display, Continued

Step	Action
8	Key in [M] in the command line and press [ENTER] key to call up the subpicture information for modification. The subpicture will show default data. <u>Example:</u> Calling up subpicture information.  The subpicture information will be modified (steps 9 thru 23) the same way for each loop selected . Each loop has 2 point tagnames for you to identify. PV TAGNAME - THIS LOOP and OP TAGNAME - THIS LOOP. Enter the same tagnames selected in Table 3-6, steps 9 and 10.
9	Move the cursor to PV TAGNAME - THIS LOOP fill-in block. Key in the tagname for PV Point for this loop. (Example [SDI1A])
10	Move the cursor to OP TAGNAME - THIS LOOP fill-in block. Key in the tagname for the OP Point for this loop. (Example [SDI1B])
11	Press the [ENTER] key, then configure the lower portion of the screen.

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for
configuring the group
display, continued

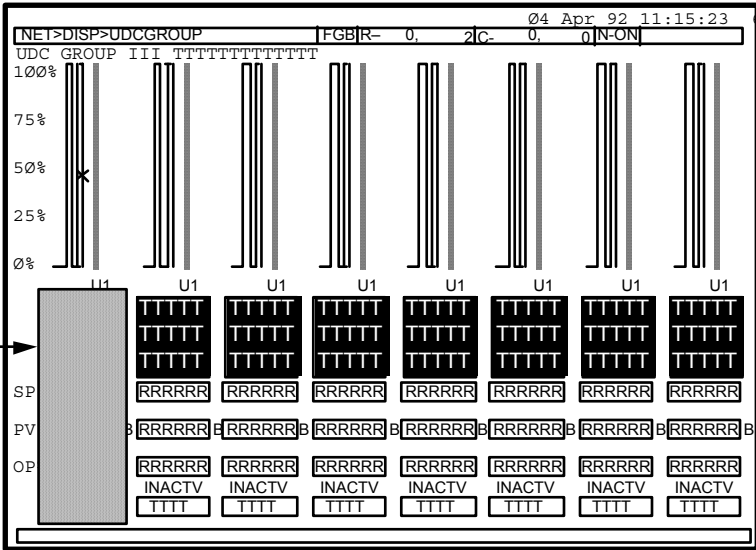
Table 3-8 Configuring the Group Display, Continued

Step	Action
12	Move cursor to the first loop portion of the lower display  Move cursor into this area

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for configuring the group display, continued

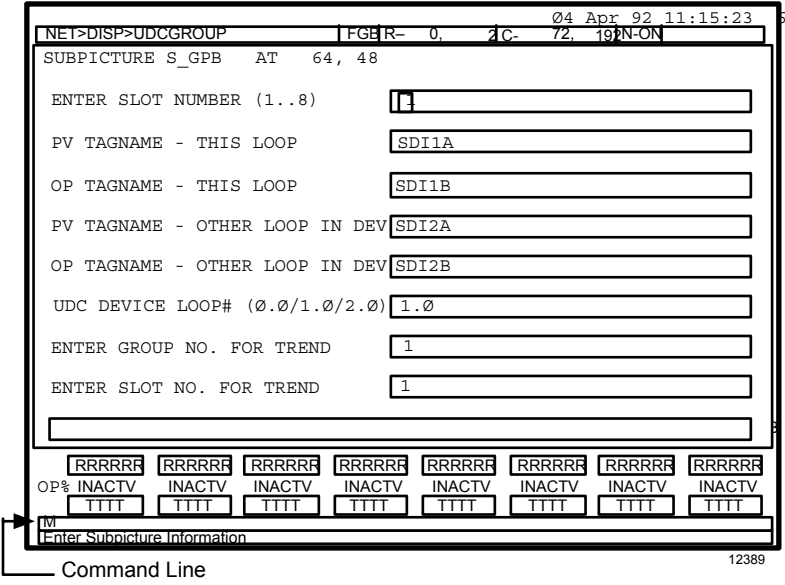
Table 3-8 Configuring the Group Display, Continued

Step	Action
13	Press [SELECT] key to select the loop you have chosen with your cursor. Press [ENTER] key to activate this field for modification. The field starts blinking. <u>Example:</u> Selecting the loop information field for modification.  12388

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for configuring the group display, continued

Table 3-8 Configuring the Group Display, Continued

Step	Action
14	Key in [M] in the command line and press [ENTER] key to call up the subpicture information for modification. The subpicture will show default data. <u>Example:</u> Calling up subpicture information. 
15	In the first fill-in block, key in the SLOT NUMBER (1..8) for the loop being modified.
16	Each loop has 2 point tagnames for you to identify. PV TAGNAME - THIS LOOP and OP TAGNAME - THIS LOOP. Enter the same tagnames selected in Table 3-6, steps 9 and 10. Move the cursor to PV TAGNAME - THIS LOOP fill-in block. Key in the tagname for PV Point for this loop. (Example [SDI1A])
17	Move the cursor to OP TAGNAME - THIS LOOP fill-in block. Key in the tagname for the OP Point for this loop. (Example [SDI1B])

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for configuring the group display, continued

Table 3-8 Configuring the Group Display, Continued

Step	Action								
18	In this step, you identify the device loop for the point tagnames you have entered in steps 16 and 17. Move cursor to UDC DEVICE LOOP# (Ø.Ø/1.Ø/2.Ø) fill-in block. <table border="1"> <thead> <tr> <th>If the points are for...</th><th>Then Key In</th></tr> </thead> <tbody> <tr> <td>the only loop in a one-loop UDC 6000/6300 controller</td><td>Ø.Ø</td></tr> <tr> <td>loop 1 in a two-loop UDC 6000/6300 controller</td><td>1.Ø</td></tr> <tr> <td>loop 2 in a two-loop UDC 6000/6300 controller</td><td>2.Ø</td></tr> </tbody> </table> In this example, key in [1.Ø] since it identifies loop 1 in a two-loop UDC 6000/6300.	If the points are for...	Then Key In	the only loop in a one-loop UDC 6000/6300 controller	Ø.Ø	loop 1 in a two-loop UDC 6000/6300 controller	1.Ø	loop 2 in a two-loop UDC 6000/6300 controller	2.Ø
If the points are for...	Then Key In								
the only loop in a one-loop UDC 6000/6300 controller	Ø.Ø								
loop 1 in a two-loop UDC 6000/6300 controller	1.Ø								
loop 2 in a two-loop UDC 6000/6300 controller	2.Ø								
19	In the next two steps, you will identify the point tagnames for the <i>other loop</i> in a 2-loop controller. These tagnames will be associated with the selections you made in steps 16 and 17. Move the cursor to PT1 TAGNAME - OTHER LOOP IN DEV. Refer to your entry in UDC DEVICE LOOP# (Ø.Ø/1.Ø/2.Ø) fill-in block. <table border="1"> <thead> <tr> <th>If the entry in step 18 was...</th><th>Then...</th></tr> </thead> <tbody> <tr> <td>Ø.Ø <i>(The only Loop in a 1-loop controller)</i></td><td>Key in the same PV TAGNAME you used for loop 1 and entered in step 16. (Example SDI1A)</td></tr> <tr> <td>1.Ø <i>(Loop 1 in a 2-loop controller)</i></td><td>Key in the Tagname for the <u>PV Point</u> for loop 2 of the same controller <i>Loop 1- PV TAGNAME was identified in step 16.</i> (Example SDI2A)</td></tr> <tr> <td>2.Ø <i>(Loop 2 in a 2-loop controller)</i></td><td>Key in the Tagname for the <u>PV Point</u> for loop 1 of the same controller- <i>Loop 2 - PVTAGNAME was identified in step 16.</i> (Example SDI1A)</td></tr> </tbody> </table>	If the entry in step 18 was...	Then...	Ø.Ø <i>(The only Loop in a 1-loop controller)</i>	Key in the same PV TAGNAME you used for loop 1 and entered in step 16. (Example SDI1A)	1.Ø <i>(Loop 1 in a 2-loop controller)</i>	Key in the Tagname for the <u>PV Point</u> for loop 2 of the same controller <i>Loop 1- PV TAGNAME was identified in step 16.</i> (Example SDI2A)	2.Ø <i>(Loop 2 in a 2-loop controller)</i>	Key in the Tagname for the <u>PV Point</u> for loop 1 of the same controller- <i>Loop 2 - PVTAGNAME was identified in step 16.</i> (Example SDI1A)
If the entry in step 18 was...	Then...								
Ø.Ø <i>(The only Loop in a 1-loop controller)</i>	Key in the same PV TAGNAME you used for loop 1 and entered in step 16. (Example SDI1A)								
1.Ø <i>(Loop 1 in a 2-loop controller)</i>	Key in the Tagname for the <u>PV Point</u> for loop 2 of the same controller <i>Loop 1- PV TAGNAME was identified in step 16.</i> (Example SDI2A)								
2.Ø <i>(Loop 2 in a 2-loop controller)</i>	Key in the Tagname for the <u>PV Point</u> for loop 1 of the same controller- <i>Loop 2 - PVTAGNAME was identified in step 16.</i> (Example SDI1A)								

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for
configuring the group
display, continued

Table 3-8 Configuring the Group Display, Continued

Step	Action								
20	Move the cursor to PT2 TAGNAME - OTHER LOOP IN DEV. Refer to your entry in UDC DEVICE LOOP# (Ø.Ø/1.Ø/2.Ø) fill-in block. <table border="1"> <thead> <tr> <th>If the entry in step 18 was...</th><th>Then...</th></tr> </thead> <tbody> <tr> <td> Ø.Ø <i>(The only Loop in a 1-loop controller)</i> </td><td> Key in the same OP TAGNAME you used for loop 1 and entered in step 17. (Example SDI1B) </td></tr> <tr> <td> 1.Ø <i>(Loop 1 in a 2-loop controller)</i> </td><td> Key in the Tagname for the <u>Output Point</u> for loop 2 of the same controller <i>Loop 1- OP TAGNAME was identified in step 17.</i> (Example SDI2B) </td></tr> <tr> <td> 2.Ø <i>(Loop 2 in a 2-loop controller)</i> </td><td> Key in the Tagname for the <u>Output Point</u> for loop 1 of the same controller <i>Loop 2 - OP TAGNAME was identified in step 17.</i> (Example SDI1B) </td></tr> </tbody> </table>	If the entry in step 18 was...	Then...	Ø.Ø <i>(The only Loop in a 1-loop controller)</i>	Key in the same OP TAGNAME you used for loop 1 and entered in step 17. (Example SDI1B)	1.Ø <i>(Loop 1 in a 2-loop controller)</i>	Key in the Tagname for the <u>Output Point</u> for loop 2 of the same controller <i>Loop 1- OP TAGNAME was identified in step 17.</i> (Example SDI2B)	2.Ø <i>(Loop 2 in a 2-loop controller)</i>	Key in the Tagname for the <u>Output Point</u> for loop 1 of the same controller <i>Loop 2 - OP TAGNAME was identified in step 17.</i> (Example SDI1B)
If the entry in step 18 was...	Then...								
Ø.Ø <i>(The only Loop in a 1-loop controller)</i>	Key in the same OP TAGNAME you used for loop 1 and entered in step 17. (Example SDI1B)								
1.Ø <i>(Loop 1 in a 2-loop controller)</i>	Key in the Tagname for the <u>Output Point</u> for loop 2 of the same controller <i>Loop 1- OP TAGNAME was identified in step 17.</i> (Example SDI2B)								
2.Ø <i>(Loop 2 in a 2-loop controller)</i>	Key in the Tagname for the <u>Output Point</u> for loop 1 of the same controller <i>Loop 2 - OP TAGNAME was identified in step 17.</i> (Example SDI1B)								
21	In the next two steps you will enter the Group and Slot number that will be used to invoke a standard TPS Group Trend Display. Move the cursor to ENTER GROUP NO FOR TREND fill-in block. Key in the group number.								
22	Move the cursor to ENTER SLOT NO FOR TREND fill-in block. Key in the slot number of the first of two points.								
23	Press [ENTER] . Go to step 6 and move the cursor to a new Bar graph and repeat steps 7 through 23 as required for the other loops. ATTENTION An alternate method for selecting a Bar graph is to select the top and bottom of the loop display together, both will blink. When you key in [M], then [ENTER], the point tagname subpicture appears first; then when that subpicture information is modified and entered, the loop subpicture will automatically appear for modification.								

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for configuring the group display, continued

Table 3-8 Configuring the Group Display, Continued

Step	Action
24	When the Group Display modifications are complete, delete any Loop from the screen that will not be used. Move the cursor to any upper and lower portion of the loop not being used. Press the [SELECT] key to select the area, then press the [ENTER] key. The line starts blinking. Press the [DELETE] key or key-in DEL. Press [ENTER].
25	Write (save) the modified Schematic. Key in [W NET>DISP>UDCGROUP] in command line. Where: W NET>DISP>GROUP NO1 <pre>graph TD A[W NET>DISP>GROUP NO1] A --- B[W] A --- C[NET] A --- D[DISP] A --- E[GROUP NO1] B --- B1[Write command] C --- C1[Directory pathname for UDC 6000 graphics (For example only - Your directory was selected when the graphics were loaded - see 2.10)] D --- D1[Group Name selected in Step 3] E --- E1[Directory pathname for UDC 6000 graphics (For example only - Your directory was selected when the graphics were loaded - see 2.10)]</pre> 12390
26	Repeat the procedure for each GROUP NAME on the summary display as required.

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for adding a loop to the group display

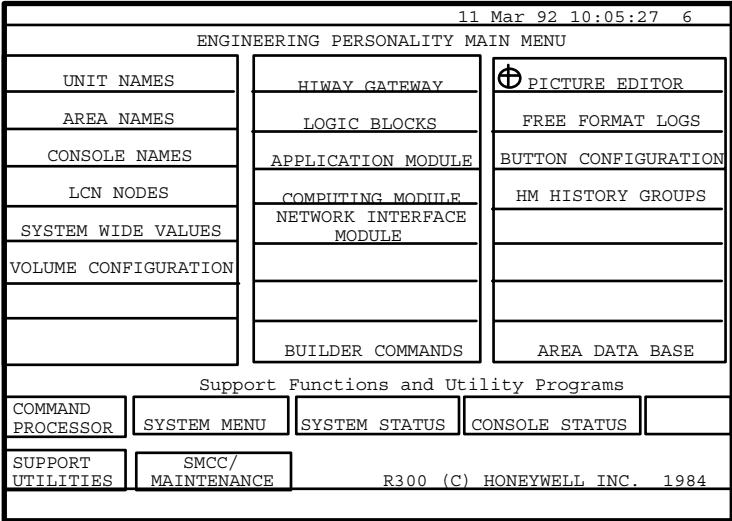
The UDC Group Display is provided, containing the subpicture(s) that lets you enter the specifications of each UDC loop that is part of a group. You can have eight loops on one group display.

If you want to add a loop to a group that contains less than eight loops, use the steps in Table 3-9 to add the loop and modify the subpicture information.

ATTENTION

ATTENTION—NET>DISP is used as the directory pathname in this procedure for example purposes only. Use the one you have selected. (See 2.10.)

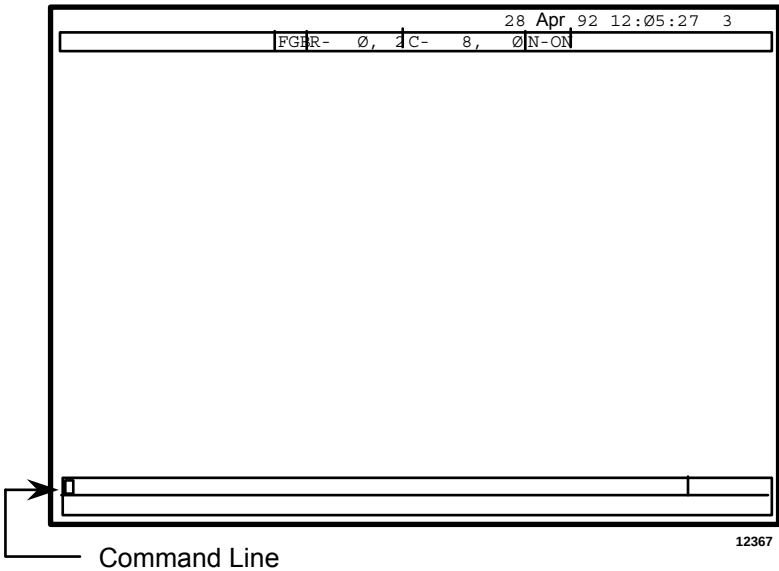
Table 3-9 Adding a Loop to the Group Display

Step	Action
1	Press [CTL] and [HELP] keys to call up Engineering Personality Main Menu. The figure below is an example of what you will see. <u>Example:</u> Calling up Engineering Personality Main Menu. 

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for adding
a loop to the group
display, continued

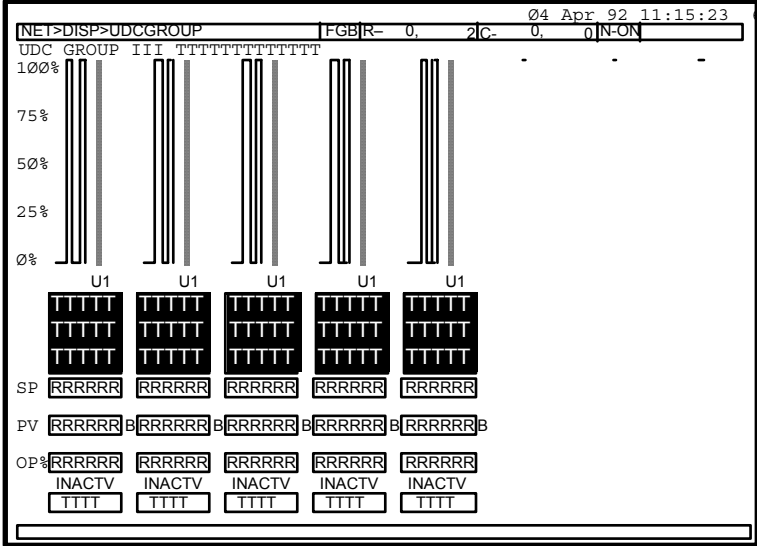
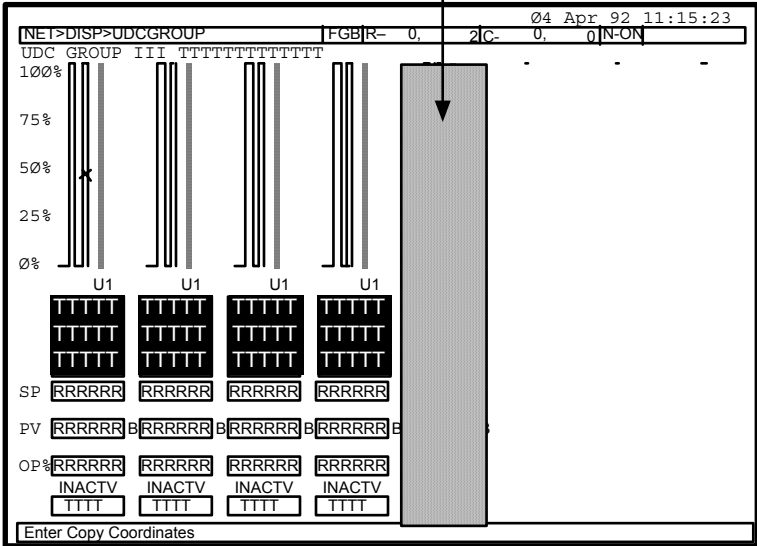
Table 3-9 Adding a Loop to the Group Display, Continued

Step	Action
2	Select PICTURE EDITOR target on the Main Menu to call up the display. The figure below is an example of what you will see. <u>Example:</u> Calling up Picture Editor. 
3	Enter your directory Pathname and Group name. Key in [R NET>DISP>UDCGROUP] in command line. Where: R NET>DISP>UDCGROUP Read command Directory pathname for UDC 6000 graphics (For example only - Your directory was selected when the graphics were loaded - see 2.10) Group Name selected in Step 31 in Table 3-6

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for adding
a loop to the group
display, continued

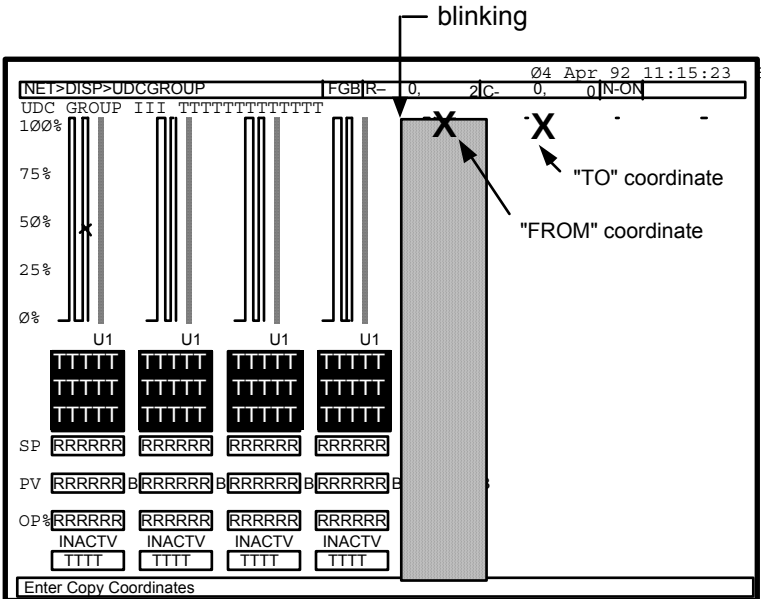
Table 3-9 Adding a Loop to the Group Display, Continued

Step	Action
4	Press [ENTER] key to call up source file for the group selected. <u>Example:</u> Calling up source file for the group selected.  12391
5	Select the top and bottom portion of the last loop on the screen and press [ENTER]. The loop display will start blinking.  12392

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for adding a loop to the group display, continued

Table 3-9 Adding a Loop to the Group Display, Continued

Step	Action
6	Press [C] to copy. Prompt in the command line states, "Enter Copy Coordinates."
7	Select "FROM" coordinate. Touch the top of the blinking display. Select "TO" coordinate. Touch the area to which you want the loop copied. 
8	Press [ENTER] and the loop is copied to the position selected.
9	Modify the subpicture information (subpicture contains information from copied loop) according to the instructions given in table 3-8, steps 5 through 26. ATTENTION make sure you enter a new slot number in the subpicture information, otherwise the slot will be duplicated.

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for Configuring an Operator Change Zone

One UDC custom Change Zone subpicture is provided which can be used for inclusion in your custom schematic. (S_UDCZSL.DS)

A “selection target” subpicture, which interfaces with the UDC Custom Change Zone, is also included. (S_UDCZSL.DS)

The Change Zone provided uses the following Display Data Base variables:

BOOL01	General Purpose
INT01	General Purpose
REAL01	General Purpose
ENT01G	Tag Name for PV Point this Loop
ENT02G	Tag Name for OP Point this Loop
ENT03G	Tag Name for PV Point other Loop
ENT04G	Tag Name for OP Point other Loop
REAL01G	Loop Type - 0.0, 0.1, 0.2

Make sure these variables are not used in the user’s schematic, otherwise, Change Zone operation may affect the functionality of the users schematic.

Use the steps in Table 3-10 to add an Operator Change Zone to a user-built schematic.

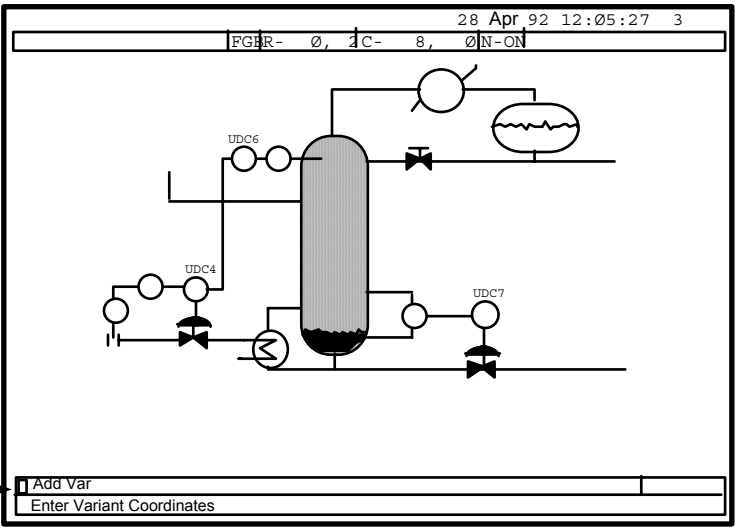
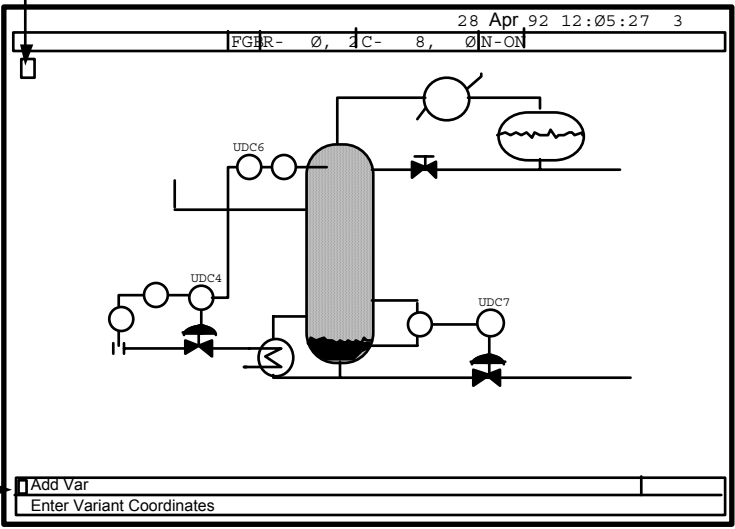
Table 3-10 Adding an Operator Change Zone

Step	Action
1	Make sure your custom Schematic is already in Picture Editor.

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for Configuring an Operator Change Zone, Continued

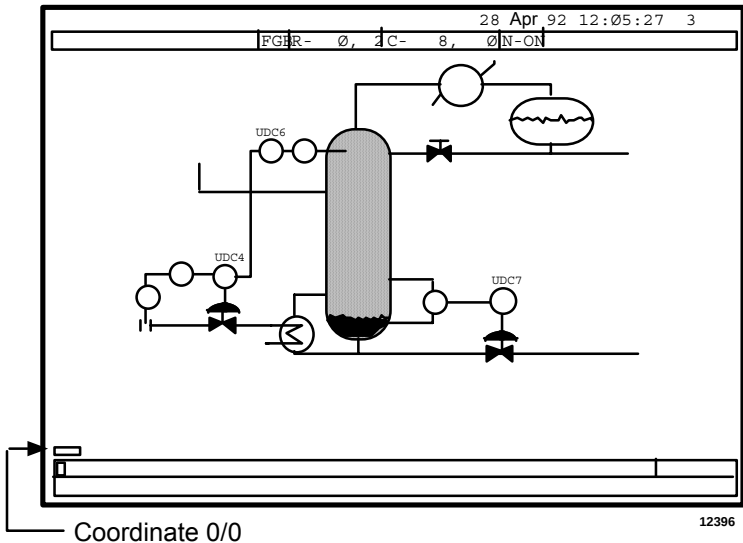
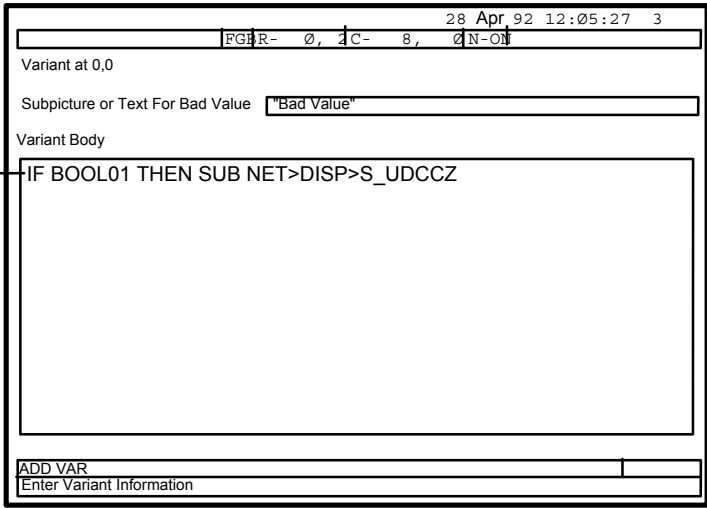
Table 3-10 Adding an Operator Change Zone, Continued

Step	Action
2	The figure below is an example of what you will see. <u>Example:</u> Calling up Picture Editor. In the command line, key-in ADD VAR.  Command Line
3	Press [ENTER] key. The display prompts “Enter Variant coordinates.” The cursor will appear at the top of the screen.  Command Line

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for Configuring an Operator Change Zone, Continued

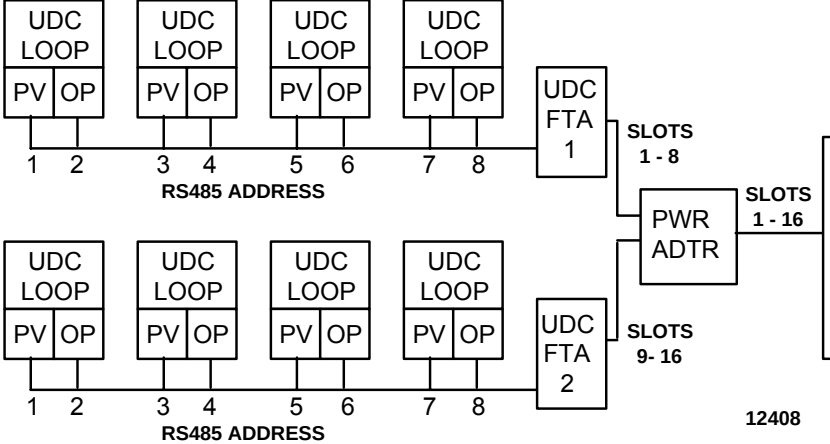
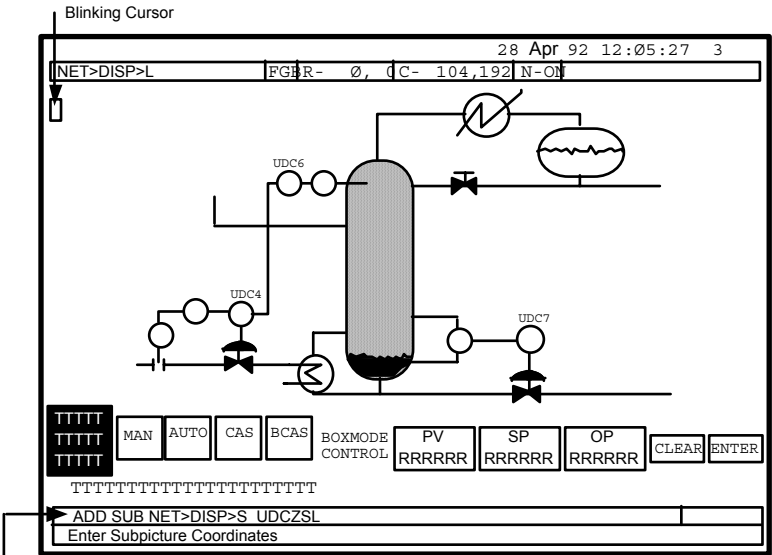
Table 3-10 Adding an Operator Change Zone, Continued

Step	Action
4	Move the cursor to coordinates 0/0 as shown below.  Coordinate 0/0
5	Press the [SELECT] key to select the coordinate, then press [ENTER]. The screen will appear as shown below. Identify the directory pathname and subpicture information. Key-in IF BOOL01 THEN SUB NET>DISP>S_UDCCZ where cursor is blinking.  REMEMBER TO USE YOUR DIRECTORY PATHNAME. NET>DISP is used here as an example.

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for Configuring an Operator Change Zone, Continued

Table 3-10 Adding an Operator Change Zone, Continued

Step	Action
6	Press [ENTER]. You will see a screen like the example below. Key-in ADD SUB NET>DISP>S_UDCZSL in the command line. 
7	Press [ENTER]. A blinking cursor will appear at the top of the screen and a prompt "Enter Subpicture Coordinates" at the bottom of the screen.  Command Line

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for Configuring an Operator Change Zone, Continued

Table 3-10 Adding an Operator Change Zone, Continued

Step	Action
8	Move the cursor to the area in which you want to have a change zone target. See figure below. ATTENTION
9	Press [SELECT], then press [ENTER] key to call up the subpicture information for modification. The subpicture will show default data. <u>Example:</u> Calling up subpicture information. REG SCHEM

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for Configuring an Operator Change Zone, Continued

Table 3-10 Adding an Operator Change Zone, Continued

Step	Action								
10	Each loop has 2 point tagnames for you to identify. PV TAGNAME - THIS LOOP and OP TAGNAME - THIS LOOP. Enter the same tagnames selected in Table 3-6, Steps 9 and 10. Move the cursor to PV TAGNAME - THIS LOOP fill-in block. Key in the tagname for PV Point for this loop. (Example [SDI1A])								
11	Move the cursor to OP TAGNAME - THIS LOOP fill-in block. Key in the tagname for the OP Point for this loop. (Example [SDI1B])								
12	In this step you identify the device loop for the point tagnames you have entered in Steps 10 and 11. Move cursor to UDC DEVICE LOOP# (Ø.Ø/1.Ø/2.Ø) fill-in block. <table border="1"> <thead> <tr> <th>If the points are for...</th><th>Then Key In</th></tr> </thead> <tbody> <tr> <td>the only loop in a 1-loop UDC 6000/6300 controller</td><td>Ø.Ø</td></tr> <tr> <td>loop 1 in a 2-loop UDC 6000/6300 controller</td><td>1.Ø</td></tr> <tr> <td>loop 2 in a 2-loop UDC 6000/6300 controller</td><td>2.Ø</td></tr> </tbody> </table> In this example, key in [1.Ø] since it identifies loop 1 in a 2-loop UDC 6000/6300.	If the points are for...	Then Key In	the only loop in a 1-loop UDC 6000/6300 controller	Ø.Ø	loop 1 in a 2-loop UDC 6000/6300 controller	1.Ø	loop 2 in a 2-loop UDC 6000/6300 controller	2.Ø
If the points are for...	Then Key In								
the only loop in a 1-loop UDC 6000/6300 controller	Ø.Ø								
loop 1 in a 2-loop UDC 6000/6300 controller	1.Ø								
loop 2 in a 2-loop UDC 6000/6300 controller	2.Ø								

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for Configuring an Operator Change Zone, Continued

Table 3-10 Adding an Operator Change Zone, Continued

Step	Action								
13	In the next two steps, you will identify the point tagnames for the <i>other loop</i> in a 2-loop controller. These tagnames will be associated with the selections you made in Steps 10 and 11. Move the cursor to PV TAGNAME - OTHER LOOP IN DEV. Refer to your entry in UDC DEVICE LOOP# (Ø.Ø/1.Ø/2.Ø) fill-in block. <table border="1"> <thead> <tr> <th>If the entry in step 12 was...</th><th>Then...</th></tr> </thead> <tbody> <tr> <td>Ø.Ø (The only Loop in a 1-loop controller)</td><td>Key in the same PV TAGNAME you used for loop 1 and entered in Step 10. (Example SDI1A)</td></tr> <tr> <td>1.Ø (Loop 1 in a 2-loop controller)</td><td>Key in the Tagname for the <u>PV Point</u> for loop 2 of the same controller <i>Loop 1- PV TAGNAME was identified in Step 10</i> (Example SDI2A)</td></tr> <tr> <td>2.Ø (Loop 2 in a 2-loop controller)</td><td>Key in the Tagname for the <u>PV Point</u> for loop 1 of the same controller- <i>Loop 2 - PV TAGNAME was identified in Step 10</i> (Example SDI1A)</td></tr> </tbody> </table>	If the entry in step 12 was...	Then...	Ø.Ø (The only Loop in a 1-loop controller)	Key in the same PV TAGNAME you used for loop 1 and entered in Step 10. (Example SDI1A)	1.Ø (Loop 1 in a 2-loop controller)	Key in the Tagname for the <u>PV Point</u> for loop 2 of the same controller <i>Loop 1- PV TAGNAME was identified in Step 10</i> (Example SDI2A)	2.Ø (Loop 2 in a 2-loop controller)	Key in the Tagname for the <u>PV Point</u> for loop 1 of the same controller- <i>Loop 2 - PV TAGNAME was identified in Step 10</i> (Example SDI1A)
If the entry in step 12 was...	Then...								
Ø.Ø (The only Loop in a 1-loop controller)	Key in the same PV TAGNAME you used for loop 1 and entered in Step 10. (Example SDI1A)								
1.Ø (Loop 1 in a 2-loop controller)	Key in the Tagname for the <u>PV Point</u> for loop 2 of the same controller <i>Loop 1- PV TAGNAME was identified in Step 10</i> (Example SDI2A)								
2.Ø (Loop 2 in a 2-loop controller)	Key in the Tagname for the <u>PV Point</u> for loop 1 of the same controller- <i>Loop 2 - PV TAGNAME was identified in Step 10</i> (Example SDI1A)								

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for Configuring an Operator Change Zone, Continued

Table 3-10 Adding an Operator Change Zone, Continued

Step	Action								
14	Move the cursor to OP TAGNAME - OTHER LOOP IN DEV. Refer to your entry in UDC DEVICE LOOP# (Ø.Ø/1.Ø/2.Ø) fill-in block.								
	<table><tr><th>If the entry in step 12 was...</th><th>Then...</th></tr><tr><td>Ø.Ø <i>(The only Loop in a 1-loop controller)</i></td><td>Key in the same OP TAGNAME you used for loop 1 and entered in Step 11. (Example SDI1B)</td></tr><tr><td>1.Ø <i>(Loop 1 in a 2-loop controller)</i></td><td>Key in the Tagname for the <u>Output Point</u> for loop 2 of the same controller <i>Loop 1- OP TAGNAME was identified in Step 11.</i> (Example SDI2B)</td></tr><tr><td>2.Ø <i>(Loop 2 in a 2-loop controller)</i></td><td>Key in the Tagname for the <u>Output Point</u> for loop 1 of the same controller <i>Loop 2 - OP TAGNAME was identified in Step 11.</i> (Example SDI1B)</td></tr></table>	If the entry in step 12 was...	Then...	Ø.Ø <i>(The only Loop in a 1-loop controller)</i>	Key in the same OP TAGNAME you used for loop 1 and entered in Step 11. (Example SDI1B)	1.Ø <i>(Loop 1 in a 2-loop controller)</i>	Key in the Tagname for the <u>Output Point</u> for loop 2 of the same controller <i>Loop 1- OP TAGNAME was identified in Step 11.</i> (Example SDI2B)	2.Ø <i>(Loop 2 in a 2-loop controller)</i>	Key in the Tagname for the <u>Output Point</u> for loop 1 of the same controller <i>Loop 2 - OP TAGNAME was identified in Step 11.</i> (Example SDI1B)
	If the entry in step 12 was...	Then...							
	Ø.Ø <i>(The only Loop in a 1-loop controller)</i>	Key in the same OP TAGNAME you used for loop 1 and entered in Step 11. (Example SDI1B)							
1.Ø <i>(Loop 1 in a 2-loop controller)</i>	Key in the Tagname for the <u>Output Point</u> for loop 2 of the same controller <i>Loop 1- OP TAGNAME was identified in Step 11.</i> (Example SDI2B)								
2.Ø <i>(Loop 2 in a 2-loop controller)</i>	Key in the Tagname for the <u>Output Point</u> for loop 1 of the same controller <i>Loop 2 - OP TAGNAME was identified in Step 11.</i> (Example SDI1B)								

3.7 Configuring UDC 6000/6300 Schematics, Continued

Procedure for Configuring an Operator Change Zone, Continued

Table 3-10 Adding an Operator Change Zone, Continued

Step	Action
15	Press [ENTER]. The target area is selected and is outlined on the screen. ATTENTION The selection target area is 9 X 5 and can be modified if required.
16	You may select more than one target area on your schematic. Repeat Steps 6 through 15 to establish each new target area.
	This ends the procedure for adding Change Zone(s) to your custom graphics. Further revision of the custom graphic depends on the customer requirements.

Button configuration

Refer to *Button Configuration Form Instructions* and *Button Configuration Data Entry* to configure a button on the US keyboard to call up the UDC 6000/6300 Summary display with a dedicated keystroke.

3.8 Preparing SDI for Operation

Background

Before you put the configured SDI points into operation, you must load the points and check the operating status of the SDI IOP.

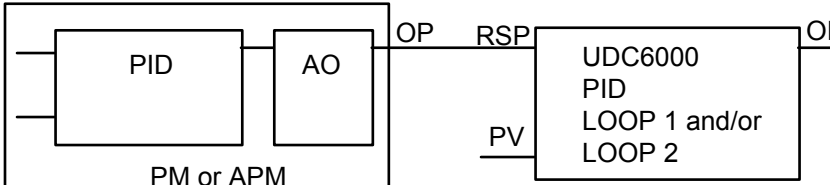
Prerequisite

- You have configured SDI points as STI points in PED.
- You have completed all the other setup procedures in this section.
- You have installed SDI IOP and SDI/UDC 6000/6300 FTA in PM/APM, and PM/APM is running.

Procedure

Perform the steps in Table 3-11 to prepare the SDI IOP for operation.

Table 3-11 Preparing SDI for Operation

Step	Action
1	With Engineering Personality Main Menu on display, select SYSTEM STATUS target to call up System Status display. <u>Example:</u> Calling up System Status display.  12411

3.8 Preparing SDI for Operation, Continued

Procedure, continued

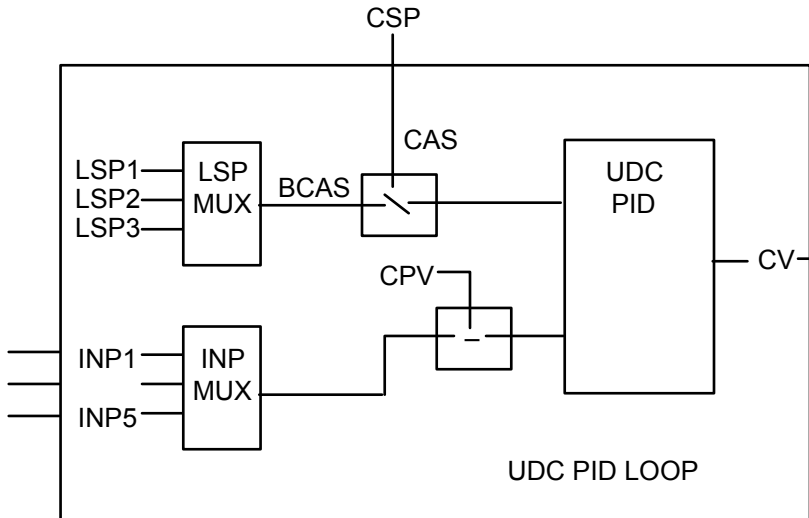
Table 3-11 Preparing SDI for Operation, Continued

Step	Action
2	Under PROCESS NETWORK STATUS, select target for appropriate network number to call up appropriate UCN Status Display. <u>Example:</u> Calling up selected UCN Status Display. ATTENTION
3	Select appropriate PM/APM target and DETAIL STATUS to call up PM/APM Status Display. <u>Example:</u> Calling up selected PM/APM Status Display. <pre>graph LR SP1[SP] --- L2[UDC6000 PID LOOP 2] PV1[PV] --- L2 L2 -- OP --- L1[UDC6000 PID LOOP 1] RSP[RSP] --- L1 PV2[PV] --- L1 L1 -- OP --- OP2[OP]</pre> 12412

3.8 Preparing SDI for Operation, Continued

Procedure, continued

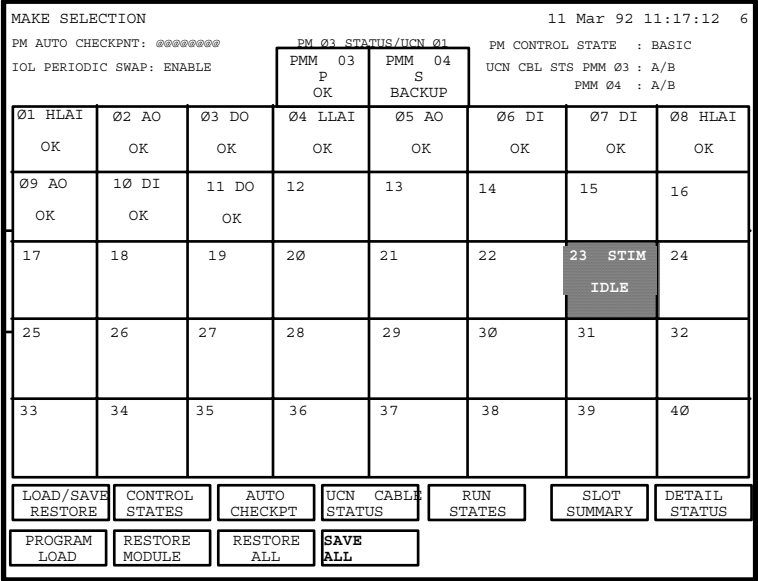
Table 3-11 Preparing SDI for Operation, Continued

Step	Action
4	Select 23 STIM IDLE target for SDI IOP assigned to module 23.
5	Select RUN STATES target to access run selections.
6	With 23 STIM IDLE target still in IDLE state, select LOAD/SAVE RESTORE target.
7	Select VALIDATE IOP DB target and press [ENTER] to validate SDI database. <u>Example:</u> Validating SDI database.  12413

3.8 Preparing SDI for Operation, Continued

Procedure, continued

Table 3-11 Preparing SDI for Operation, Continued

Step	Action						
8	Select SAVE ALL target and press [ENTER] to save SDI database to checkpoint file. <u>Example:</u> Saving SDI database.  12407						
9	With 23 STIM IDLE target still in IDLE state, select RUN STATES target.						
10	Select STARTUP target and press [ENTER] to start the SDI IOP.						
11	Check that module state changes from IDLE to OK state. <table border="1"> <thead> <tr> <th>If...</th><th>Then...</th></tr> </thead> <tbody> <tr> <td>state changes to OK</td><td>continue with operation</td></tr> <tr> <td>error message appears</td><td>recheck your work and refer to <i>Process Manager Service</i> binder for troubleshooting data</td></tr> </tbody> </table>	If...	Then...	state changes to OK	continue with operation	error message appears	recheck your work and refer to <i>Process Manager Service</i> binder for troubleshooting data
If...	Then...						
state changes to OK	continue with operation						
error message appears	recheck your work and refer to <i>Process Manager Service</i> binder for troubleshooting data						

Section 4 – PM/APM - UDC 6000/6300 Cascade Control Integration

4.1 PM/APM - UDC 6000/6300 Cascade Interface

Overview

The PM/APM - UDC 6000/6300 integration provides for cascaded control architecture between the Process Manager (PM) or the Advanced Process Manager (APM) and the UDC 6000/6300 Controller. The cascade interface is designed to provide the following:

- DDC Cascade support for non redundant PM, APM, and AM control loops.
- Split Cascade to allow fast secondary loops for flow control.
- Manual/Auto station support to allow field operators to get local manual control.
- Analog setpoint cascade to allow cascaded interface to existing analog setpoints.

This section provides the detailed information required for cascaded interfaces between the PM, APM, AM and the UDC 6000/6300 Controller.

4.2 UDC 6000/6300 SDI Point Connections

Introduction

The following information reviews the basics of the PM/APM - UDC 6000/6300 interface. A general understanding of the basic interface is required before attempting to configure cascade control.

AM/APM - UDC 6000/6300 interface connections

The PM/APM - UDC 6000/6300 interface provides connection between the PM (Process Manager) or APM (Advanced Process Manager) and the UDC 6000/6300 Process Control Loop through two SDI (Serial Device Interface), Smart Transmitter Interface Module (STIM), Analog Input (AI) points.

The parameters of the SDI STIM AI points maintain read and write parameter sets available for general I/O connection in the PM/APM.

SDI STIM AI points

Two SDI STIM AI points are used for interfacing each UDC 6000/6300 loop to the PM/APM.

The first point maintains the UDC 6000/6300 loops Process Variable (PV) as the AI PV value and is referred to as the PV Point.

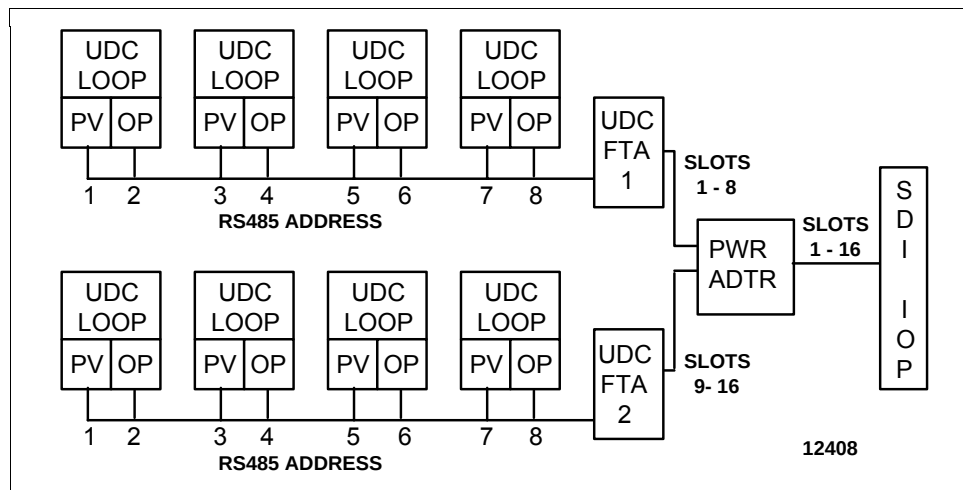
The second point maintains the UDC 6000/6300 loops Output as the AI PV value and is referred to as the OP point.

PV and OP point configuration

The PV point and OP point assignments are configured as RS485 addresses through the front panel of the UDC 6000/6300 during set up of RS485 TDC communications.

Figure 4-1 shows a diagram, which illustrates the mapping of UDC 6000/6300 RS485 addresses to PM/APM STIM slots through the SDI IOP.

Figure 4-1 UDC 6000/6300 to PM/APM Point Interface (Example)



4.2 UDC 6000/6300 SDI Point Connections, Continued

PV and OP point configuration, continued

The destination AI point slot in the PM/APM is related to the RS485 addresses as summarized in Table 4-1.

Table 4-1 UDC 6000/6300 RS485 Address to PM/APM IOP Point Mapping

FTA#	RS485 Address	PM/APM IOP Slot
1	1	1
1	2	2
1	3	3
1	4	4
1	5	5
1	6	6
1	7	7
1	8	8
2	1	9
2	2	10
2	3	11
2	4	12
2	5	13
2	6	14
2	7	15
2	8	16

What happens when communication is established

When communication is established between the FTA and a UDC loop, all point connection information (PV or OP point type, UDC name, BOXMODE, etc) is transferred from the UDC 6000/6300 to the FTA.

All point connection information is stored in the nonvolatile EARAM of the UDC 6000/6300 controller. The point connection information can be “transplanted” from one UDC 6000/6300 controller to another by transferring the EARAM.

4.3 UDC 6000/6300 - PM/APM Interface Verification

Introduction	Some SDI point parameters are reserved to aid in the verification of UDC to SDI IOP connections.
Device tag name	The UDC 6000/6300 device tag name, that was configured at the UDC 6000/6300 keyboard and visible on the UDC 6000/6300 interface, is transferred to the STITAG parameter of all SDI points, which have their device databases, connected to that UDC. The device tag name is configured at the UDC 6000/6300 only.
Loop number and point slot type	In addition to the device tag names, the UDC loop number and point slot type are transferred to the SDI STISWVER parameter. Both the STITAG and STISWVER parameters are viewable on the configuration display of SDI AI points.
Example	If an SDI point connection is made to a UDC 6000/6300 whose device TAGNAME is “FLOW 001” at an address configured (at the UDC) for transfer of PV information for loop 1, the SDI STITAG will show “FLOW 001” and the STISWVER will show “LOOP 1PV.”
ATTENTION	ATTENTION—The UDC 6000/6300 provides for a nine character name. Only the first eight characters are transferred to the STITAG with the ninth character being ignored.

4.4 UDC SDI Interface Parameters

Control loop interface There are four basic SDI parameters used to provide control loop interface through SDI AI points. Table 4-2 shows these parameters based on the point type, Output (OP), or Process Variable (PV) configured.

Table 4-2 Basic SDI Parameters

Point Type	Parameter Name	Usage
PV	PVRAW	Current PV
	PVTV	Operator Setpoint
	SECVAR	Current Setpoint
	CJTACT	SP Lock
OP	PVRAW	Current Output
	PVTV	Operator Output
	SECVAR	PV (RFB)
	CJTACT	Control Enable (OP Lock)

Cascade and automatic interface

The SDI interface also maintains three SDI parameters specifically for cascade and automatic interface. These parameters are based on the point type configured. The parameters and their definitions are listed in Table 4-3.

Table 4-3 SDI Parameters Used for Cascade

Point Type	Parameter Name	Usage	Definition	Owner
PV	C1*	CSP	Computer Setpoint - Refers to a setpoint sent from the TDC to override the UDC loop's setpoint.	TDC
	C2	CPV	Computer Process Variable - Refers to a process variable sent from the TDC to override the UDC loops PV.	TDC
	PIUOTDCF	MONITOR FLAG	Monitor Flag - ON indicates that the UDC is not controlled by the TDC. Access is allowed at the UDC.	TDC
OP	C1*	COP	Computer Output - Refers to an output sent from the TDC to override the UDC 6000/6300 loop's output. Normally COP is used in DDC type cascade control.	TDC
	C2	C MODE	Computer Mode - Mode of primary during cascade.	
	PIUOTDCF	OP TRACK FLAG	Cascade Initialization Flag - ON indicates to primary that cascade is not active and that the cascade primary should back-initialize its output.	TDC

* Overrides to the C1 parameter are monitored by a 4 second watch dog at the SDI FTA. If the FTA does not receive a write to the C1 parameter at least once in 4 seconds, the watchdog will timeout and the override will be cancelled.

4.5 UDC 6000/6300 Cascade Connections to PIDERFB Regulatory Points

Introduction

The PIDERFB (PID with External Reset Feedback) is the only PM, APM, or AM regulatory control algorithm, which provides general I/O connections, allowing for cascade from the PM/APM to the UDC 6000/6300.

The PIDERFB algorithm allows output connection to general I/O point parameters like those of the SDI points, and provides input connections for preventing reset windup.

PIDERFB input and output connections

The PIDERFB provides three input connections by way of “Tag.param” reference. These connections are configured during point build of the PIDERFB regulatory point.

The PIDERFB algorithm also provides a control output available for general connection by way of “Tag.param” specification.

Table 4-4 shows the PIDERFB input and output connections and their functions.

Table 4-4 PIDERFB Input and Output Connections

Connection	Function
PIDERFB Input Connections	PVAUTO (Automatic Process Variable) - typically connected to the PVCALC of an AI point. The AI point need not be that from a UDC 6000/6300.
	RFB (Reset Feedback) - typically connected to the secondary's PV. In the case of TDC to UDC cascade, this parameter is normally connected to PVRAW (or PVCALC) of the PV point (or SECVAR of the OP Point)of the UDC loop, which is acting as the secondary.
	TRFB (Tracking Value) - typically connected to the parameter, which the output of the PIDERFB should track when cascade is inactive. In the case of the TDC to UDC cascade, TRFB is normally connected to the PVRAW parameter of UDC loop's OP point whose C1 parameter is connected to the PIDERFB output.
PIDERFB Output Connection	OUTPUT - In the case of UDC SDI AI points, the output is connected to the C1 parameter of the UDC loop's PV or OP point.

4.5 UDC 6000/6300 Cascade Connections to PIDERFB Regulatory Points, Continued

PIDERFB related parameters

Other parameters of the PIDERFB algorithm that are important to understand while developing cascade interfaces with the UDC 6000/6300 include S1, MODE, the Tuning constants K, T1, and T2 and the Reset Limit Gain K1.

Table 4-5 shows the PIDERFB related parameters and their functions.

Table 4-5 PIDERFB Related Parameters

Connection	Function
PIDERFB Related Parameters	S1 - Switch input which, when set to ON, enables the output to track the TRFB connection. In the PM and APM, this signal requires connection by use of logic or CL. In the AM, the S1 can be configured as a general input connection.
	MODE - Provides an enumeration of the current control mode of the PIDERFB as defined in the parameter reference dictionary.
	K - Loop PID Gain constant.
	T1 - Loop PID Integral constant.
	T2 - Loop PID Derivative constant.
	K1 - Reset Gain Constant. When K1 is used, the accumulated integral value is limited to $K1 * (SP-PV)$. This prevents reset windup in SPC cascade configuration while the secondary is in manual.

4.6 PM/APM - UDC 6000/6300 Cascade Options Overview

Introduction

The PM/APM - UDC 6000/6300 SDI Interface provides interface parameters, which allow split and backup cascade control architectures. The available cascade options are:

- SPCRSP - Supervisory using remote (UDC) setpoint
- SPCCSP - Supervisory using computer (TDC) setpoint
- DDCLSP - Direct control with local (UDC) setpoint
- DDCRSP - Direct control with remote (UDC) setpoint
- DDCCSP - Direct control with computer (TDC) setpoint

Usage of terms

In this discussion of PM/APM - UDC 6000/6300 Cascade, certain terms are used. Table 4-6 lists these terms and their definitions.

Table 4-6 Cascade Terms and Definitions

Term	Definition	Usage
SPC	Supervisory Control	In SPC, the setpoint of the secondary is controlled by a higher level (Primary) controller.
DDC	Direct Digital Control	In DDC, the output of the secondary is controlled by the output of the higher level (Primary) controller.
RSP	UDC 6000/6300 Remote Setpoint	The Remote Setpoint (RSP) may be connected to a UDC 6000/6300 analog input (Input 2, 3, 4, or 5), an Input algorithm (IN ALG 1 or IN ALG 2), or to the second loop of a 2-loop controller. The connection is dependent on the configuration of the UDC 6000/6300. While the remote setpoint is active, "RSP" is displayed on the UDC operator interface. It is also indicated as the SPSRC(setpoint source), which is shown on the FIRST page and OP DATA page of the Custom UDC 6000/6300 Regulatory Schematic Display. RSP is enabled only during CAS and BCAS modes of operation.
LSP	Current UDC 6000/6300 Local Setpoint	The UDC can have up to three local setpoints. The number of setpoints that are active depends on the configuration of the UDC 6000/6300. The current setpoint is selectable from the second (OP DATA) page of the Custom UD C 6000/6300 Regulatory Schematic Display. When the local setpoint is the active setpoint, SP, SP2, or SP3 will be displayed on the operator interface of the UDC 6000/6300. Also, LSP is indicated as the SPSRC (setpoint source), which is shown on the First page and OPDATA page of the Custom UDC 6000/6300 Regulatory Schematic Display.

4.6 PM/APM - UDC 6000/6300 Cascade Options Overview,

Continued

Usage of terms,
continued

Table 4-6 Cascade Terms and Definitions, continued

Term	Definition	Usage
CSP	Setpoint sent from the PM/APM through communications	The CSP may either be a cascaded setpoint from a PM, APM, or AM automatic control algorithm or an operator entered value. While the CSP is selected as the current setpoint, "CSP" will be displayed on the operator interface of the UDC 6000/6300. Also, CSP is indicated as the SPSRC (setpoint source), which is shown on the First page and OPDATA page of the Custom UDC 6000/6300 Regulatory Schematic Display. The automatic CSP is interfaced to the SDI IOP through the PV Points STIM C1 parameter.
CPV	Process Variable override value sent to the UDC Communications	The CPV may be used with any of the cascade options The CPV is interfaced to the PM/APM through the PV points STIM C2 parameter.
COP	Output override value sent to the UDC through Communications	The COP provides override of the UDC output for DDC applications. The COP is interfaced to the PM/APM through the OP Points STIM C1 parameter.

4.7 PM/APM - UDC 6000/6300 Cascade Options Configuration

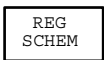
Introduction

Cascade options can be configured by setting the BOXMODE parameter to FULL and making the desired CASOPT selection.

Procedure

Follow the procedure listed in Table 4-7 to configure the desired Cascade option.

Table 4-7 Configuring the Cascade Option

Step	Action
1	Select  on the UDC 6000/6300 Custom Detail summary or group display.
2	To enable the UDC to accept TDC Cascade, the UDC BOXMODE parameter must be set to full. Select the target [BOXMODE] on the first page of the display and change the selection to "FULL." This will allow cascaded writes from the PM/APM to the UDC 6000/6300.
3	Go to the OP DATA page of the Regulatory Schematic and select the target [CASOPT]. Select the desired cascade option: • SPCRSP - Supervisory with Remote (UDC) Setpoint • SPCCSP - Supervisory with Computer (TDC) Setpoint • DDCLSP - Direct Control with Local (UDC) Setpoint • DDCRSP - Direct Control with Remote (UDC) Setpoint • DDCCSP - Direct Control with Computer (TDC) Setpoint
4	Refer to the sections that follow for all the information required to use each specific option.

ATTENTION

- Changing the BOXMODE parameter to BASIC while in cascade will cause the UDC 6000/6300 (both loops of a two loop controller) to Shed with the control loop transferred to the UDC.
- Configuration changes are not permitted at the front panel of the UDC 6000/6300 if the UDC's BOXMODE is set to FULL and any point connection to that UDC 6000/6300 is active.

4.8 SPCRSP Cascade (Supervisory using remote UDC setpoint)

Introduction

SPCRSP cascade refers to a control loop resident in a UDC 6000/6300 that is cascaded from its remote setpoint (RSP) in supervisory type architecture.

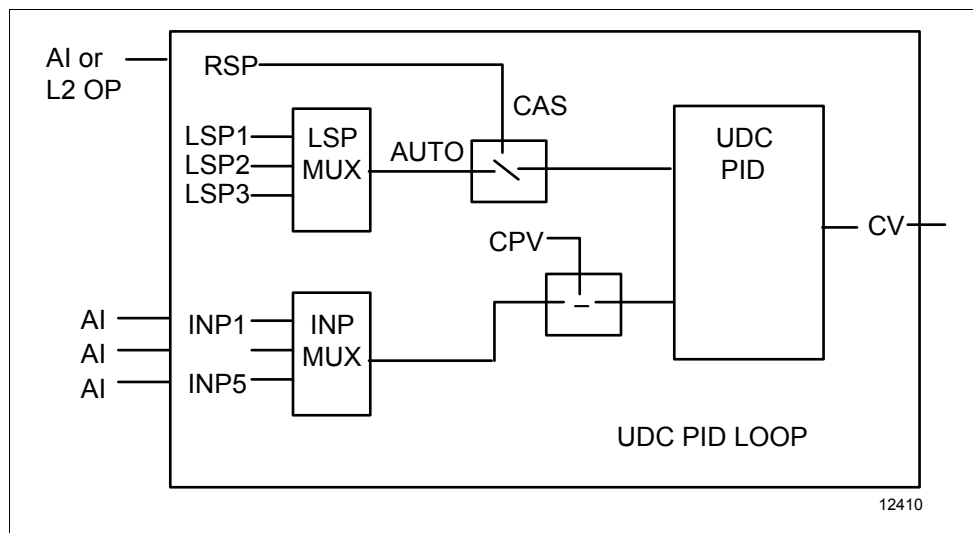
This mode allows control and monitor of cascaded UDC 6000/6300 secondary loops where the primary is either a second loop in the UDC 6000/6300 or by way of an analog connection at the UDC 6000/6300 Analog Input terminals.

Select SPCRSP under the CASOPT target on the OPDATA page of the UDC 6000/6300 Regulatory Schematic display.

Description

A simplified block diagram of a UDC loop configured for SPCRSP cascade is shown in Figure 4-2. While in SPCRSP, the front panel of the UDC 6000/6300 will display CAS and RSP.

Figure 4-2 UDC 6000/6300 Loop in SPCRSP Configuration



When CAS mode is enabled, RSP is selected as the loops setpoint and the loop is placed into AUTO when CAS mode is enabled. RSP is deselected (LSP is selected) when the AUTO mode is selected. BCAS is not implemented in SPCRSP and will default to AUTO when selected.

4.8 SPCRSP Cascade (Supervisory using remote UDC setpoint),

Continued

RSP limit checks

A remote setpoint, when selected from an analog input, is the characterized (Engineering Unit) value of the analog signal present on the analog input of the UDC. The characterization of the analog signal is limited to the range of the analog input calibrated range.

Limited checking using the setpoint limits of the PID loop is done prior to transferring the characterized input value to the loop's remote setpoint.

When using two UDC 6000/6300 cascaded loops, the output of loop two, the primary loop, is in percent and is limited to the output limits of the primary loop.

The output percent value of loop two is scaled by the loop PV range of loop one into engineering units.

Setpoint range limit checking is done prior to transferring loop two's scaled output value to the loop one's remote setpoint.

SPCRSP control mode definitions

Table 4-8 lists the control mode definitions for SPCRSP Cascade.

Table 4-8 Control Mode Definitions for SPCRSP Cascade (Boxmode=Full)

Mode	Definition
MAN	In SPCRSP MANUAL, the operator may adjust the UDC's output from the Universal Station using the custom schematic or the PVTV (setpoint) of the Output (OP) point.
AUTO	In the SPVRSP AUTO mode, a UDC local setpoint is used as the UDC loop's setpoint. The local setpoint can be modified by engineering access on the OPDATA page of the UDC regulatory schematic. If the second loop of a two loop UDC is being used as the primary, PV tracking can be configured to prevent integral windup on the second loop while CAS is active.
CAS	If the second loop of a two loop UDC is used as the primary, CAS mode will enable the first loop to receive the output of the second loop as its Remote Setpoint value. While in CAS/SPCRSP, CAS will be displayed on the operator interface of the UDC 6000/6300.
BCAS	BCAS is not valid in SPCRSP and defaults to AUTO.

4.8 SPCRSP Cascade (Supervisory using remote UDC setpoint),

Continued

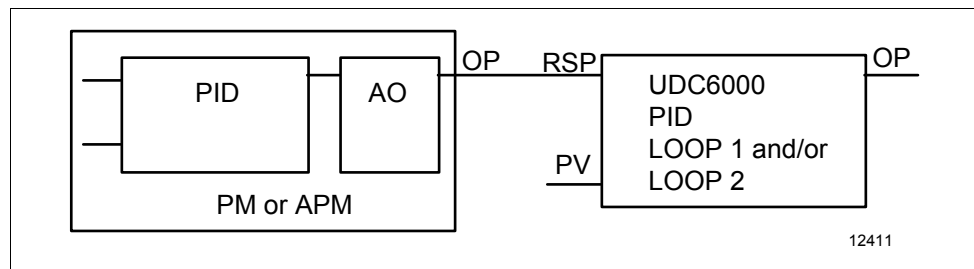
SPCRSP PIDERFB point connections

When using SPCRSP, all cascade connections are made at the UDC 6000/6300. No SDI point connections are required in the PM/APM.

SPCRSP cascade with PM/APM primary

Figure 4-3 shows an example of a UDC loop with SPCRSP cascade to an analog input. This analog input can be transmitted from a PM/APM analog output, an analog M/A station, or another UDC 6000/6300.

Figure 4-3 UDC 6000/6300 SPCRSP Cascade with PM/APM Primary Block Diagram



UDC configuration with PM/APM primary

In this example, the remote setpoint is to be sent to the UDC 6000/6300 at an analog input on the UDC rear terminal panel.

Configuration of the UDC remote setpoint source can be done at the operator interface of the UDC 6000/6300 or from an operator station using the configuration page of the UDC 6000/6300 regulatory schematic.

ATTENTION

Configuration changes are not permitted at the front panel of the UDC 6000/6300 if the UDC's BOXMODE parameter is FULL and the OP point connection to that UDC loop is active with Control (OP CJTACT) enabled.

Configuration of the UDC 6000/6300 for an analog remote setpoint is shown in Table 4-9. Refer to the UDC 6000 or UDC 6300 product manual for details.

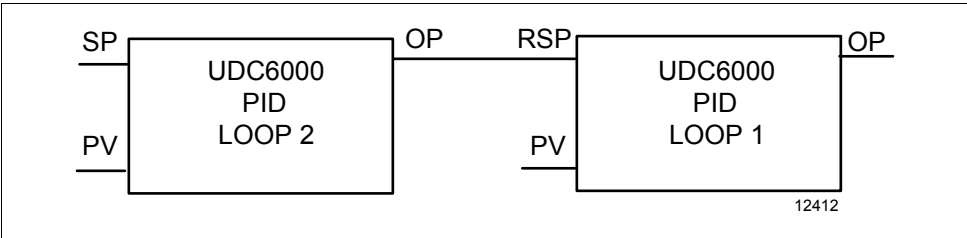
Table 4-9 Configure the UDC with PM/APM Primary

Step	Action
1	Select the UDC 6000/6300 Remote setpoint source. Go to Set Up prompt "CONTROL," Function prompt "RSP SRC" and make a selection (Input 1, Input 2, INALG1, etc.)
2	Go to Set Up Group "ALGORITHM," Function prompt "PID LOOPS" and select 1LOOP or 2LOOPS.

4.8 SPCRSP Cascade (Supervisory using remote UDC setpoint), Continued

SPCRSP cascade with UDC 6000/6300 primary Figure 4-4 shows an example of SPCRSP cascade between two loops within a single UDC 6000/6300 Controller.

Figure 4-4 UDC 6000/6300 SPCRSP Cascade with UDC Primary Block Diagram



UDC configuration with UDC primary The UDC 6000/6300 allows two loops within a controller to be cascaded. In this mode, LOOP1 is always the secondary and LOOP2 is always the primary. If you want the UDC 6000/6300 to be cascaded to its second internal loop, configure the UDC 6000/6300 as shown in Table 4-10. Refer to the UDC 6000 or UDC 6300 product manual for details.

Table 4-10 Configure the UDC with UDC Primary

Step	Action
1	Go to Set Up Group "ALGORITHM," Function prompt "PID LOOPS" and select "CASCADE."

4.9 SPCCSP Cascade (Supervisory using computer [TDC] setpoint)

Introduction

SPCCSP cascade refers to a control loop in the UDC 6000/6300 having its setpoint generated from the output of an algorithm in the PM, APM, or AM.

While SPCCSP cascade is active, the UDC 6000/6300 operator interface will display CAS and CSP.

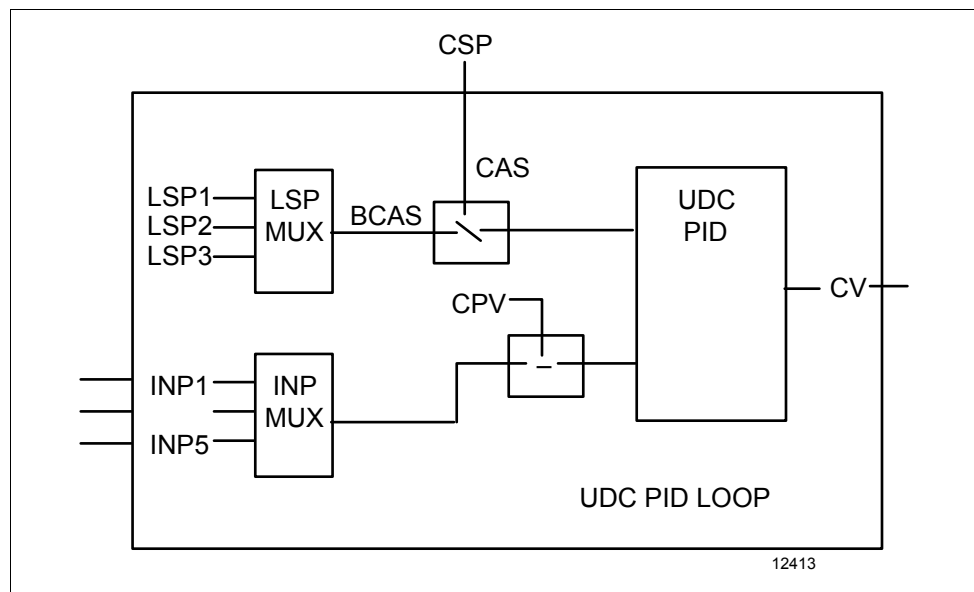
Select SPCCSP by way of the CASOPT target on the OPDATA page of the UDC 6000/6300 Regulatory Schematic display.

Description

Figure 4-5 is a simplified block diagram of a UDC loop configured for SPCCSP cascade.

The primary for the cascade may be a PIDERFB (PID with external reset feedback or other algorithms with general output connections) located in the PM, APM, or AM.

Figure 4-5 UDC 6000/6300 Loop in SPCCSP Configuration



Setpoint limit checks

The SPCCSP setpoint is sent to the UDC 6000/6300 from the PM/APM by the C1 parameter of the PV point.

Limit checking for the setpoint is performed using the PVEUHI and PVEULO STIM parameters. The UDC 6000/6300 also performs limit checking of the setpoint value using the setpoint limits configured in the UDC 6000/6300.

4.9 SPCCSP Cascade (Supervisory using computer [TDC] setpoint),

Continued

Limiting integral saturation

Integral windup can be limited by setting the PIDERFB's K1 parameter. When K1 is set, it will scale the maximum integral value to K1 times the error (Primary's SP-PV).

Output limits

When using a PIDERFB as a SPC primary algorithm in the PM, APM, or AM, the output EU limits of the primary should be made to be within the setpoint limits of the UDC 6000/6300 setpoint. This will prevent the output from exceeding the UDC's setpoint limits and avoid exceeding the UDC's SP range limits and allow the PIDERFB to perform normal output limiting.

SPCCSP control mode definitions

Table 4-11 lists the control mode definitions for SPCCSP Cascade.

Table 4-11 Control Mode Definitions for SPCCSP Cascade

Mode	Definition
MAN	In manual, the operator may adjust the UDC's output. During Manual, the UDC PID (secondary loop) is reset with the output initialized to the manual value. AM/APM/PM PIDERFB primary loops should be configured to limit integral windup while the secondary is in manual.
AUTO	In the AUTO mode, the UDC loop's setpoint is the last setpoint received from the PM. If there is no valid PM setpoint, the UDC will use its current local setpoint. When using a PM/APM setpoint, the UDC will display CSP as the setpoint source. An operator can modify the CSP from the first page of the UDC Regulatory Schematic, from a UDC Group schematic or from the SP(PVTV) of a UDC SDI PV point. PM/APM/AM PIDERFB primary loops should be configured to limit integral windup while a secondary is in AUTO.
CAS	In CAS, the cascaded setpoint from the AM/APM/PM primary loop is used as the UDC's setpoint. Operator changes to the setpoint value are ignored while in SPCCSP CAS.
BCAS	In BCAS, the configured setpoint at the UDC 6000/6300 is used as the secondary's setpoint. BCAS applies if the UDC is configured for an RSP or if the UDC is configured for internal cascade loops (ALGORITHM/PID LOOPS = CASCADE) allowing for a backup cascade strategy. BCAS will enable the backup cascade strategy.

4.9 SPCCSP Cascade (Supervisory using computer [TDC] setpoint), Continued

PIDERFB point configuration

Table 4-12 shows the point configurations that are required for connecting an AM/APM/PM PIDERFB regulatory control loop to a UDC 6000/6300 PID loop using the SPCCSP option.

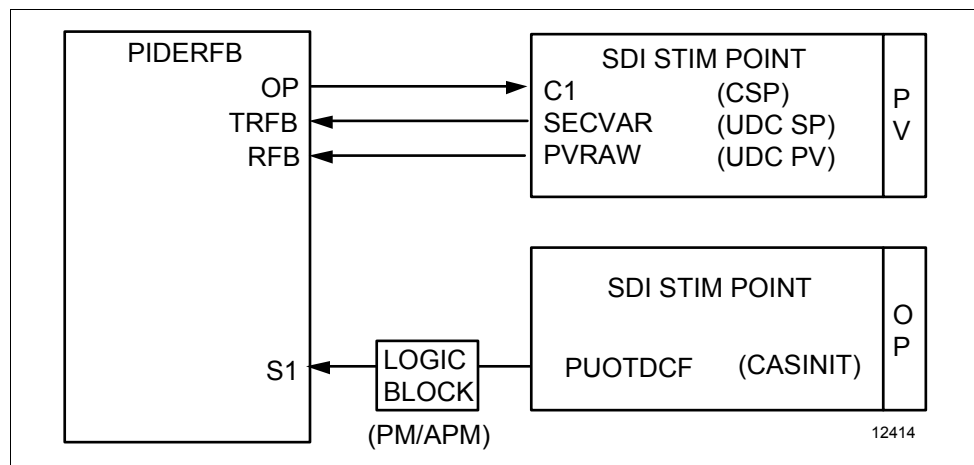
Table 4-12 SPCCSP - PIDERFB Point Build Connections

Point	Configuration
CVEUHI	Set to a value within the UDC setpoint range.
CVEULO	Set to a value within the UDC setpoint range.
OP	Connect to UDC 6000/6300 SDI STIM PV points C1.
RFB	Normally connected to the UDC 6000/6300 SDI STIM PV points PVRAW (PV).
TRFB	Not Used
SI	Not Used

SPCCSP STIM SDI point connections block diagram

Figure 4-6 is a block diagram of a typical SPCCSP PIDERFB to UDC 6000/6300 SDI STIM PV and OP point interface connections.

Figure 4-6 UDC 6000/6300 - SPCCSP / PIDERFB Interface



4.10 DDCLSP Cascade (Direct Control with local [UDC] setpoint)

Introduction

DDCLSP cascade allows a PM, APM, or AM to control the UDC 6000/6300 loop's output directly while using the UDC's local setpoint as the backup setpoint source.

This mode of control can be used for backup control and/or as an Auto/Manual station for non redundant PM, APM, or AM control loops with static setpoints.

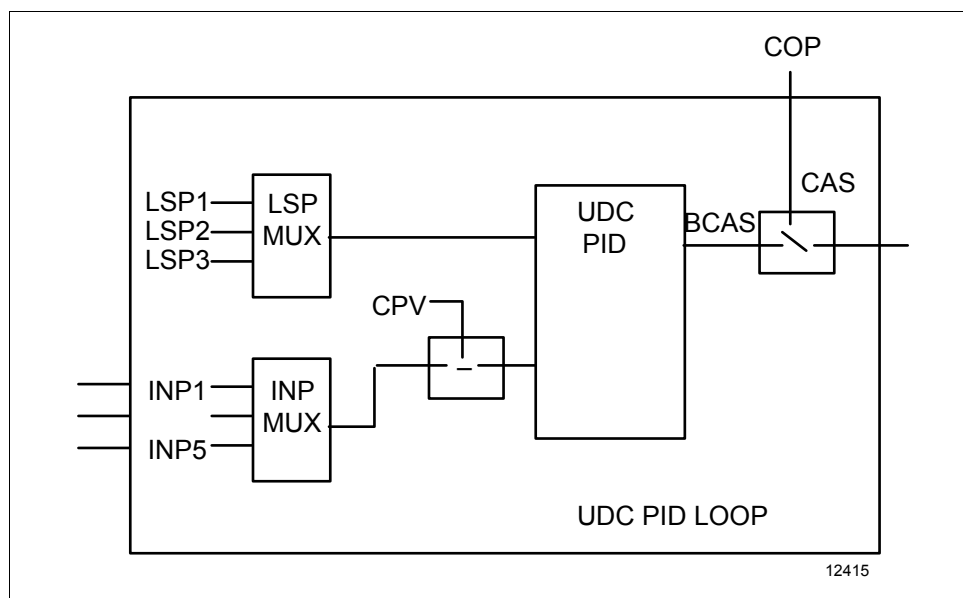
Select DDCLSP by way of the CASOPT target on the OPDATA page of the UDC 6000/6300 Regulatory Schematic display.

Description

Figure 4-7 is a simplified block diagram of a UDC loop configured for DDCLSP cascade. In DDCLSP configuration, the output of the UDC PID is overridden by a TDC control loop while CAS mode is active.

Emergency manual (LMAN) mode at the UDC is allowed during CAS mode.

Figure 4-7 UDC 6000/6300 Loop in DDCLSP Configuration



Output adjustments at the UDC 6000/6300

Using the UDC 6000/6300 local manual feature (Emergency Manual) allows an operator at the operator interface of the UDC 6000/6300 to place the PM, APM, or AM loop into "INIT" while manual adjustments are made to the output at the UDC 6000/6300.

Deselecting local manual will return the loop to the previous state.

4.10 DDCLSP Cascade (Direct Control with local [UDC] setpoint),

Continued

Output limit checks

In DDCLSP, the output of the UDC 6000/6300 PID loop is overridden by a value sent from the AM, PM, or APM. The range of the value must be within the output range limits of the UDC PID loop.

If the range limits are exceeded, the value will be rejected and the TRACK flag (UDC 6000/6300 SDI OP points PIUOTDCF parameter) will be set to ON.

The UDC will maintain the last valid value received from the PM as the output.

The output limits of the AM/APM/PM control algorithm should be set equal to or within the output limits of the UDC loop.

DDCLSP control mode definitions

Table 4-13 lists the control mode definitions for DDCLSP Cascade.

Table 4-13 Control Mode Definitions for DDCLSP Cascade

Mode	Definition
MAN	In manual mode, the operator can adjust the UDC's output. During Manual, the TRACK flag (UDC SDI OP points PIUOTDCF parameter) is set to ON to indicate to an overriding AM/APM/PM control loop, that it should back-initialize its output. If back-initialization is not done and the output value is changed, there will be a "bump" when CAS mode is resumed. Connection to the TRACK flag is done from PIDERFB general I/O connections in the AM and by logic in the PM/APM.
AUTO	When using the DDCLSP option, AUTO mode will enable the PID algorithm in the UDC 6000/6300 to generate the output. Overrides to the output are ignored. During AUTO mode, the TRACK flag will be ON indicating that the overriding control strategy should back-initialize its output. The UDC setpoint during AUTO for DDCLSP is a UDC local setpoint (LSP). The LSP value can be modified using engineering access on the second page of the UDC regulatory schematic (Boxmode Full). LSP values are stored in the UDC nonvolatile database.
CAS	DDCLSP CAS mode enables the UDC 6000/6300 loop's output to be overridden by an AM/APM/PM control strategy. While in CAS, the UDC 6000/6300 PID automatically back-initializes its auto output from the overriding output.
BCAS	When using the DDCLSP option, BCAS will be identical to AUTO.

4.10 DDCLSP Cascade (Direct Control with local [UDC] setpoint),

Continued

PIDERFB point configuration

Table 4-14 shows the point configurations that are required for PIDERFB connection to UDC 6000/6300 SDI STIM PV and OP points.

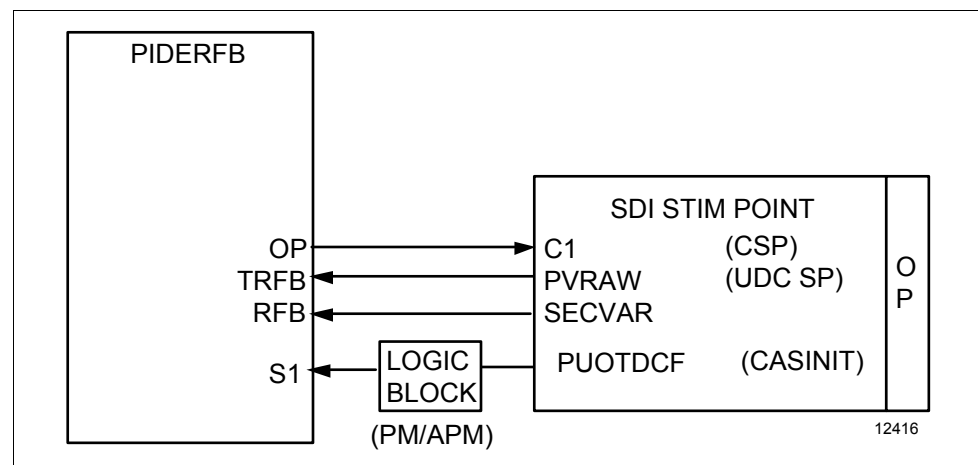
Table 4-14 DDCLSP - PIDERFB Point Build Connections

Point	Configuration
CVEUHI	Set to a value within the UDC output range.
CVEULO	Set to a value within the UDC output range.
OP	Connect to UDC 6000/6300 SDI STIM OP points C1.
RFB	Normally connected to the UDC 6000/6300 SDI STIM OP points SECVAR(PV).
TRFB	Normally connected to UDC 6000/6300 SDI STIM OP points SECVAR (PV) or PVRAW (OP).
SI (AM)	Make general input connection to UDC 6000/6300 SDI STIM OP points PIUOTDCF.
S1 (PM/APM)	Connect through logic block connection from the UDC 6000/6300 SDI STIM OP points PIUOTDCF parameter to the S1 parameter of the PM/APM PIDERFB algorithm.

DDCLSP STIM SDI point connections block diagram

Figure 4-8 is a block diagram of a typical DDCLSP PIDERFB to UDC 6000/6300 SDI STIM PV and OP point interface connections.

Figure 4-8 UDC 6000/6300 - DDCLSP STIM SDI Point Connections



4.11 DDCRSP Cascade (Direct Control with remote [UDC] setpoint)

Introduction

DDCRSP cascade can be used as a backup strategy of non redundant cascaded PM, APM, or AM loops.

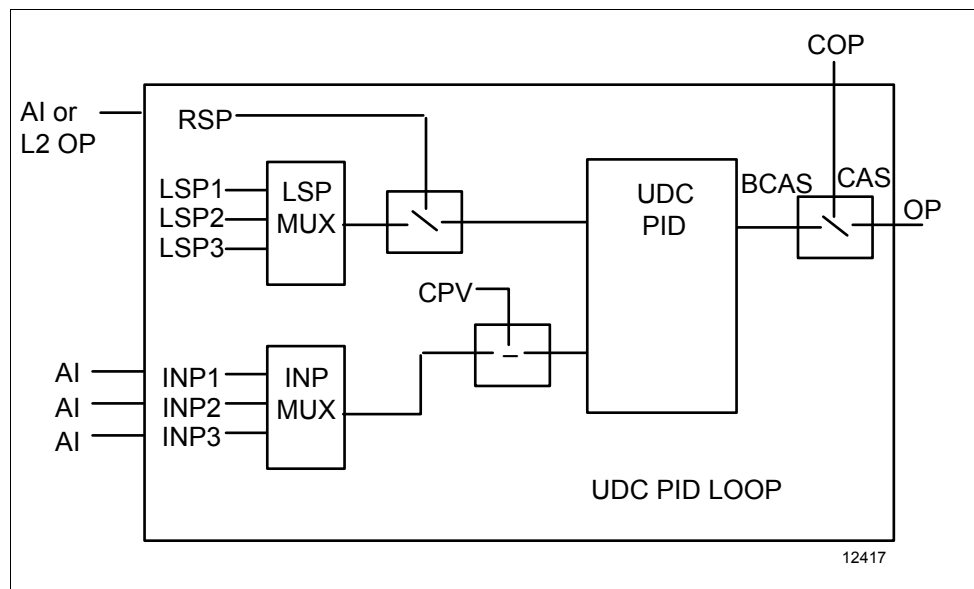
In DDCRSP, while cascade mode is active, the output of the UDC 6000/6300 is controlled directly by the PM/APM secondary's output, with the UDC 6000/6300 loop receiving a remote setpoint.

Select DDCRSP under the CASOPT target on the OPDATA page of the UDC 6000/6300 Regulatory Schematic display.

Description

A simplified block diagram of a UDC loop configured for DDCRSP cascade option is shown in Figure 4-9. Note that the RSP can be either an analog input or internally cascaded to the UDC 6000/6300 second loop.

Figure 4-9 UDC 6000/6300 Loop in DDCRSP Configuration



Output limit checks

When using DDCRSP, the PM/APM is sending an output override value to the UDC 6000/6300. The override value will be rejected by the UDC 6000/6300 if it exceeds the high or low output limits of the UDC 6000/6300 database.

4.11 DDCRSP Cascade (Direct Control with remote [UDC] setpoint),

Continued

DDCRSP control mode definitions

Table 4-15 lists the control mode definitions for DDCRSP Cascade.

Table 4-15 Control Mode Definitions for DDCRSP Cascade

Mode	Definition
MAN	In manual, the operator can adjust the UDC's output. While in manual, the TRACK flag (OP points PIUOTDCF) will be active indicating to the overriding control loop to track the current output of the UDC (OP point PVRAW).
AUTO	In AUTO, the UDC6000/6300's current local setpoint is used as the UDC loop's setpoint. While in AUTO, the TRACK flag (OP point PIUOTDCF) is set to ON to signal the overriding loop that the output is being generated at the UDC and that tracking feedback (TRFB) should occur. Tracking feedback prevents "bumps" in control when switching between AUTO and CAS modes. The UDC local setpoint can be changed, using engineering access, from the UDC regulatory schematic.
CAS	During CAS, the output from the TDC loop is used as the UDC's output. While in CAS the TRACK flag (OP point PIUOTDCF) is OFF. PV tracking should be configured at the UDC 6000/6300 to prevent "bumps" in control when switching from CAS back to AUTO.
BCAS	In BCAS, the configured remote setpoint at the UDC 6000/6300 is used as the UDC loop's setpoint and the UDC setpoint source is set to RSP. In configurations where the second loop of a UDC is cascaded to the first, the second loop will be the backup cascade primary. Alternatively, an analog signal can be configured to provide the backup cascade setpoint to a UDC PID loop.

4.11 DDCRSP Cascade (Direct Control with remote [UDC] setpoint),

Continued

PIDERFB point configuration

Table 4-16 shows the point configurations that are required for DDCRSP connection from a PIDERFB algorithm to UDC 6000/6300 SDI STIM PV and OP points.

The DDCRSP point build connections are identical to those of the DDCLSP cascade option. An additional AO point build may be necessary if a PM/APM analog output is to be used as the UDC 6000/6300 remote setpoint.

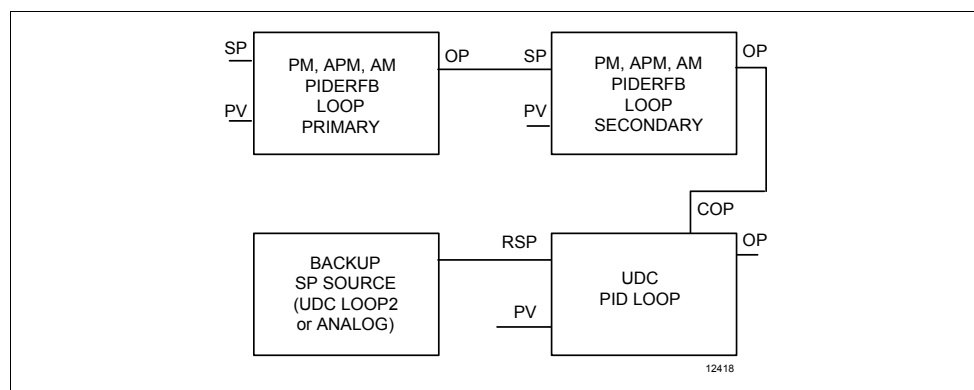
Table 4-16 DDCRSP - PIDERFB Point Build Connections

Point	Configuration
CVEUHI	Set to a value within the UDC output range.
CVEULO	Set to a value within the UDC output range.
OP	Connect to UDC 6000/6300 SDI STIM OP points C1.
RFB	Normally connected to the UDC 6000/6300 SDI STIM OP points SECVAR (PV).
TRFB	Normally connected to UDC 6000/6300 SDI STIM OP points PVRAW (OP) or SECVAR(PV).
S1 (AM)	Make general input connection to UDC 6000/6300 SDI STIM OP points PIUOTDCF (TRACK).
S1 (PM/APM)	Connect through logic block connection from the UDC 6000/6300 SDI STIM OP points PIUOTDCF parameter to the S1 parameter of the PM/APM S1 input (TRACK).

DDCRSP with BCAS of PM/APM primary

Figure 4-10 is a block diagram of the UDC 6000/6300 in DDCRSP cascade. In this example the output of the UDC control loop is being overridden by the output of a PM/APM PIDERFB secondary loop in the PM/APM. A backup setpoint source is connected to the RSP of the UDC PID loop.

Figure 4-10 Simplified UDC 6000/6300 DDCRSP Cascade Block Diagram



4.12 DDCCSP Cascade (Direct Control with computer [TDC] setpoint)

Introduction

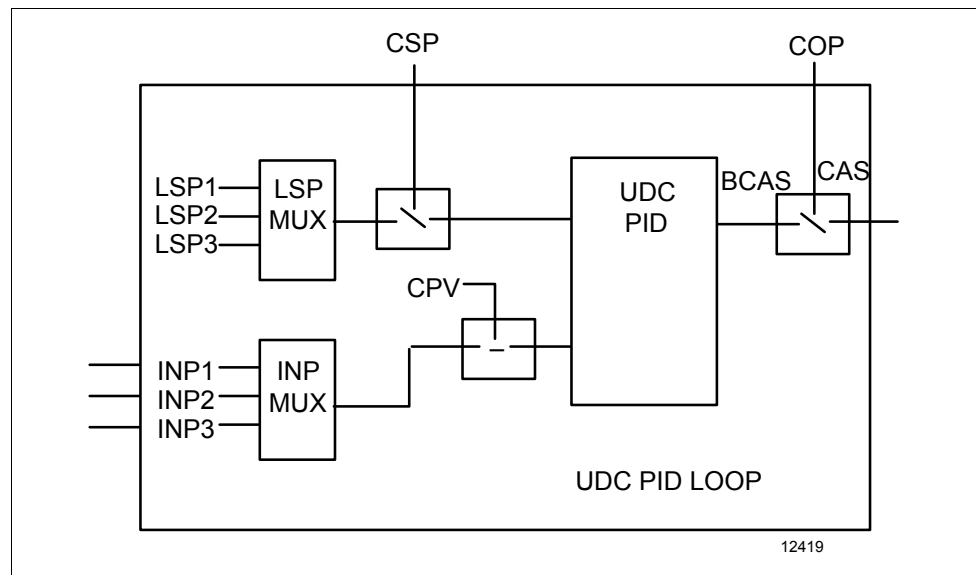
DDCCSP cascade allows the PM, APM, or AM to directly control the UDC6000/6300's output while providing a computer (TDC) setpoint.

Select DDCCSP under the CASOPT target on the OPDATA page of the UDC 6000/6300 Regulatory Schematic display.

Description

A simplified block diagram of a UDC PID loop configured for DDCCSP cascade option is shown in Figure 4-11.

Figure 4-11 UDC 6000/6300 Loop in DDCCSP Configuration



Setpoint and output limit checks

When using DDCCSP, the PM/APM is sending an output override value and a setpoint override value to the UDC 6000/6300. The output and setpoint override values will be rejected by the UDC 6000/6300 if they exceed the respective high and low output and setpoint limits which are configured in the UDC 6000/6300 database.

Limiting integral windup

If the output limits of the UDC 6000/6300 are exceeded, integral windup can result. PIDERFB algorithms provide a scaled integral windup limit equal to $K1$ times the loop error. Windup can be avoided if the output limits of the overriding loop are within or equal to the configured limits of the UDC output.

4.12 DDCCSP Cascade (Direct Control with computer [TDC] setpoint), Continued

DDCCSP control mode definitions

Table 4-17 lists the control mode definitions for DDCCSP Cascade.

Table 4-17 Control Mode Definitions for DDCCSP Cascade

Mode	Definition
MAN	In manual, the operator can adjust the UDC's output. While in manual, the CASINIT flag will be active indicating to the overriding control loop to track the current output of the UDC (OP point PVRAW).
AUTO	The UDC setpoint is overridden by the setpoint override (CSP) value sent from the AM/APM/PM. While in AUTO, the TRACK flag (OP point PIUOTDCF) is set to ON to signal to the control strategy attempting to override the UDC output that the output is being generated locally and that tracking feedback (TRFB) should occur. Tracking feedback prevents "bumps" in control when switching between AUTO and CAS modes. During AUTO, the UDC setpoint override value (CSP) can be modified by operator access from the first page of the UDC regulatory schematic or the SP of the UDC PV point.
CAS	In CAS, the output from the TDC loop is used as the UDC's output. While in CAS, the TRACK flag (OP point PIUOTDCF) is OFF. PV tracking should be configured at the UDC 6000/6300 to prevent integral windup of the UDC PID.
BCAS	For DDCCSP, a backup primary in the AM/PM/APM provides the setpoint for the UDC PID loop. The backup primary interfaces to the UDC through the C1 parameter of the PV point. While in BCAS, the UDC 6000/6300 PID loop is generating the output value and the TRACK flag (UDC SDI OP points PIUOTDCF) is set to ON to indicate to the automatic control, which is overriding the UDC loop's output that it should track.

4.12 DDCCSP Cascade (Direct Control with computer [TDC] setpoint),

Continued

PIDERFB point configuration

The DDCCSP can be connected to two AM/APM/PM PIDERFB regulatory points.

One regulatory point writes to the Output of the UDC 6000/6300. A second point can be used to write to the setpoint of the UDC 6000/6300.

Table 4-18 and 4-19 show the connections to these regulatory points.

Table 4-18 refers to the point writing to the UDC output as a DDC connection and Table 4-19 refers to the regulatory point writing to the UDC setpoint as a SPC connection.

Table 4-18 DDCCSP - PIDERFB Point Build DDC (Output) Connections

Point	Configuration
CVEUHI	Set to a value within the UDC output range.
CVEULO	Set to a value within the UDC output range.
OP	Connect to UDC 6000/6300 SDI STIM OP points C1.
RFB	Normally connected to the UDC 6000/6300 SDI STIM PV points PVRAW (PV).
TRFB	Normally connected to UDC 6000/6300 SDI STIM OP points SECVAR (PV) or PVRAW (OP).
SI (AM)	Make general input connection to UDC 6000/6300 SDI STIM OP points PIUOTDCF.
S1 (PM/APM)	Connect through logic block connection from the UDC 6000/6300 SDI STIM OP points PIUOTDCF parameter to the S1 parameter of the PM/APM S1 input.

Table 4-19 DDCCSP - PIDERFB Point Build SPC (Setpoint) Connections

Point	Configuration
CVEUHI	Set to a value within the UDC setpoint range.
CVEULO	Set to a value within the UDC setpoint range.
OP	Connect to UDC 6000/6300 SDI STIM PV points C1.
RFB	Normally connected to the UDC 6000/6300 SDI STIM PV points PVRAW (PV).
TRFB	Not Used
SI	Not Used

4.12 DDCCSP Cascade (Direct Control with computer [TDC] setpoint),

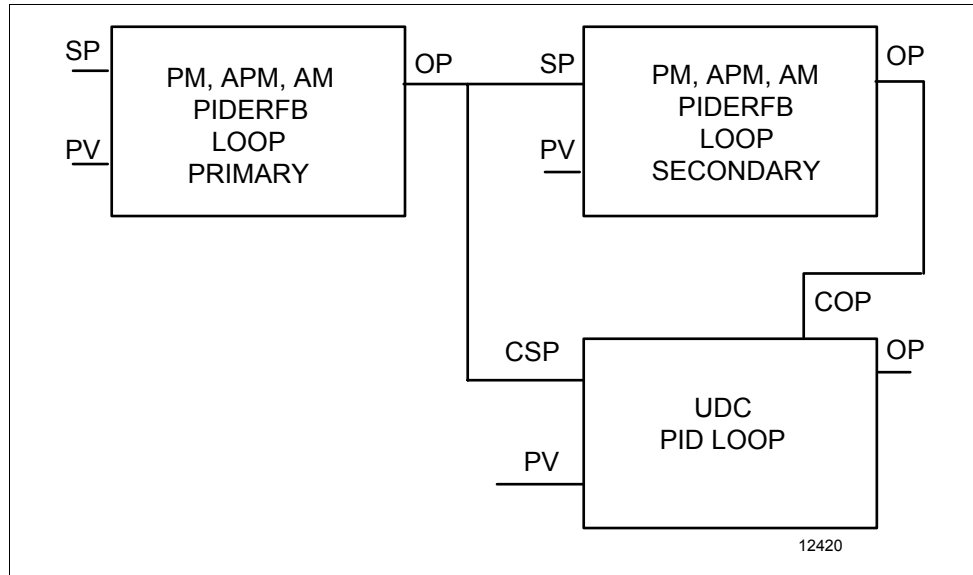
Continued

UDC 6000/6300 DDCCSP/PIDERFB cascade

Figure 4-12 shows a UDC loop with DDCCSP cascade. The CSP source can be generated from the primary output as shown in the figure.

Alternatively, the CSP can be generated from a CL program, a logic point, or from the UDC 6000/6300 detail custom schematics.

Figure 4-12 Simplified UDC 6000/6300 DDCCSP Cascade Block Diagram



Section 5 – Operation

5.1 Overview

Purpose of this section

This section describes the various US displays through which an operator can interact with a given UDC 6000/6300 as an SDI point.

Reader assumptions

- You know how to interact with the **TotalPlant** Solution (TPS) system using the US touch screens and keyboard.
 - The US station is in the Operator mode.
 - You have completed the setup tasks in Section 3.
 - If you are using cascade control, you have read Section 4.
 - You know how to interact with the UDC 6000/6300 controller using its front face displays and keys.
-

5.2 UDC 6000/6300 Summary Display

Description

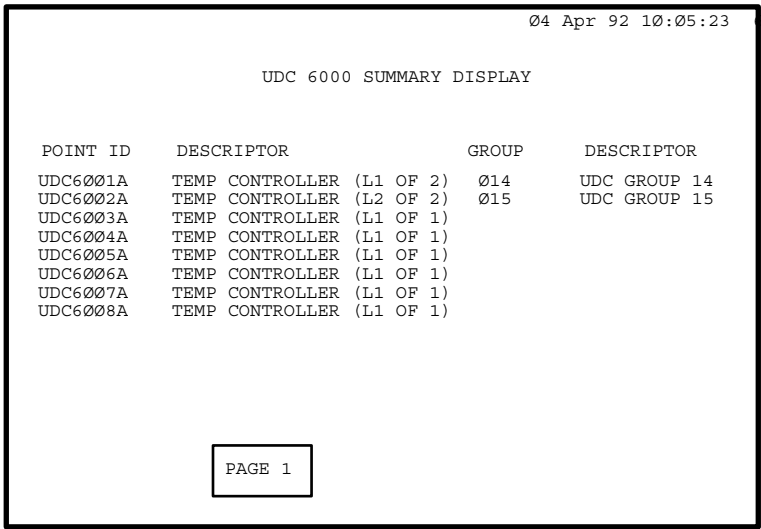
The UDC 6000/6300 Summary display is a custom schematic that provides a list of all configured UDC 6000/6300 SDI points by point ID (tag name) and point descriptor. It also lists any configured UDC 6000/6300 Group displays by number and group descriptor.

This display provides access to the UDC 6000/6300 Detail display for a selected loop and the UDC 6000/6300 Group display for a selected group.

Display callup and selections

Use the steps in Table 5-1 to call up the UDC 6000/6300 Summary display and select either a UDC 6000/6300 Detail display or a UDC 6000/6300 Group display.

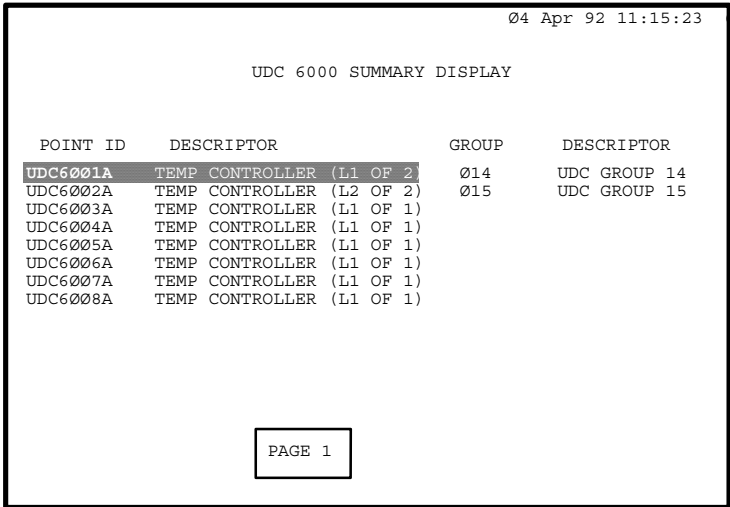
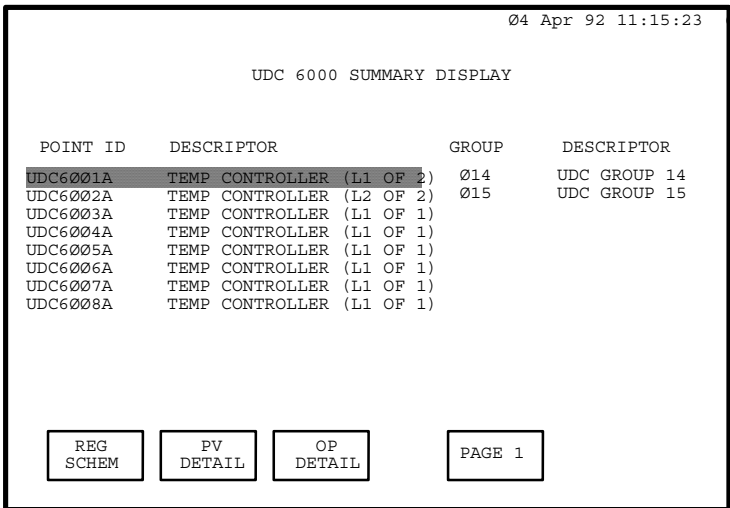
Table 5-1 Calling Up Summary Display for Detail or Group Selection

Step	Action								
1	Press key designated for UDC 6000/6300 schematics during setup to call up Summary display. <u>Example:</u> Calling up Summary display. 								
2	Do you want to access a UDC 6000/6300 Detail display or Group display? <table border="1"> <thead> <tr> <th>If you want to access...</th><th>Then go to ...</th></tr> </thead> <tbody> <tr> <td>a UDC 6000/6300 Detail display</td><td>Step 3</td></tr> <tr> <td>a UDC 6000/6300 Group display</td><td>Step 4</td></tr> <tr> <td>neither display</td><td>another procedure of your choice</td></tr> </tbody> </table>	If you want to access...	Then go to ...	a UDC 6000/6300 Detail display	Step 3	a UDC 6000/6300 Group display	Step 4	neither display	another procedure of your choice
If you want to access...	Then go to ...								
a UDC 6000/6300 Detail display	Step 3								
a UDC 6000/6300 Group display	Step 4								
neither display	another procedure of your choice								

5.2 UDC 6000/6300 Summary Display, Continued

Display callup and selections, continued

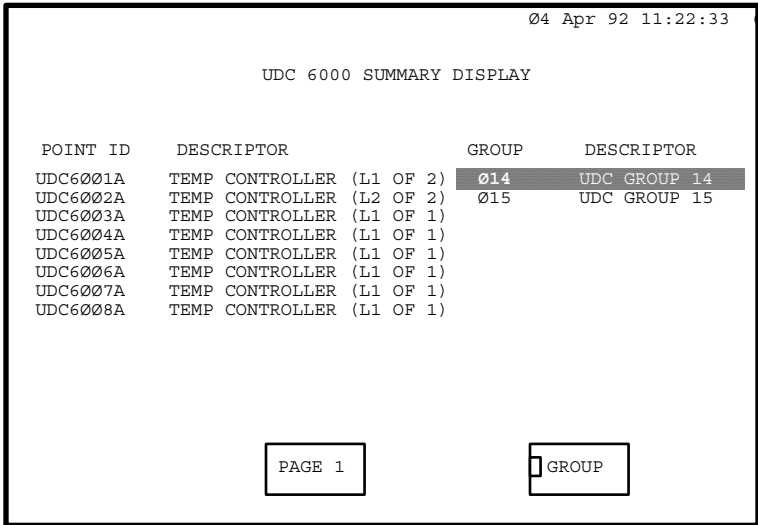
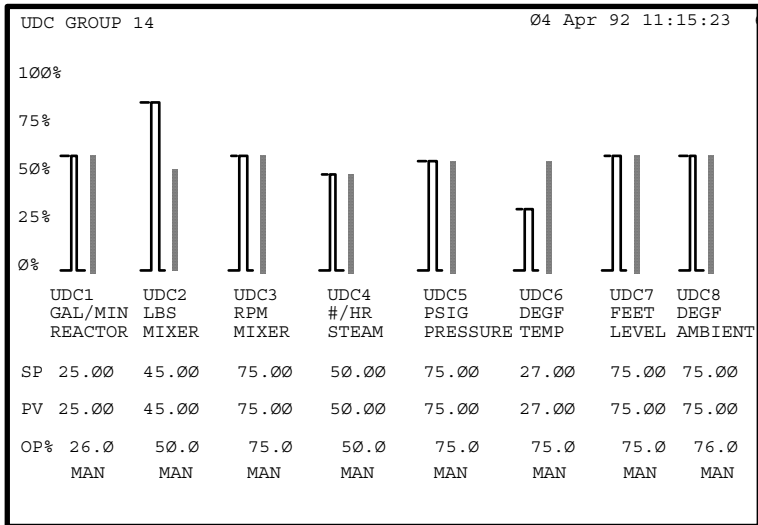
Table 5-1 Calling Up Summary Display, Continued

Step	Action
3	To call up a detail display, select (touch) the desired point ID. <u>Example:</u> Selecting desired point ID.  12422 A display will appear that lets you select one of three displays for that loop.  12423 Refer to “UDC 6000/6300 DETAIL DISPLAY” for display details.

5.2 UDC 6000/6300 Summary Display, Continued

Display callup and selections, continued

Table 5-1 Calling Up Summary Display, Continued

Step	Action
4	To call up a Group display, select (touch) the Group ID descriptor. <u>Example: Selecting desired group.</u>  12424 The selected UDC 6000/6300 Group Display will appear.  12425 Refer to “UDC 6000/6300 DETAIL DISPLAY” for display details.
5	Make another dedicated keystroke to exit UDC 6000/6300 schematics.

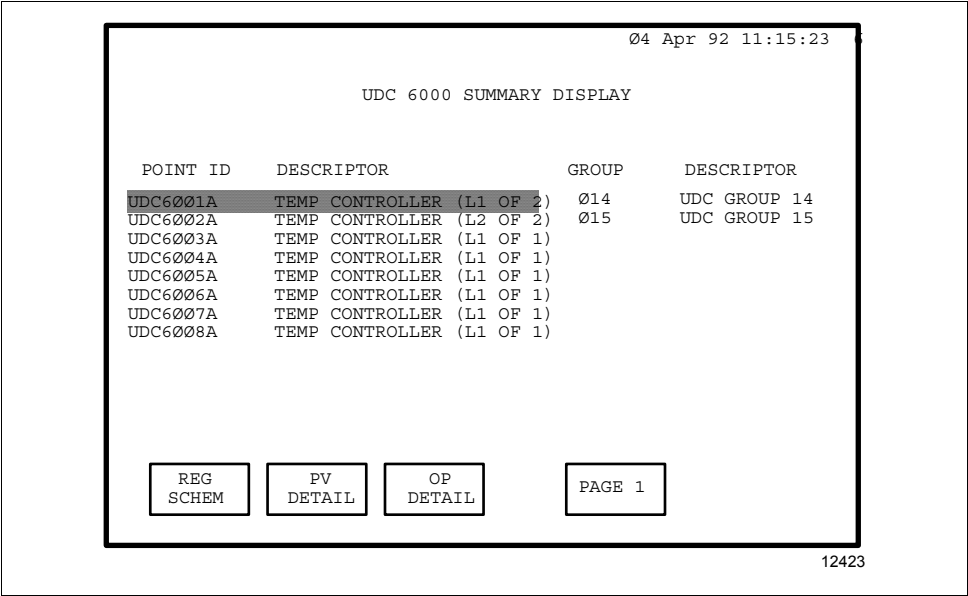
5.3 UDC 6000/6300 Detail Display

Description

The UDC 6000/6300 Detail display is a custom schematic, which provides operator interaction with the UDC 6000/6300 loop tied to the selected SDI point.

Figure 5-1 shows a detail display for the Point ID selected from the summary page (Table 5-1).

Figure 5-1 Detail Display Selections



Target selections

Table 5-2 lists the four selection targets on the Detail Display screen.

Table 5-2 Display Targets

Target	Display Selection
OP DETAIL	select to display the Output(OP) point data for the loop selected. See “OP POINT DETAIL DISPLAY.”
PV DETAIL	select to display the Process Variable(PV) point data for the loop selected. See “PV POINT DETAIL DISPLAY.”
REG SCHEM	select to display the regulatory schematic for the loop selected. See “REGULATORY SCHEMATIC DISPLAY.”
PAGE 1	select to move to the next page of Point ID or Group descriptors.

5.3 UDC 6000/6300 Detail Display, Continued

OP point detail display

Perform the step in Table 5-3 to call up the OP POINT DETAIL DISPLAY.

Table 5-3 Selecting OP Point Detail Display

Step	Action
1	Select OP DETAIL on the Detail display. This display shows all the OP (Output) point data for the loop selected. ATTENTION This display can also be accessed from the "OP DATA PAGE" of the "REGULATORY SCHEMATIC DISPLAY" <pre> SDI1B UDC6000 LOOP 1 OUTPUT 04 Apr 92 11:15:23 6 FIRST PAGE POINT DATA PVAUTO 50.000 ALARM LIMITS PVSOURCE AUTO PVHIHITP 90.000 PTEXECST ACTIVE PVCALC 50.000 PVHITP 80.000 ALENBST ENABLE LAST PV 50.000 PVLOTP 20.000 OVERVAL 25 PVLLTP 10.000 PNTFORM FULL PVROCPTP 20.000 PVRAW 46.0820 RANGE LIMITS PV COEFFICIENTS PVEXEUHI 102.0 TF 0.00000 PVEUHI 100.0 C1 ----- PVEULO 0.0 C2 ----- PVEXEULO -2.0 SDI1B CONTCUT OFF PVRAWHI ----- PERCENT PT TYPE ANINNIM PVRWLO ----- LOOP1 LCN NODE 31 URL 1000.0 SP 50.00 PROC NET 02 URV 100.0 UCN NODE 05 LRV 0.0 PV 50.00 DEV TYPE PM LRL 0.0 MOD NUM 001 OP% 50.0 MOD TYPE STIM SLOT NUM 0002 </pre> 12426
2	Refer to Table 5-4 for a list of parameters that can be changed.

5.3 UDC 6000/6300 Detail Display, Continued

Changeable parameters

Table 5-4 lists the parameters that an operator can change when the point (PTEXECST) is active. Any others cannot be changed.

Table 5-4 OP Point Detail Display Changeable Parameters

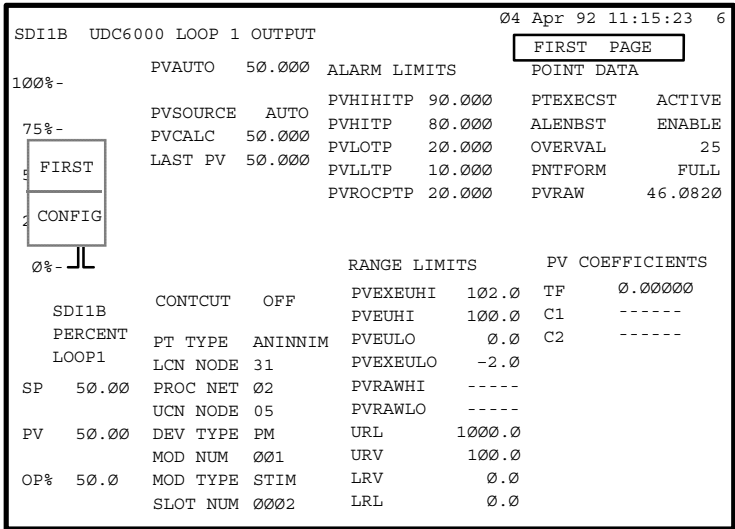
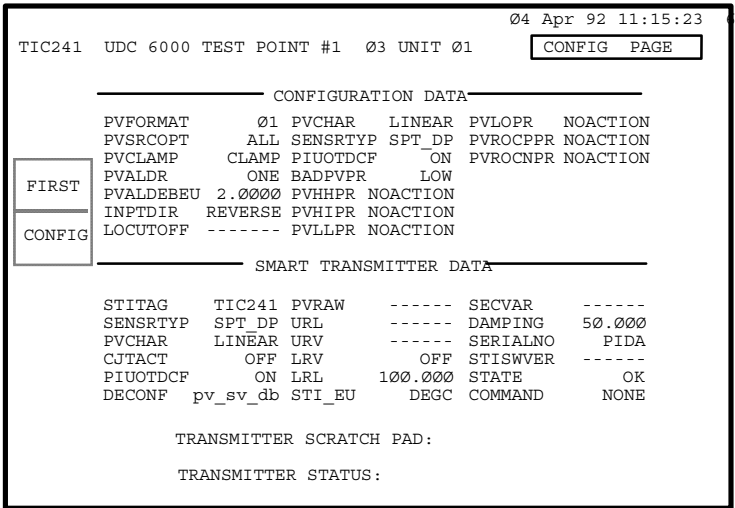
Parameter	Access	Value/Selection	Changeable
PVSOURCE		AUTO, MAN, SUB, TRACK	Yes
PVHIHITP	Supervisor	Within range limits	Yes
PVHITP	Supervisor	Within range limits	Yes
PVLOTP	Supervisor	Within range limits	Yes
PVLLTP	Supervisor	Within range limits	Yes
PVROCPTP	Supervisor	Within range limits	Yes
PVROCNT	Supervisor	Within range limits	Yes
PTEXECST	Supervisor	NOT CONFIG, ACTIVE, INACTIVE, READY,DONE, WAITING	Yes
ALENBST	Engineer	ENABLE, DISABLE, INHIBIT	Yes
PVEXEUHI	Engineer	Within range limits	Yes
PVEUHI	Engineer	Within range limits	Yes
PVEULO	Engineer	Within range limits	Yes
PVEXEULO	Engineer	Within range limits	Yes
URL	Engineer	Within range limits	Yes
URV	Supervisor	Within range limits	Yes
LRV	Supervisor	Within range limits	Yes
C1	Supervisor	Reserved for Cascade	
C2	Supervisor	Reserved for Cascade	
FIRST PAGE		FIRST or CONFIG	Yes

5.3 UDC 6000/6300 Detail Display, Continued

Standard detail display selection

Perform the steps in Table 5-5 to call up the STANDARD DETAIL DISPLAY from the OP Point Detail Display.

Table 5-5 Calling Up the Standard Detail Display

Step	Action
1	Select the target FIRST PAGE on the OP Point Detail Display.
2	Select CONFIG from the selections that appear on the display.  SDI1B UDC6000 LOOP 1 OUTPUT 04 Apr 92 11:15:23 6 FIRST PAGE POINT DATA 100%- PVAUTO 50.000 ALARM LIMITS 75%- PVSOURCE AUTO PVHIHITP 90.000 PTEXECST ACTIVE PVCALC 50.000 PVHITP 80.000 ALENBST ENABLE LAST PV 50.000 PVLOTP 20.000 OVERVAL 25 PVLLTP 10.000 PNTFORM FULL PVROCPTP 20.000 PVRAW 46.0820 RANGE LIMITS PV COEFFICIENTS SDI1B CONTCUT OFF PVEXEUI 102.0 TF 0.00000 PERCENT PVEUHI 100.0 C1 ----- LOOP1 PT TYPE ANINNIM PVEULO 0.0 C2 ----- SP 50.00 LCN NODE 31 PVEXEULO -2.0 PROC NET 02 PVRAWHI ----- UCN NODE 05 PVRAWLO ----- PV 50.00 DEV TYPE PM URL 1000.0 MOD NUM 001 URV 100.0 OP% 50.0 MOD TYPE STIM LRV 0.0 SLOT NUM 0002 LRL 0.0 12427
3	The screen that appears is the standard detail display.  TIC241 UDC 6000 TEST POINT #1 03 UNIT 01 04 Apr 92 11:15:23 CONFIG PAGE CONFIGURATION DATA PVFORMAT 01 PVCHAR LINEAR PVLOPR NOACTION PVSRCOPT ALL SENSRTYP SPT DP PVROCPPR NOACTION PVCLAMP CLAMP PIUOTDCF ON PVROCNPR NOACTION PVALDR ONE BADPVPR LOW PVALDEBEU 2.0000 PVHPR NOACTION INPTDIR REVERSE PVHIPR NOACTION LOCUTOFF ----- PVLLPR NOACTION SMART TRANSMITTER DATA STITAG TIC241 PVRAW ----- SECVAR ----- SENSRTYP SPT_DP URL ----- DAMPING 50.000 PVCHAR LINEAR URV ----- SERIALNO PIDA CJTACT OFF LRV OFF STISWVER ----- PIUOTDCF ON LRL 100.000 STATE OK DECONF pv_sv_db STI_EU DEGC COMMAND NONE TRANSMITTER SCRATCH PAD: TRANSMITTER STATUS: 12428 Refer to “STANDARD DETAIL DISPLAY” section for display details.
4	Select FIRST from the selection that appears on the display to return to the OP Point Detail Display.

5.3 UDC 6000/6300 Detail Display, Continued

PV point detail display

Select

PV
DETAIL

 on the Detail display.

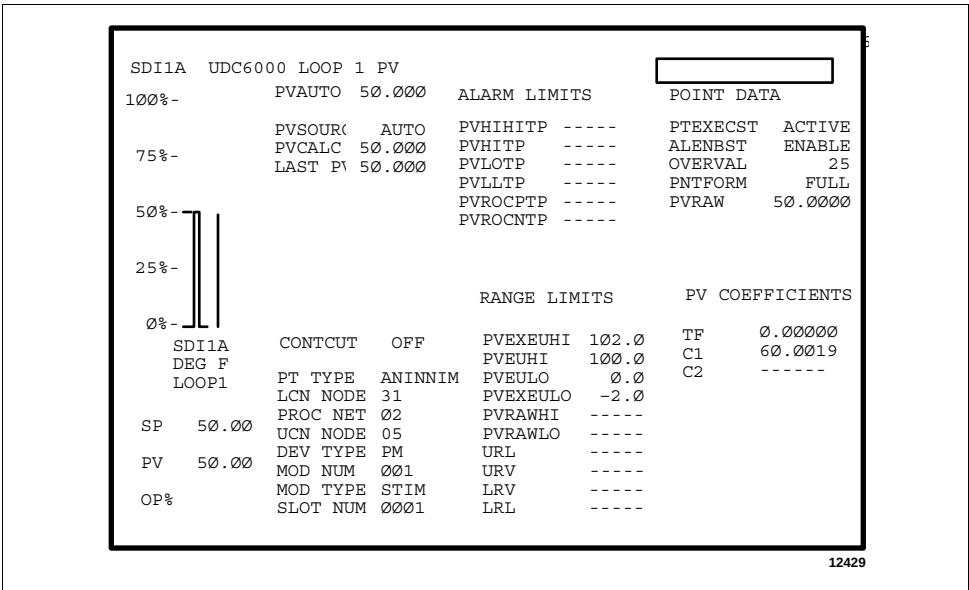
This display shows all the PV (Process Variable) point data for the loop selected.

Figure 5-2 is an example of a PV Point Detail display.

ATTENTION

ATTENTION—This display can also be accessed from the “OP DATA PAGE” of the “REGULATORY SCHEMATIC DISPLAY.”

Figure 5-2 PV Point Detail Display



5.3 UDC 6000/6300 Detail Display, Continued

Changeable parameters

Table 5-6 lists the parameters that an operator can change when the point (PTEXECST) is active. Any others cannot be changed.

Table 5-6 PV Point Detail Display Changeable Parameters

Parameter	Access	Value/Selection	Changeable
PVSOURCE		AUTO, MAN, SUB, TRACK	Yes
PVHIHITP	Supervisor	Within range limits	Yes
PVHITP	Supervisor	Within range limits	Yes
PVLOTP	Supervisor	Within range limits	Yes
PVLLTP	Supervisor	Within range limits	Yes
PVROCPTP	Supervisor	Within range limits	Yes
PVROCNTP	Supervisor	Within range limits	Yes
PTEXECST	Supervisor	NOT CONFIG, ACTIVE, INACTIVE, READY,DONE, WAITING	Yes
ALENBST	Engineer	ENABLE, DISABLE, INHIBIT	Yes
PVEXEUHI	Engineer	Within limits	Yes
PVEUHI	Engineer	Within range limits	Yes
PVEULO	Engineer	Within range limits	Yes
PVEXEULO	Engineer	Within limits	Yes
URL	Engineer	Within range limits	Yes
URV	Supervisor	Within range limits	Yes
LRV	Supervisor	Within range limits	Yes
C1	Supervisor	Reserved for Cascade	Yes
C2	Supervisor	Reserved for Cascade	Yes
FIRST PAGE		FIRST or CONFIG	Yes

5.3 UDC 6000/6300 Detail Display, Continued

Standard detail display selection

Perform the steps in Table 5-7 to call up the STANDARD DETAIL DISPLAY from the PV Point Detail Display.

Table 5-7 Calling Up the Standard Detail Display

Step	Action
1	Select the target FIRST PAGE on the PV Point Detail Display.
2	Select CONFIG from the selections that appear on the display.
3	The screen that appears is the standard detail display.
4	Select FIRST from the selection that appears on the display to return to the OP Point Detail Display.

5.3 UDC 6000/6300 Detail Display, Continued

Regulatory Schematic display

The UDC 6000/6300 Regulatory Schematic display is a custom schematic, which provides operator interaction with the UDC 6000/6300 loop tied to the selected SDI point. It is partitioned into these three pages:

- FIRST,
- OP DATA, and
- CONFIG.

Purpose of pages

Table 5-8 summarizes the purpose of the FIRST, OP DATA, and CONFIG pages.

Table 5-8 FIRST, OP DATA, and CONFIG Pages Summary

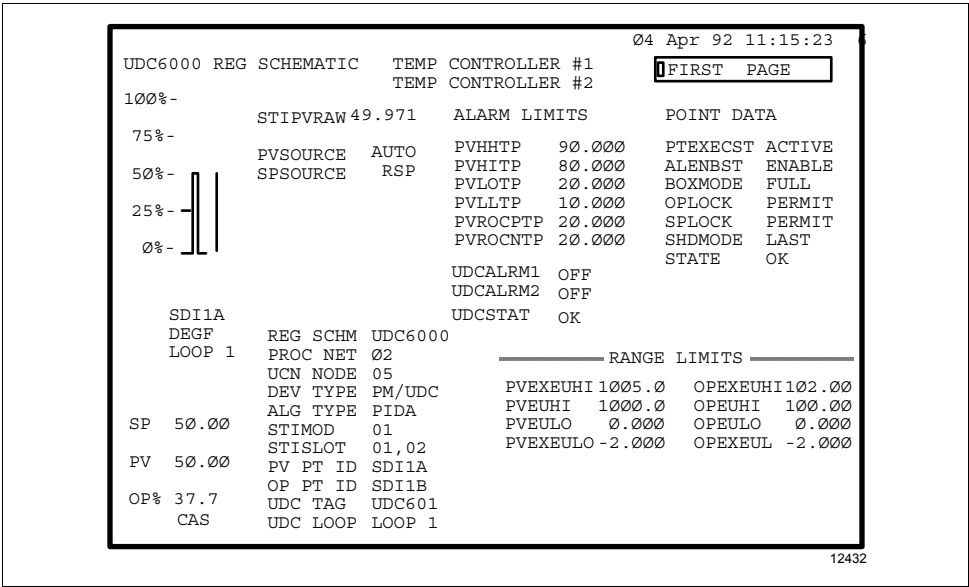
Detail Display Page	Purpose
FIRST	Lets operators monitor essential operating parameters and change certain parameters as applicable.
OP DATA	Lets operators view selected input and setpoint values.
CONFIG	Lets supervisors view configuration parameters and lets engineers change configuration parameters stored in UDC 6000/6300 database through supervisor access.

5.3 UDC 6000/6300 Detail Display, Continued

Regulatory Schematic Display You call up the UDC 6000/6300 Regulatory Schematic display through the UDC 6000/6300 Summary display. Refer to the previous Summary display call- up procedure in Table 5-1 for details.

FIRST page call up Select REG SCHEM on the Detail display.
This display shows all the data for the loop selected.

Figure 5-3 is an example of a Regulatory Schematic Detail display.
Figure 5-3 Regulatory Schematic Display



Select a Loop The targets located at the top of the display

TEMP CONTROLLER #1 or TEMP CONTROLLER #2

let you select Loop 1 or Loop 2.

The examples shown are for Loop #1.

5.3 UDC 6000/6300 Detail Display, Continued

parameters on FIRST
page of Reg
Schematic

Table 5-9 lists the parameters on the first page that an operator will see. Some of these parameters can be changed when the point (PTEXECST) is active and BOXMODE is Full. Others cannot be changed.

Table 5-9 Parameters on FIRST Page

Parameter	Access	Value/Selection	Changeable
SP		Within setpoint limits	Yes
PV		Within process variable limits	Yes
OP%		Within output limits only when control mode is MAN	Yes
control mode		MAN, AUTO, CAS	Yes
PVSOURCE		AUTO, MAN, or SUB	Yes
PVHIHITP	Supervisor	Within range limits	Yes
PVHITP	Supervisor	Within range limits	Yes
PVLOTP	Supervisor	Within range limits	Yes
PVLLTP	Supervisor	Within range limits	Yes
PVROCPTP	Supervisor	Within range limits	Yes
PVROCNTTP	Supervisor	Within range limits	Yes
PTEXECST	Supervisor	ACTIVE, or INACTIVE	Yes
ALENBST	Engineer	ENABLE, or DISABLE	Yes
BOXMODE	Engineer	Full or Basic	Yes
SHEDMODE	Engineer	LAST, MAN, SAFE, AUTO	YES
SPSOURC	View	LSP, RSP, CSP,CAS	No
STATE	View	Normal, OK, DB Change, Loading, Loadcomp	No
UDC ALARM 1	View	On, Off	No
UDC ALARM 2	View	On, Off	No
UDC STAT	View	Busy, Time out, OK, DB Change	No
FIRST PAGE	View	FIRST, OP DATA, or CONFIG	Yes

5.3 UDC 6000/6300 Detail Display, Continued

Parameters on FIRST
page of Reg
Schematic, continued

Table 5-9 Parameters on FIRST Page, Continued

Parameter	Access	Value/Selection	Changeable
PVEUHI		Within range limits	No
PVEULO		Within range limits	No
SPHILIM		Within range limits	No
SPLOLIM		Within range limits	No
OPHILIM		Within range limits	No
OPLOLIM		Within range limits	No

Typical usage of
FIRST page

A typical use of the FIRST page could be to adjust the output manually. Use the steps in Table 5-10 to make a manual output adjustment assuming that the present control mode is CAS.

As for all the changeable parameters listed in Table 5-9, prompts such as those shown in Table 5-10 will step your way through the procedure for changing any parameter.

Table 5-10 Adjusting Output Manually

Step	Action
1	Select control mode CAS.
2	Select MAN from mode selections that appear on display. <u>Example: Selecting MAN control mode.</u>

UDC6000 REG SCHEMATIC TEMP CONTROLLER #1 04 Apr 92 11:15:23

100%-

75%-

MODE

MAN

AUTO

CAS

BCAS

1A

P 1

REG SCHM UDC6000

PROC NET 02

UCN NODE 05

DEV TYPE PM/UDC

ALG TYPE PIDA

STIMOD 01

STISLOT 01,02

PV PT ID SDI1A

OP PT ID SDI1B

UDC TAG UDC601

UDC LOOP LOOP 1

ALARM LIMITS

PVHHTP 90.000

PVHITP 80.000

PVLOTP 20.000

PVLLTP 10.000

PVROCPTP 20.000

PVROCNT 20.000

UDCALRM1 OFF

UDCALRM2 OFF

UDCSTAT OK

POINT DATA

PTEXECST ACTIVE

ALENBST ENABLE

BOXMODE FULL

SHDMODE LAST

STATE OK

RANGE LIMITS

PVEXEUHI 1005.0

PVEUHI 1000.0

PVEULO 0.000

PVEXEULO -2.000

OPEXEUHI 102.00

OPEUHI 100.00

OPEULO 0.000

OPEXEULO -2.000

OP% 37.7

CAS

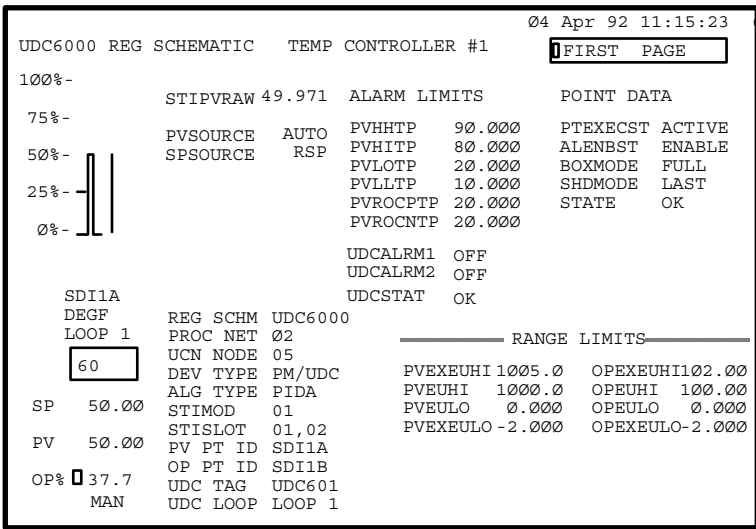
ENTER

12510

5.3 UDC 6000/6300 Detail Display, Continued

Typical usage of
FIRST page, continued

Table 5-10 Adjusting Output Manually, Continued

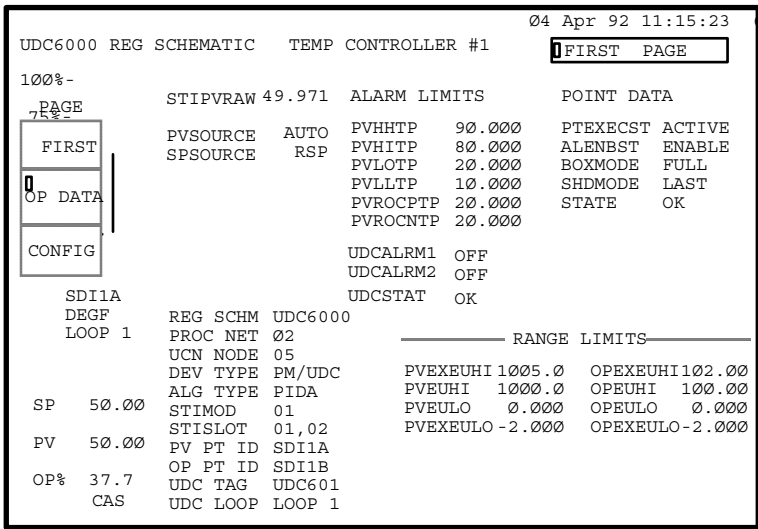
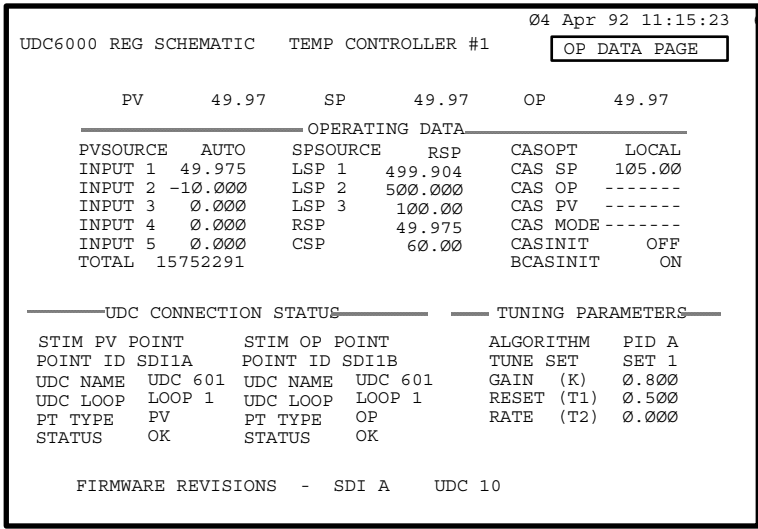
Step	Action
3	Select [ENTER] target and watch for mode to change to MAN.
4	Select output value.
5	Key in desired output value [such as, 60] in percent in fill-in box that opens on display. <u>Example:</u> Keying in desired output value in manual mode.  The screenshot shows the UDC6000 REG SCHEMATIC display. At the top, it says 'UDC6000 REG SCHEMATIC TEMP CONTROLLER #1' and '04 Apr 92 11:15:23'. Below this, there's a 'FIRST PAGE' button. The display is divided into several sections: '100%-', '75%', '50%', '25%', and '0%'. On the left, there's a vertical scale with a bar graph. In the center, there's a table of parameters: STIPVRAW 49.971, ALARM LIMITS, POINT DATA, PVSOURCE AUTO, PVHITP 90.000, PTEEXECST ACTIVE, SPSOURCE RSP, PVHITP 80.000, ALENBST ENABLE, PVLOTP 20.000, BOXMODE FULL, PVLLTP 10.000, SHDMODE LAST, PVROCPTP 20.000, STATE OK, PVROCNT 20.000, UDCALRM1 OFF, UDCALRM2 OFF, UDCSTAT OK. Below this, there's a section for 'SDI1A DEGF LOOP 1' with a fill-in box containing '60'. To the right of this, there's a 'RANGE LIMITS' section with values like PVEXEUHI 1005.0, OPEXEUHI 102.00, PVEUHI 1000.0, OPEUHI 100.00, PVEULO 0.000, OPEULO 0.000, PVEXEULO -2.000, OPEXEULO -2.000. At the bottom, there's a section for 'OP% 37.7' and 'MAN'.
6	Press [ENTER] key and watch for output value to change to 60.0.

5.3 UDC 6000/6300 Detail Display, Continued

OP DATA page call up

Perform the steps in Table 5-11 to call up the OP DATA page from the FIRST page of the custom detail display.

Table 5-11 Calling Up OP DATA Page

Step	Action
1	Select FIRST PAGE .
2	Select OP DATA from page selections that appear on display. <u>Example: Selecting OP DATA page.</u>  The screenshot shows the 'FIRST PAGE' menu with the following options: FIRST, OP DATA (highlighted), and CONFIG. The background displays various process parameters including PV, SP, and OP values, as well as alarm limits and point data.
3	The screen that appears is the OP DATA page.  The screenshot shows the 'OP DATA PAGE' with detailed operating data, UDC connection status, tuning parameters, and firmware revisions.

Refer to Table 5-12 for a list of the parameters that can be changed.

5.3 UDC 6000/6300 Detail Display, Continued

Changeable parameters on OP DATA page of Reg Schematic

Table 5-12 lists the parameters on the OP Data page that an operator can change. These parameters can be changed when the point (PTEXECST) is active and BOXMODE is Full. Any not listed cannot be changed.

Table 5-12 Changeable Parameters on OP DATA Page

Parameter	Access	Value/Selection	Changeable
TOTAL	Engineer	allows reset of the Totalizer - RESET, NORESET	Yes
SPSOURCE	Engineer	LSP, RSP, CSP,CAS	Yes
LSP 1	Engineer	Within Setpoint limits	Yes
LSP 2	Engineer	Within Setpoint limits	Yes
LSP 3	Engineer	Within Setpoint limits	Yes
CASOPT	Engineer	SPCRSP, SPCCSP, DDCLSP, DDCCSP, DDCRSP ATTENTION For definition and selection of Cascade option, refer to subsection 3.9 <i>“Configuring PM or APM for Control Integration.”</i>	Yes
GAIN(K)	Engineer	0.001 to 1000	Yes
RESET	Engineer	0.02 to 50.00	Yes
RATE	Engineer	0.08 to 10.00 (0.0= Off)	Yes
OP DATA PAGE		FIRST, OP DATA, or CONFIG	Yes

Target to PV Point or OP Point Detail Display

You can go to the PV Point or OP Point detail display from the OP DATA PAGE. Select one of the following targets:

STIM PV POINT
POINT ID ----

to go to the PV Point detail display, see
“PV Point Detail Display” in this section.

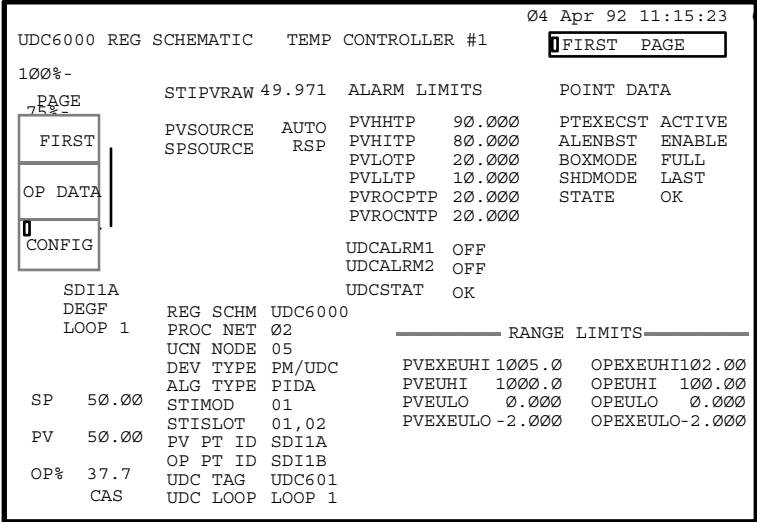
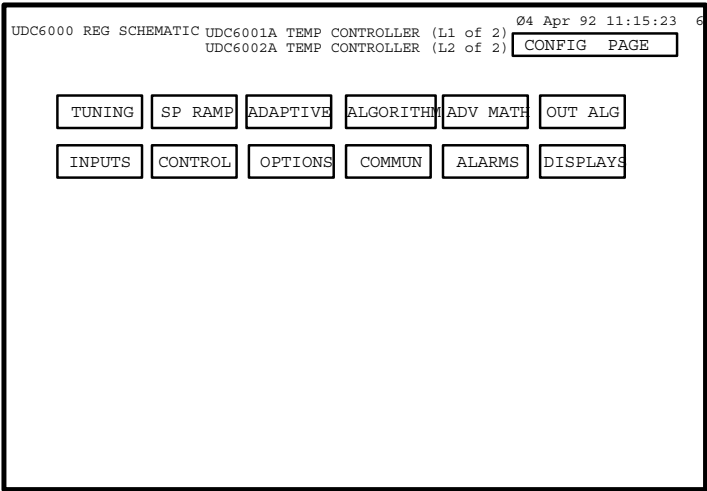
STIM OP POINT
POINT ID ----

to go to the OP Point detail display, see
“OP Point Detail Display” in this section.

5.3 UDC 6000/6300 Detail Display, Continued

CONFIG page call up Perform the steps in Table 5-13 to call up the CONFIG page from the FIRST page of the custom detail display.

Table 5-13 Calling Up CONFIG page

Step	Action
1	Select FIRST PAGE .
2	Select CONFIG from page selections that appear on display. <u>Example:</u> Selecting CONFIG Page.  The screenshot shows the 'UDC6000 REG SCHEMATIC' screen. At the top right is the date and time '04 Apr 92 11:15:23'. Below it, a box contains '1 FIRST PAGE'. The screen is divided into several sections: '100%-', 'PAGE 75%', 'STIPVRAW 49.971', 'ALARM LIMITS', 'POINT DATA', 'PVSOURCE AUTO', 'PVHHTP 90.000', 'PTEXCST ACTIVE', 'SPSOURCE RSP', 'PVHITP 80.000', 'ALENBST ENABLE', 'PVLOTP 20.000', 'BOXMODE FULL', 'PVLLTP 10.000', 'SHDMODE LAST', 'PVROCTPT 20.000', 'STATE OK', 'PVROCNTP 20.000', 'UDCALRM1 OFF', 'UDCALRM2 OFF', 'UDCSTAT OK', 'SDI1A', 'DEGF', 'LOOP 1', 'REG SCHM UDC6000', 'PROC NET 02', 'UCN NODE 05', 'DEV TYPE PM/UDC', 'PVEUHI 1005.0', 'OPEUHI 102.00', 'ALG TYPE PIDA', 'PVEUHI 1000.0', 'OPEUHI 100.00', 'STIMOD 01', 'PVEULO 0.000', 'OPEULO 0.000', 'STISLOT 01,02', 'PVEEULO -2.000', 'OPEEULO -2.000', 'PV PT ID SDI1A', 'OP PT ID SDI1B', 'OP% 37.7', 'UDC TAG UDC601', 'CAS UDC LOOP LOOP 1'. On the left side, a vertical menu shows 'FIRST', 'OP DATA', and 'CONFIG' (highlighted with a cursor). The bottom right corner of the screen displays '12513'.
3	The screen that appears is the Configuration Page.  The screenshot shows the 'UDC6000 REG SCHEMATIC' screen. At the top right is the date and time '04 Apr 92 11:15:23'. Below it, a box contains 'CONFIG PAGE'. The screen displays two rows of menu options: 'TUNING', 'SP RAMP', 'ADAPTIVE', 'ALGORITHM', 'ADV MATH', 'OUT ALG' and 'INPUTS', 'CONTROL', 'OPTIONS', 'COMMUN', 'ALARMS', 'DISPLAYS'. The bottom right corner of the screen displays '12434'. ATTENTION You can use this procedure to call up the CONFIG page if you select OP DATA page in step 2 by pressing OP DATA PAGE on that screen.

5.3 UDC 6000/6300 Detail Display, Continued

CONFIG page for loop 2 call up

If the point is for loop 1 of a 2-loop UDC 6000/6300 controller, use the steps in Table 5-14 to call up the CONFIG page with parameters that are exclusive to the configuration of loop 2. Note that configuration for both loops is done through the point associated with loop 1 of the controller.

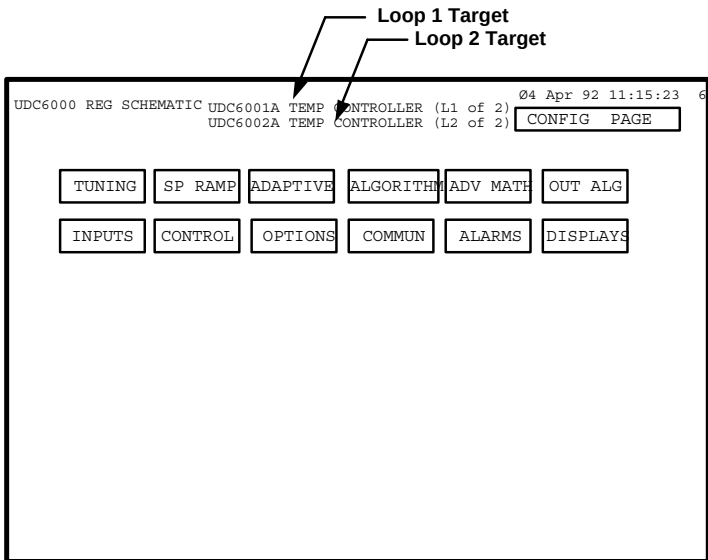
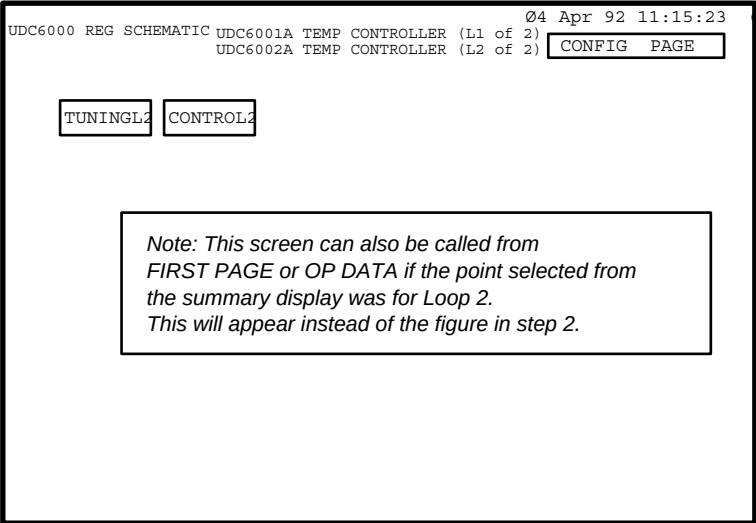
Table 5-14 Calling Up CONFIG Page for Loop 2

Step	Action
1	Call up CONFIG page per steps in Table 5-13.

5.3 UDC 6000/6300 Detail Display, Continued

CONFIG page for
loop 2 call up,
continued

Table 5-14 Calling Up CONFIG Page for Loop 2, Continued

Step	Action
2	Select Target (for example, UDC6002A TEMP CONTROLLER (L2 of 2)) listed for loop 2 (L2) <u>Example:</u> Moving cursor to point name for loop 2.  12514
3	The screen that appears is the CONFIG page for loop 2. <u>Example:</u> Calling up CONFIG page of loop 2.  12435
4	Move cursor to point name (for example, UDC6001A TEMP CONTROLLER (L1 of 2)) listed of loop 1 to return to CONFIG page for loop 1.

5.3 UDC 6000/6300 Detail Display, Continued

Typical usage of CONFIG page

A typical use of the CONFIG page could be to change the value of the alarm hysteresis configuration parameter. Use the steps in Table 5-15 to change an alarm configuration parameter.

While the steps in Table 5-15 are specific to alarm configuration, the general process outlined by the steps applies to any named configuration group. These groups and parameters parallel those that are used to configure the UDC 6000/6300 configuration database through its front face displays and keys. Thus, refer to the configuration section of the UDC 6000 or UDC 6300 product manual to find additional information about an equivalent parameter.

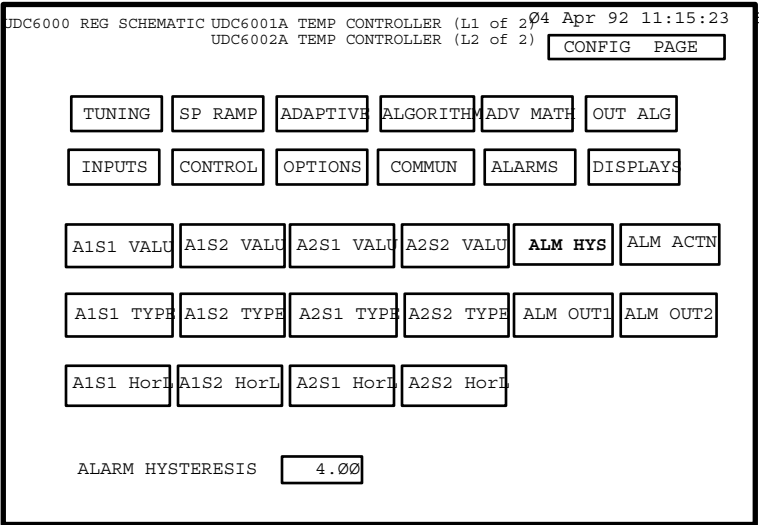
Table 5-15 Changing an Alarm Configuration Parameter

Step	Action
1	Select ALARMS target to call up associated configuration parameters.

5.3 UDC 6000/6300 Detail Display, Continued

Typical usage of
CONFIG page,
continued

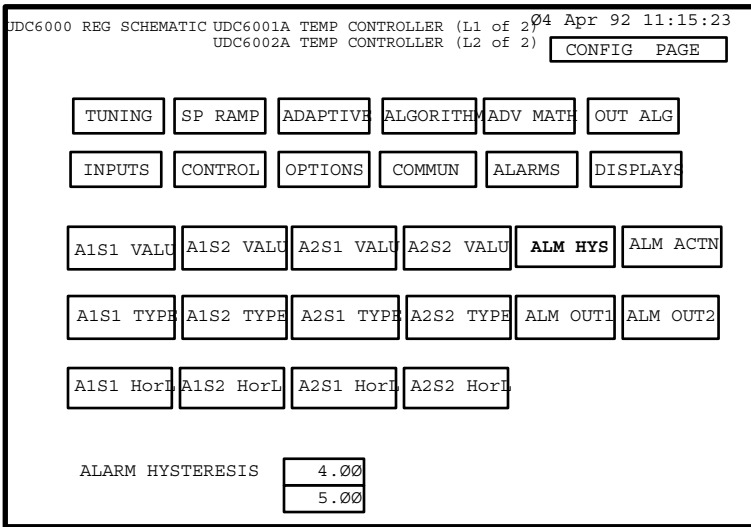
Table 5-15 Changing an Alarm Configuration Parameter, Continued

Step	Action
2	Select ALM HYS target to call up present configuration value in UDC 6000/6300. <u>Example:</u> Calling up a configuration parameter value.  The screenshot shows the UDC 6000 REG SCHEMATIC UDC6001A TEMP CONTROLLER (L1 of 2) and UDC6002A TEMP CONTROLLER (L2 of 2). The CONFIG PAGE is displayed with various menu options. The ALM HYS parameter is highlighted, and its value is shown as 4.00. 12515
3	Select current configuration value to open value entry box. ATTENTION The point (PTEXECST) listed on the first page must be inactive to make a change in a configuration parameter.

5.3 UDC 6000/6300 Detail Display, Continued

Typical usage of
CONFIG page,
continued

Table 5-15 Changing an Alarm Configuration Parameter, Continued

Step	Action
4	Key in desired alarm hysteresis value and press [ENTER] key. <u>Example:</u> Keying in new configuration value.  The screenshot shows the UDC 6000 REG SCHEMATIC UDC6001A TEMP CONTROLLER (L1 of 2) and UDC6002A TEMP CONTROLLER (L2 of 2) interface. The top right corner displays the date and time: 04 Apr 92 11:15:23. The CONFIG PAGE is selected. The menu structure is as follows: - TUNING - SP RAMP - ADAPTIVE - ALGORITHM - ADV MATH - OUT ALG - INPUTS - CONTROL - OPTIONS - COMMUN - ALARMS - DISPLAYS - A1S1 VALU - A1S2 VALU - A2S1 VALU - A2S2 VALU - ALM HYS - ALM ACTN - A1S1 TYPE - A1S2 TYPE - A2S1 TYPE - A2S2 TYPE - ALM OUT1 - ALM OUT2 - A1S1 HorL - A1S2 HorL - A2S1 HorL - A2S2 HorL The ALM HYS menu is highlighted, showing the following options: - ALARM HYSTERESIS 4.00 5.00 12516
5	ATTENTION Refer to configuration section in the UDC 6000 or UDC 6300 product manual for applicable value entry ranges. Watch for new value to appear after it is written in UDC 6000/6300 database.
6	Select CONFIG PAGE to call up FIRST or OP DATA page.

5.4 Standard Detail Display

Description

The standard Detail display includes the same type of data as found on the custom detail display, but it is generated for an STI IOP point. Since the UDC 6000/6300 SDI IOP point is configured as an STI IOP point, you can access the standard detail display by entering a valid SDI IOP point identification (STITAG).

The second page or config page of the standard detail display lists standard STI configuration parameters, which are mapped SDI parameters, and provides access to the command display. Refer to Table 3-3 for cross-reference parameters.

While you can interact with the UDC 6000/6300 through the config page parameters and command display, we recommend that you use the custom schematics for all interaction with a UDC 6000/6300 loop.

Display call up

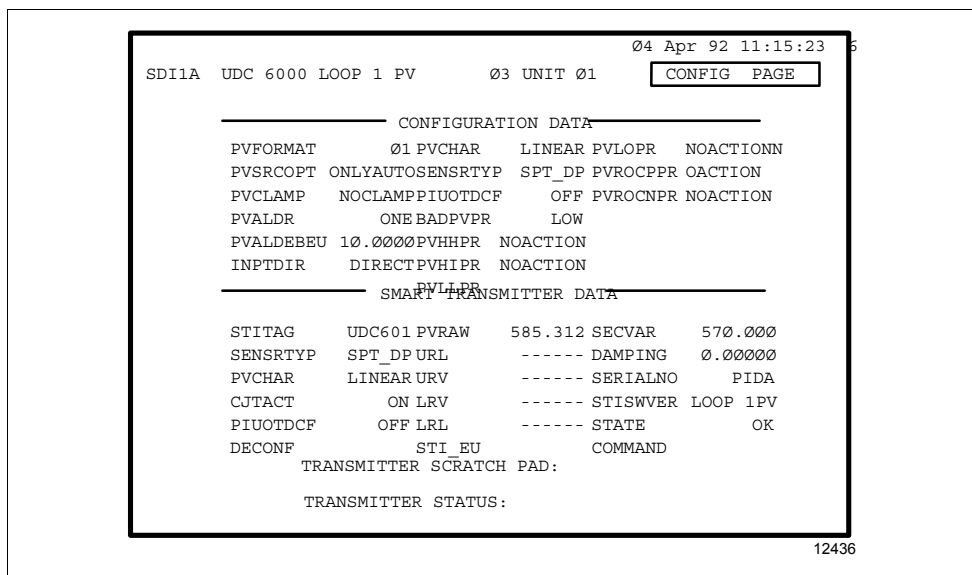
You can call up the first page of the standard detail display by pressing the dedicated [detail] key on the US and entering the desired point tag in response to the onscreen prompt .

You can also access the Standard Detail Display through the “*OP Point Detail Display*” or the “*PV Point Detail Display*.”

Page 2 reference

Figure 5-4 shows a typical page 2 (CONFIG PAGE) of a standard detail display for reference. The Configuration Data and Smart Transmitter Data are applicable to mapped SDI parameters for the identified UDC 6000/6300 loop built as an STI point. Refer to Tables 3-4 (PV Point Data) and 3-5 (OP Point Data) for definitions of mapped SDI parameters.

Figure 5-4 Typical CONFIG PAGE for Standard Detail Display



5.4 Standard Detail Display, Continued

Command display reference

Figure 5-5 shows a command display called up on the CONFIG PAGE for reference. These commands are mapped through the SDI IOP to provide operator interaction with a UDC 6000/6300 loop through custom schematics. Thus, use the custom schematics for all interface with UDC 6000/6300 loop parameters.

Refer to Table 5-16 for a definition of the mapped command parameters.

Figure 5-5 Command Display for Reference

04 Apr 92 11:15:23

SDI1A UDC 6000 LOOP 1 PV
03 UNIT 01

CONFIG PAGE

COMMAND
 NONE
 DNLOADDB
 UPLOADDB
 SET_LRV
 SET_URV
 COR_LRV
 COR_URV
 COR_INPT
 MORE

CONFIGURATION DATA

FORMAT	01	PVCHAR	LINEAR	PVLOPR	NOACTION
SRCOPT	ONLYAUTO	SENSRTYP	SPT_DP	PVROCPPR	NOACTION
CLAMP	NOCLAMP	PIUOTDCF	OFF	PVROCNPR	NOACTION
ALDR	ONE	BADPVPR	LOW		
ALDEBEU	10.0000	PVHHPR	NOACTION		
PTDIR	DIRECT	PVHIPR	NOACTION		
CUTOFF	-----	PVLLPR	NOACTION		

SMART TRANSMITTER DATA

ITAG	UDC601	PVRAW	585.312	SECVAR	570.000
NSRTYP	SPT_DP	URL	-----	DAMPING	0.00000
CHAR	LINEAR	URV	-----	SERIALNO	PIDA
FACT	ON	LRV	-----	STISWVER	LOOP 1PV
JOTDCF	OFF	LRL	-----	STATE	OK
CONF	pv_sv_db	STI_EU	INH20	COMMAND	NONE

TRANSMITTER SCRATCH PAD:

 TRANSMITTER STATUS:

12437

Command definitions

Table 5-16 provides a definition of the mapped SDI interface commands used in the command display shown in Figure 5-5 for reference.

Table 5-16 Command Definitions for Mapped SDI Interface

Command	Definition
None	Serves as a non busy signal to coordinate actions among multiple users.
DNLOADDB	Works in conjunction with URV to write a configuration parameter value to UDC 6000/6300.
UPLOADDB	Read a parameter value from UDC 6000/6300.
SET_LRV	CAS
SET_URV	AUTO
COR_LRV	MAN
COR_URV	BCAS
COR_INPT	Monitor UDC inputs.
RST_COR	Undefined

164

PM/APM - UDC 6000 Integration

08/04

5.5 UDC 6000/6300 Group Display

Description

The UDC 6000/6300 Group display is a custom schematic, which provides up to eight UDC 6000/6300 loops for simultaneous display and adjustment.

Display call up

You call up the UDC 6000/6300 Group display through the UDC 6000/6300 Summary display. Refer to the previous summary display call up procedure in Table 5-1 for details.

Changeable parameters

Table 5-17 lists the parameters that an operator can change when the point (PTEXECST) is active and BOXMODE is control.

Table 5-17 Changeable Parameters on Group Display

Parameter	Value/Selection
SP	Within setpoint limits
PV	Within process variable limits only when PVSOURCE is MAN
OP%	Within output limits only when control mode is MAN
control mode	MANr AUTO, CAS

Typical usage of Group display

A typical use of the Group display could be to change the value of the setpoint for a given loop.

Use the steps in Table 5-18 to make a change in a setpoint value.

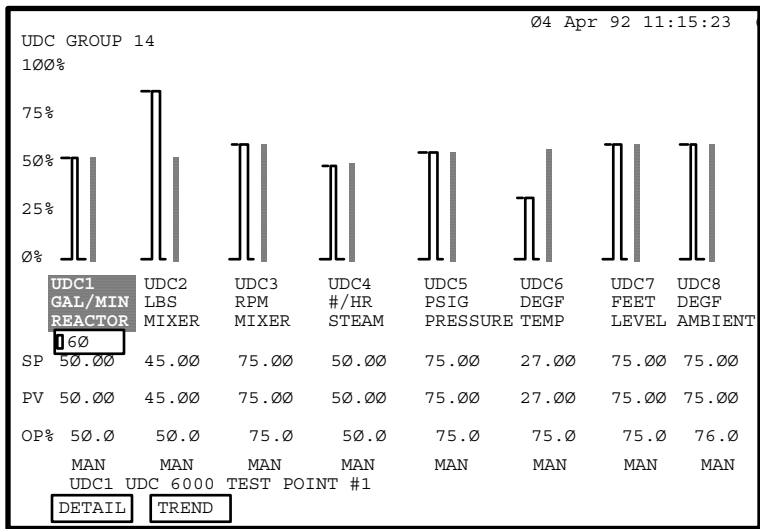
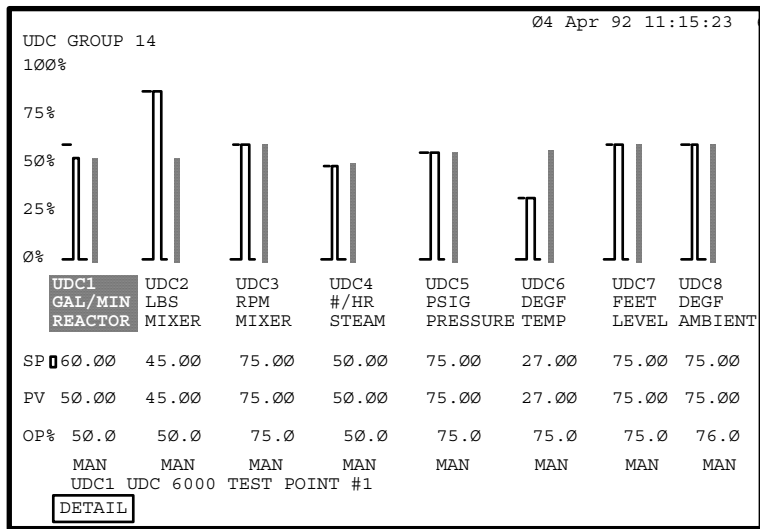
Table 5-18 Changing Set Point Value

Step	Action
1	Select SP value for loop one (UDC1 in sample Group display).

5.5 UDC 6000/6300 Group Display, Continued

Typical usage of
Group display,
continued

Table 5-18 Changing Set Point Value, Continued

Step	Action
2	Key in [60] as desired SP value. <u>Example:</u> Keying in new setpoint value.  The screenshot shows the UDC GROUP 14 display with a bar chart at the top. Below the chart, the UDC1 REACTOR is highlighted. The SP value is shown as 60.00, with a box around the '60' indicating it was just entered. The display also shows PV, OP%, and MAN values for all UDCs.
3	Press [ENTER] key and watch for SP value and display to change. <u>Example:</u> Checking display for changes  The screenshot shows the same UDC GROUP 14 display, but now the SP value for UDC1 is 60.00, and the '60' is no longer highlighted. The display also shows PV, OP%, and MAN values for all UDCs.
4	Select DETAIL target to call up custom detail display for selected loop or press [PREV DISPLAY] key return to summary display.

5.6 UDC 6000/6300 Operator Change Zone Display

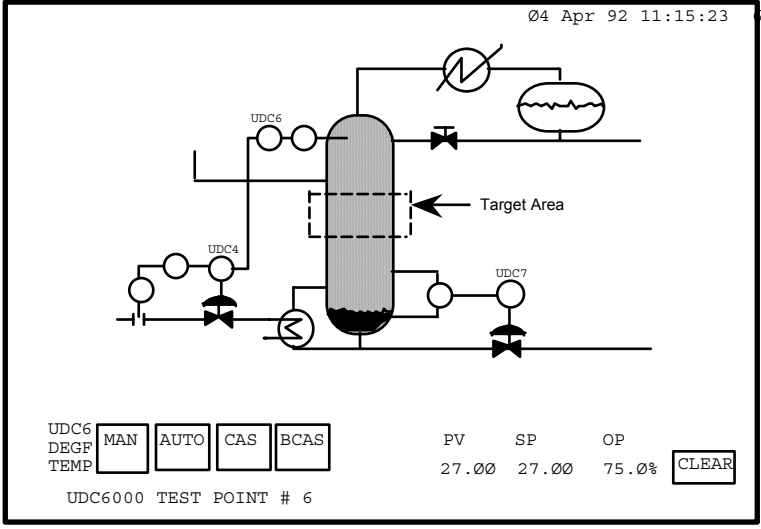
Description

The UDC 6000/6300 Operator Change Zone display is a custom schematic that provides operator interaction with a UDC 6000/6300 directly from a graphic process display.

Display call up

Table 5-19 outlines the typical steps for calling up a custom operator change zone display.

Table 5-19 Calling Up a Custom Operator Change Zone Display

Step	Action
1	Call up desired plant graphic display.
2	Select UDC loop related graphic target to call up custom operator change zone display. Example: Calling up custom operator change zone display. 

Changeable parameters

Table 5-20 lists the parameters that an operator can change when the point (PTEXECST) is active and BOXMODE is Full.

Table 5-20 Changeable Parameters on Change Zone Display

Parameter	Value/Selection
SP	Within setpoint limits
PV	Within process variable limits only when PVSOURCE is MAN
OP%	Within output limits only when control mode is MAN
control mode	MAN, AUTO, Cascade, Backup Cascade

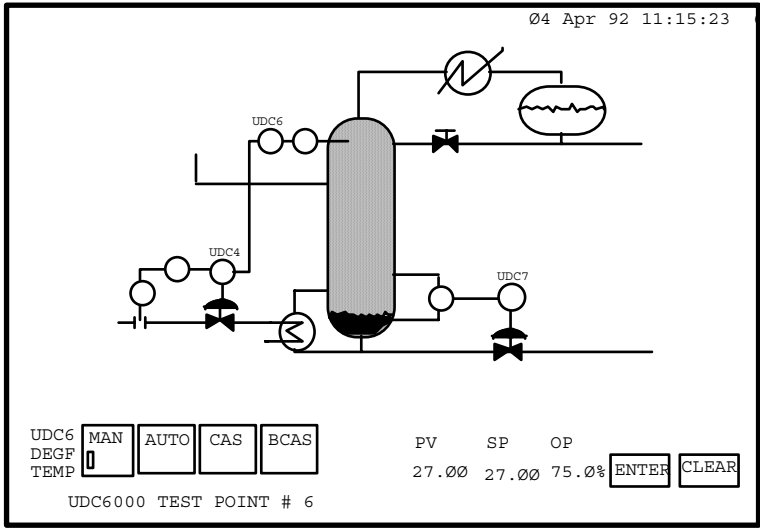
5.6 UDC 6000/6300 Operator Change Zone Display, Continued

Typical usage of change zone display

A typical use of the UDC 6000/6300 operator change zone display could be to adjust the output manually.

Use the steps in Table 5-21 to make a manual output adjustment assuming that the present control mode is AUTO.

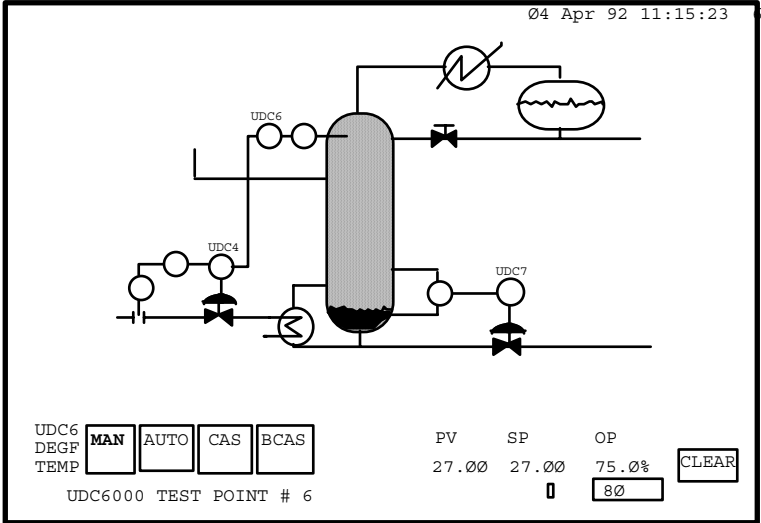
Table 5-21 Adjusting Output Manually from Change Zone Display

Step	Action
1	Select control mode MAN .
2	Select flashing ENTER target and wait for MAN target to be highlighted. <u>Example:</u> Changing control mode. 
3	Select OP value.

5.6 UDC 6000/6300 Operator Change Zone Display, Continued

Typical usage of
change zone display,
continued

Table 5-21 Adjusting Output Manually from Change Zone Display, Continued

Step	Action
4	Key in [80] as desired output in percent and press [ENTER] key. <u>Example:</u> Keying in desired output value. 
5	Watch for OP value to change to 80.0% on display.
6	Select CLEAR to remove change zone from display, or select tag to call up custom detail display.

Section 6 – Service

6.1 Overview

Purpose of this section

This section provides servicing information and diagnostics for the software associated with PM/APM - 6000/6300 Integration.

Reader assumptions

You are familiar with diagnostic and troubleshooting sections of the **TotalPlant** Solution (TPS) system and the UDC 6000/6300 controllers.

6.2 Diagnostics

Introduction

This section covers setup and operation diagnostics for the PM/APM - UDC 6000/6300 integration.

Background

The UDC 6000/6300, FTA, SDI IOP and PM/APM perform individual internal diagnostics to verify their own operating status. The tables in this section contain information on how to troubleshoot these particular devices as well as STIM Point Detail and Custom Schematic problems.

Universal Station— system status

Table 6-1 contains diagnostics for the Universal Station—System Status.

Table 6-1 Universal Station—System Status

Indication	Problem	Action
System Status indicates a No Response (NR) failure for STIM module.	No Response failures are generated when the PM/APM is not able to communicate to the STIM IOP.	Check whether STIM IOP is inserted in configured IOP file.
System failure indicates a soft failure (SF) for STIM module.	The IOP has determined that the database for an FTA is not communication or a slot data is invalid.	Check Device slot failures will be present if the PM detects a problem with SDI IOP card.
System status indicates Input Slot soft failures (SF) for STIM slot.	FTA is not receiving data from UDC.	Check Transmitter status on second page of detail display of slot with soft failure.

SDI IOP diagnostics

Table 6-2 contains the SDI IOP diagnostics.

Table 6-2 SDI IOP Diagnostics

Indication	Problem	Action
SDI IOP STATUS LED is OFF.	IOP failure.	Check for device soft failures.
SDI IOP STATUS LED is ON.	IOP is communicating to all active slots.	
SDI IOP STATUS LED is blinking.	IOP is not receiving data from at least one active slot.	Check FTA, power adapter, and status indications. Check for STIM slot soft failures.

ATTENTION

Slot soft failure indications are cleared by the SDI IOP, one at a time, with a 20-second delay between each. Therefore, when multiple soft failures are detected at an IOP, the clearing of all soft failure indications may take a few minutes.

6.2 Diagnostics, Continued

UDC 6000/6300 FTA diagnostics

Table 6-3 contains the UDC 6000/6300 FTA diagnostics.

Table 6-3 UDC 6000/6300 FTA Diagnostics

Indication	Problem	Action
FTA PWR LED is ON.	FTA is receiving power from IOP.	
FTA PWR LED is OFF.	FTA is not receiving power from IOP.	1. Check fuses on PM motherboard for SDI IOP slot. 2. Check cable connection from IOP to power adapter. 3. Check cable connection from power adapter to FTA. 4. Check fuse on SDI IOP module (PWB). 5. Replace power adapter. 6. Replace FTA.
FTA STATUS LED is OFF.	FTA module not operating.	Replace FTA module.
FTA STATUS LED is blinking at a slow rate (once per second).	IOP to FTA communication error.	1. Check cable connections between IOP and power adapter. 2. Check cable connections between power adapter and FTA. 3. Check that IOP is Serial Device Interface (SDI) module. 4. Replace SDI IOP.
FTA STATUS LED is blinking at a fast rate (four times per second).	FTA to UDC 6000/6300 communication failure. At least one active SDI point is not in communication with a UDC 6000/6300 assigned to this FTA.	1. Check for IOP slot soft failures to determine which slots are in failure. Soft failures will appear for active slots only. 2. Check Detail page of soft failure points in failure for diagnostic messages.
FTA STATUS LED is ON (solid).	All active points are in communication with UDC 6000/6300(s).	Soft failures should clear. If soft failure persists, reset IOP and recheck soft failures.

6.2 Diagnostics, Continued

STIM point detail diagnostics

Table 6-4 contains diagnostics for the STIM Point Detail.

Table 6-4 STIM Point Detail Diagnostics

Indication	Problem	Action
STATE is OK.	No diagnostic errors are detected.	Soft failure should clear.
STATE is LOADING.	FTA is in the process of loading database configuration information from the UDC 6000/6300.	Should clear.
STATE is LOADCOMP.	FTA has successfully loaded UDC database information.	Should clear.
STATE is LOADFAIL	FTA has not been able to load configuration data from the UDC 6000/6300.	Check UDC 6000/6300 for status errors. Refer to device diagnostics in UDC 6000/6300 user manual.
STATE is DBCHANGE.	The FTA does not have valid data or was unable to load valid configuration data from the UDC 6000/6300 mapped to this slot.	Check TRANSMITTER STATUS for database discrepancy diagnostic message.
TRANSMITTER STATUS indicates DATABASE DISCREPANCY: "SDI RS485 COMM TIMEOUT"	There has been no response to this slot from any device on the RS485 link.	1. Check RS485 physical connection from the FTA to the UDC 6000/6300. 2. Check that the UDC 6000/6300 communication option is set to "RS485 TDC." 3. Check that PV and OP address assignments at the UDC 6000/6300 are correct. 4. Check that all address assignments are not mapped to address in other UDC 6000/6300s connected on the same RS485 link. 5. Check IOP and FTA STATUS LEDs. Custom regulatory schematic indicates "POINT CONNECTION CONFLICT."

6.2 Diagnostics, Continued

Custom schematic diagnostics

Table 6-5 contains Custom Schematic diagnostics.

Table 6-5 Custom Schematic Diagnostics

Indication	Problem	Action
Custom Schematics UDC 6000/6300 Detail display indicates CONNECTION CONFLICT.	The custom schematic has detected that the configuration data from the SDI points assignments selected for this loop are not mapped correctly.	1. Check CONNECTION STATUS on OP DATA page of Custom Schematics. 2. Check that all PV and OP points assigned to this schematic do not have DECONF = ANALOG.
CONNECTION STATUS indicates that the PV and OP point TAGNAMES are different.	The PV and OP points have not established connection to the same UDC 6000/6300.	Check that both the PV and OP point are active and that their states are OK.
CONNECTION STATUS indicates that the LOOP is incorrect.	The UDC loop assignments are not consistent.	Check UDC address to IOP slot assignments.
CONNECTION STATUS indicates that a PV or OP POINT type is incorrect.	The type of connection to a UDC address assignment for the PV or OP point is not that expected.	Check UDC address to SDI IOP slot assignments.

6.3 Removing a UDC 6000/6300 from an Active Link

Introduction

At times, it may be necessary to remove a UDC 6000/6300 from an active link for servicing.

In order to do this, you must deactivate all points associated with that UDC 6000/6300. By doing this, you will avoid soft failure alarms.

Up to 4 points can be mapped to a UDC so you should know in advance the number of points associated with this particular UDC.

Procedure

Follow the procedure listed in Table 6-6 to deactivate all points associated with the UDC to be serviced.

Table 6-6 Removing a UDC from an Active Link

Step	Action																																				
1	Press key designated for UDC 6000/6300 schematics during setup to call up Summary display. <u>Example:</u> Calling up Summary display. 04 Apr 92 10:05:23 UDC 6000 SUMMARY DISPLAY <table><tr><th>POINT ID</th><th>DESCRIPTOR</th><th>GROUP</th><th>DESCRIPTOR</th></tr><tr><td>UDC6001A</td><td>TEMP CONTROLLER (L1 OF 2)</td><td>014</td><td>UDC GROUP 14</td></tr><tr><td>UDC6002A</td><td>TEMP CONTROLLER (L2 OF 2)</td><td>015</td><td>UDC GROUP 15</td></tr><tr><td>UDC6003A</td><td>TEMP CONTROLLER (L1 OF 1)</td><td></td><td></td></tr><tr><td>UDC6004A</td><td>TEMP CONTROLLER (L1 OF 1)</td><td></td><td></td></tr><tr><td>UDC6005A</td><td>TEMP CONTROLLER (L1 OF 1)</td><td></td><td></td></tr><tr><td>UDC6006A</td><td>TEMP CONTROLLER (L1 OF 1)</td><td></td><td></td></tr><tr><td>UDC6007A</td><td>TEMP CONTROLLER (L1 OF 1)</td><td></td><td></td></tr><tr><td>UDC6008A</td><td>TEMP CONTROLLER (L1 OF 1)</td><td></td><td></td></tr></table> PAGE 1	POINT ID	DESCRIPTOR	GROUP	DESCRIPTOR	UDC6001A	TEMP CONTROLLER (L1 OF 2)	014	UDC GROUP 14	UDC6002A	TEMP CONTROLLER (L2 OF 2)	015	UDC GROUP 15	UDC6003A	TEMP CONTROLLER (L1 OF 1)			UDC6004A	TEMP CONTROLLER (L1 OF 1)			UDC6005A	TEMP CONTROLLER (L1 OF 1)			UDC6006A	TEMP CONTROLLER (L1 OF 1)			UDC6007A	TEMP CONTROLLER (L1 OF 1)			UDC6008A	TEMP CONTROLLER (L1 OF 1)		
POINT ID	DESCRIPTOR	GROUP	DESCRIPTOR																																		
UDC6001A	TEMP CONTROLLER (L1 OF 2)	014	UDC GROUP 14																																		
UDC6002A	TEMP CONTROLLER (L2 OF 2)	015	UDC GROUP 15																																		
UDC6003A	TEMP CONTROLLER (L1 OF 1)																																				
UDC6004A	TEMP CONTROLLER (L1 OF 1)																																				
UDC6005A	TEMP CONTROLLER (L1 OF 1)																																				
UDC6006A	TEMP CONTROLLER (L1 OF 1)																																				
UDC6007A	TEMP CONTROLLER (L1 OF 1)																																				
UDC6008A	TEMP CONTROLLER (L1 OF 1)																																				

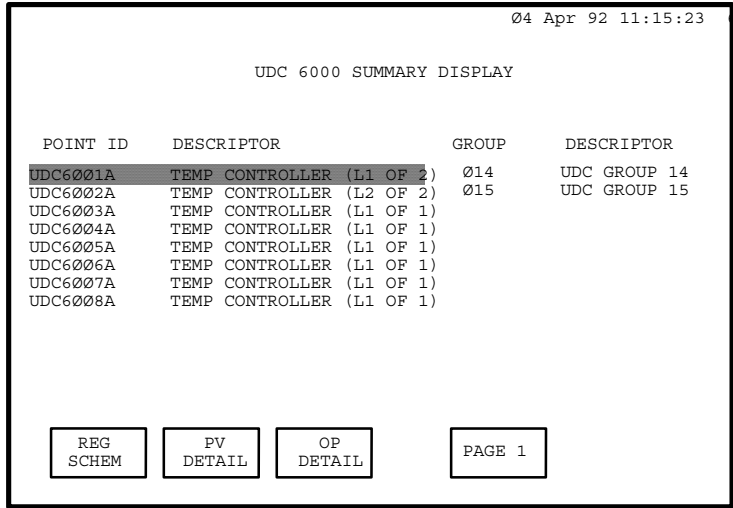
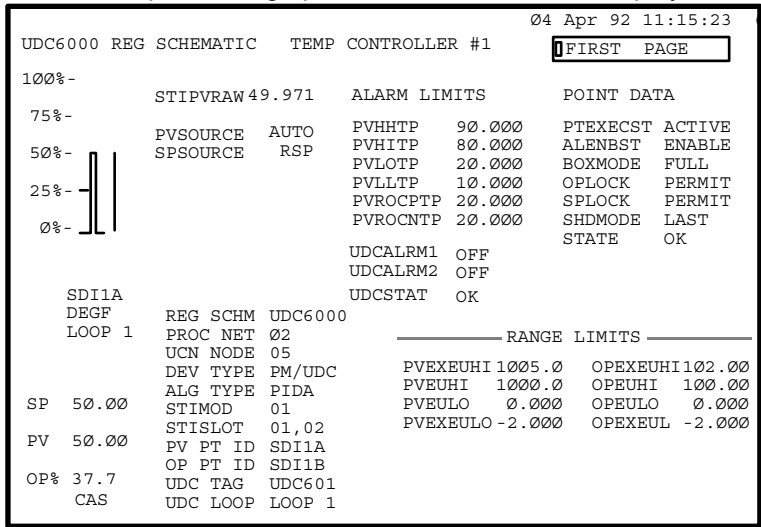
12443

6.3 Removing a UDC 6000/6300 from an Active Link,

Continued

Procedure, continued

Table 6-6 Removing a UDC from an Active Link, Continued

Step	Action
2	Select desired point ID. <u>Example: Selecting desired point ID.</u>  12423
3	Select DETAIL target to call up detail display for selected point. <u>Example: Calling up UDC 6000/6300 Detail display.</u>  12432
4	Select PTEXECST on the detail display. Change the selection to INACTIVE ATTENTION You must repeat this procedure for each point associated with the UDC being removed from the link. To reinstall the UDC onto the link, Change PTEXECST to ACTIVE for all points.

Index

A

- adding a loop to the group display, 93
- adding a point ID or group descriptor to the summary display, 76
- Address vs Slot Configuration, 43
- alarm function data, 62
- ALENBST, 145, 148, 152
- AM/APM - UDC 6000/6300
 - interface connections, 112

B

- Basic mode, 7
- box data point, 35;
 - build, 35;
 - Procedure, 35
- BOXMODE, 152, 156, 165, 167
- Button configuration, 105

C

- C1: SP Point SDI parameter definition, 44
- C2: SP Point SDI parameter definition, 44
- Cascade Options Configuration, 120
- Cascade Terms and Definitions, 118
- CASOPT, 156
- Changeable parameters on FIRST
 - page of Reg Schematic, 152
- Changeable parameters on OP DATA
 - page of Reg Schematic, 156
- CJTACT: SP Point SDI parameter definition, 44
- COMMAND: DC Point SDI parameter definition, 50
 - SP Point SDI parameter definition, 45
- communications link: definition, 2
 - multidrop configuration, 2
- Communications Link: integration hierarchy, 3
- CONFIG page, 157
 - call up, 157
 - loop 2 call up, 158
 - Typical usage, 160
- configuration process, 34
- control mode, 152, 167
- COR_URV, 164
- COR_INPT, 164
- COR_LRV, 164
- custom graphics software: Loading, 31

D

- DAMPING: DC Point SDI parameter definition, 50
 - SP Point SDI parameter definition, 45
- data exchange functions: AI points, 6
 - Basic and Full modes, 7
 - SDI IOP parameters, 5
- Data Exchange Functions, 4

- DDCCSP cascade: control mode definitions, 135
 - Description, 134
- Limiting integral windup, 134
 - output limit checks, 134
- PIDERFB cascade, 137
 - PIDERFB point configuration, 136
 - Setpoint limit checks, 134
- DDCCSP cascade, 134
- DDCLSP cascade: control mode definitions, 129
 - Output limit checks, 129
 - PIDERFB point configuration, 130
 - STIM SDI point connections block diagram, 130
 - UDC 6000/6300 output adjustments, 128
- DDCLSP cascade, 128
- DDCRSP cascade, 131
 - BCAS of PM/APM primary, 133
 - control mode definitions, 132
 - Description, 131
 - output limit checks, 131
 - point onfiguration, 133
- DECONF: DC Point SDI parameter definition, 50
 - SP Point SDI parameter definition, 45
- Detail display: config page for loop 2, 158
- Detail Display: config page callup, 157
 - config page usage, 160;
 - typical usage, 153
- Device tag name, 114
- diagnostics, 172
- digital communications link, 10, 24
 - Cable selection guidelines, 10
 - guidelines, 26
 - wire, 24
- directory pathname, 30, 31
- directory/volume, 27
 - selection, 27
- DNLOADDB, 164

E

- EUDESC: selection, 56

F

- FIRST PAGE, 145, 148, 152
- FTA, 17
 - install, 17
- Full mode, 7

Index

G

GAIN(K), 156
Group Descriptors, 73
Group display: Typical usage, 165
Group Display: configuring loops, 83
 configuring, 81
 entering tag names, 86
 loop subpicture information, 88
 point tag names, 89
 subpicture information, 86
 tag name, 86
 trend display information, 91

H

HM volume, 28
 selecting, 28

I

Identifying Mapped SDI Parameters, 41
INPT DIR: SP Point SDI parameter definition, 46
INPTDIR: selection, 60
installation tasks, 14
integration components: identification, 8
Integration Components: 9
 part numbers, 9
 summary, 7
IOM configuration data, 39
IOP Slot vs RS485 Address, 42

K

KEYWORD: selection, 56

L

Loop number, 114
LRL: SP Point SDI parameter definition, 46
LRV: SP Point SDI parameter definition, 46
LSP 1, 156
LSP 2, 156
LSP 3, 156

M

Mapped Data Types, 42
mapped SDI parameters, 41

N

NIM Build Type Select Menu, 36
 call up, 36
NIM Point Assignment, 56
Node Specific Point, 40
NODENUM: selection, 56
NODETYP: selection, 56
NTWKNUM: selection, 56

O

OP DATA PAGE, 156
OP point SDI definitions, 49
OP%, 152, 165, 167
Operation, 139
operator change zone: add, 97
Operator Change Zone, 97
 configure, 97
 entering subpicture coordinates, 100
 entering tag names, 102
 subpicture information, 101
 target area selection, 105
Operator Change Zone display: Typical usage, 168
Operator Change Zone display: call up, 167
 Changeable parameters, 167
Operator Change Zone: , 99, 100, 101, 102, 103, 104, 105
OPHILIM, 153
output adjustment, 153

P

Parameter Entry Display (PED), 36
PIDERFB, 116
 input connections, 116
 output connections, 116
 related parameters, 117
PIUOTDCF: SP Point SDI parameter definition, 46
PM Status Display: call up, 107
PM/APM - UDC 6000/6300 Integration, 2
PM/APM Interface Verification, 114
PM/APM Status Display, 107
PM/UDC 6000/6300 Integration: hierarchy, 3
PNTFORM: selection, 56
PNTMODTYP: selection, 57
point name, 37
 checking, 37
point slot type, 114
power adapter, 16, 19
 install, 16
 wiring guidelines, 22
 wiring to FTA, 19

Index

Preparing SDI for Operation, 106
 procedure, 106
 validating SDI database, 108
PTDESC: selection, 56
PTEXECST, 145, 148, 152
Purpose of this section, 1
PV, 152, 165, 167
PV and OP point configuration, 112
PV point SDI definitions, 44
PVCHAR: selection, 58
PVCLAMP: selection, 60
PVEUHI, 153
 selection, 60
 SP Point SDI parameter definition, 46
PVEULO, 153
 checking, 60
PVHIHITP, 145, 148, 152
PVHITP, 145, 148, 152
PVLLTP, 145, 148, 152
PVLOTP, 145, 148, 152
PVROCNP, 145, 148, 152
PVROCPTP, 145, 148, 152
PVSOURCE, 145, 148, 152
 selection, 61
PVSRCOPT: selection, 60

R

RATE, 156
reconstitute function, 40
Removing a UDC 6000/6300 from an Active Link, 176
Removing a UDC from an Active Link, 176
RESET, 156
RST_COR, 164

S

SDI database, 108; save, 109; validate, 108
SDI IOP, 15, 18, 109
 install, 15
 start, 109
 wire, 18
SDI IOP as STI IOP, 5
SDI IOP point: Procedure, 55
SDI IOP POINT: loading, 63
SDI IOP point building: overview, 54
 procedure, 55
SDI IOP points, 43
SDI IOP points, 54
SDI points, 106; load, 106
SDI STIM AI points, 112
Selecting group target, 142
SENSRTYP: selection, 58
Service, 171
set up, 33
SET_LRV, 164

SET_URV, 164
SLOTNUM: selection, 57
SP, 152, 165, 167
SPCCSP cascade: control mode definitions, 126
 Description, 125
 Limiting integral saturation, 126
 Output limits, 126
 PIDERFB point configuration, 127
 Setpoint limit checks, 125
 STIM SDI point connections block diagram, 127
SPCCSP cascade, 125
SPCRSP cascade: cascade with PM/APM primary, 123
 cascade with UDC 6000/6300 primary, 124
 control mode definitions, 122
 Description, 121
 PIDERFB point connections, 123
 RSP limit checks, 122
SPCRSP cascade, 121
SPSOURCE, 156
Standard Detail display, 163
 call up, 163
 Command definitions, 164
 Page 2 reference, 163
STATE, 152
 DC Point SDI parameter definition, 53
 SP Point SDI parameter definition, 48
STI EU: selection, 58
STI IOP points, 54
STI TAG: DC Point SDI parameter definition, 53
 SP Point SDI parameter definition, 48
STI_EU: DC Point SDI parameter definition, 53
 SP Point SDI parameter definition, 48
STISWVER: DC Point SDI parameter definition, 53
 SP Point SDI parameter definition, 48
subpicture information: call up, 68, 74, 86, 89, 101
Summary display, 71
 call up, 71
 selecting detail target, 141
Summary Display: compiling, 75; deleting lines, 70
 group subpicture information, 74
 modifying group descriptors, 73
 point tag names, 68
 saving subpicture, 71
 selecting group, 142
 selecting point ID for detail, 141
 subpicture information, 68
 variant body, 72
Summary display file, 75
 compile, 75
System Status display, 106
 call up, 106
the subpicture information, 68, 74, 86, 89, 101

Index

T

TOTAL, 156

U

UCN Status Display, 107
 call up, 107
UDC 6000 Detail displays, 65
UDC 6000 Summary displays, 65
 configure the Group displays, 81
 Procedure for configuring the group display, 81
 Procedure for configuring the summary display, 65
UDC 6000/6300 Box Mode, 7
UDC 6000/6300 controller, 23
 configuration, 64
 install, 23
UDC 6000/6300 controller, 64
UDC 6000/6300 Detail display, 143, 150
 OP DATA, 155
 CONFIG, 157
 FIRST page, 153
UDC 6000/6300 Detail displays, 65
UDC 6000/6300 graphics software, 27
UDC 6000/6300 Group display, 165
 call up, 165
 Changeable parameters, 165
UDC 6000/6300 Operator Change Zone display, 167
UDC 6000/6300 Regulatory Schematic display, 150
UDC 6000/6300 SDI Point Connections, 112
UDC 6000/6300 Summary display, 140
 callup, 140
 selections, 140
UDC 6000/6300 Summary displays:
 UDC Summary Display, 65
UDC 6000/6300/SDI Interface, 42
UDC ALARM 1, 152
UDC ALARM 2, 152
UDC STAT, 152
UNIT selection, 56
UPLOADDB, 164
URL: STI DC Point SDI parameter definition, 53
 STI SP Point SDI parameter definition, 48
URV: STI DC Point SDI parameter definition, 53
 STI SP Point SDI parameter definition, 48

V

variant at 0, 336
 call up, 72
variant at 0, 96, 72

